



**DIVISION OF
ASTRONOMY AND ASTROPHYSICS**
Faculty of Mathematics, Physics and Informatics
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Bratislava



(NIE LEN) VESMÍRNE MISIE 2026

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RNDr. Roman Nagy, PhD.

Space Lab Summer School 2026



PREHLAD



PREHLAD

○ Ako vidíme vesmír?



PREHLAD

- Ako vidíme vesmír?
- To najlepšie, čo (doma) máme



PREHLAD

- Ako vidíme vesmír?
- To najlepšie, čo (doma) máme
- Dáta, archívy a AI (?)





ČASŤ 1

Ako vidíme vesmír?



RÔZNORODÉ VESMÍRNE OČI



RÔZNORODÉ VESMÍRNE OČI

- Fotometria



RÔZNORODÉ VESMÍRNE OČI

- Fotometria
- Astrometria



RÔZNORODÉ VESMÍRNE OČI

- Fotometria
- Astrometria
- Spektroskopia



RÔZNORODÉ VESMÍRNE OČI

- Fotometria
- Astrometria
- Spektroskopia
- Gravitačné šošovky



RÔZNORODÉ VESMÍRNE OČI



- Fotometria
- Astrometria
- Spektroskopia
- Gravitačné šošovky
- Gravitačné vlny

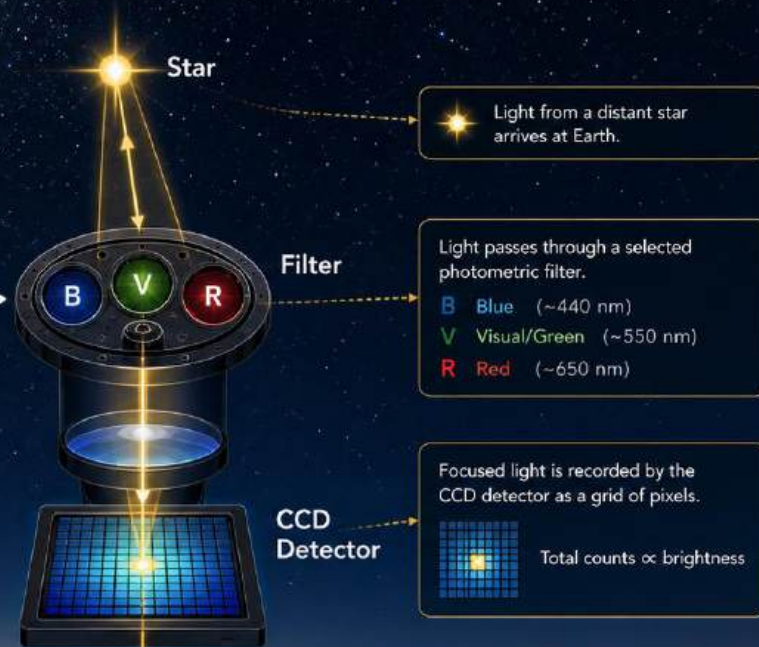


FOTOMETRIA



PHOTOMETRY

Measuring the brightness of astronomical objects as a function of time.



Light Curve



Brightness vs. time reveals variations such as a transit or eclipsing binary.

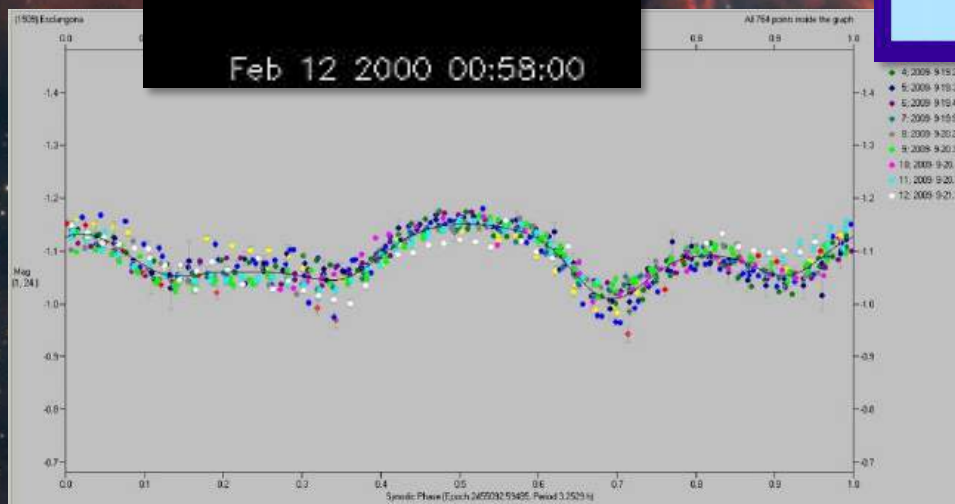
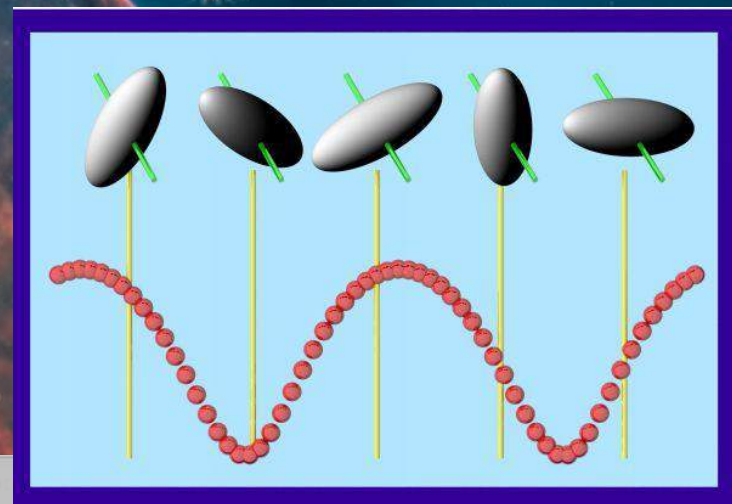


PHOTOMETRY IN A NUTSHELL

Measure the number of photons from an object over time.
Convert detector counts to calibrated brightness.
Study how brightness changes to learn about the object.



FOTOMETRIA - ASTEROIDY



FOTOMETRIA - EXOPLANÉTY



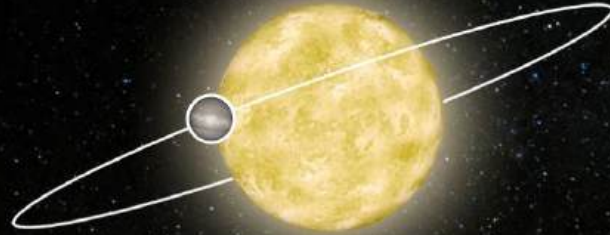
FOTOMETRIA - EXOPLANÉTY



VIEW FROM ABOVE



VIEW FROM TELESCOPE



LIGHT CURVE



Artist conception, not to scale

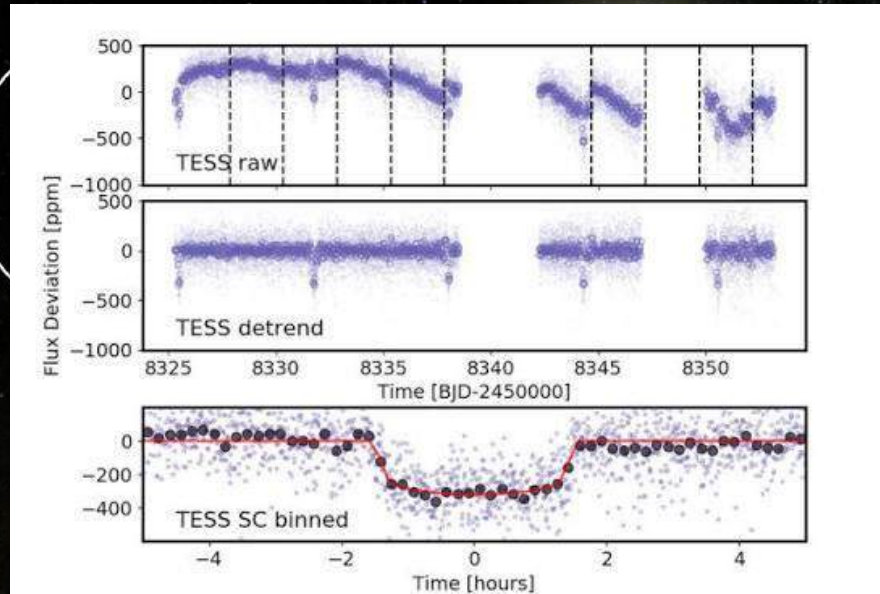


FOTOMETRIA - EXOPLANÉTY



VIEW FROM ABOVE

VIEW FROM TELESCOPE



APP/BRIG

TIME

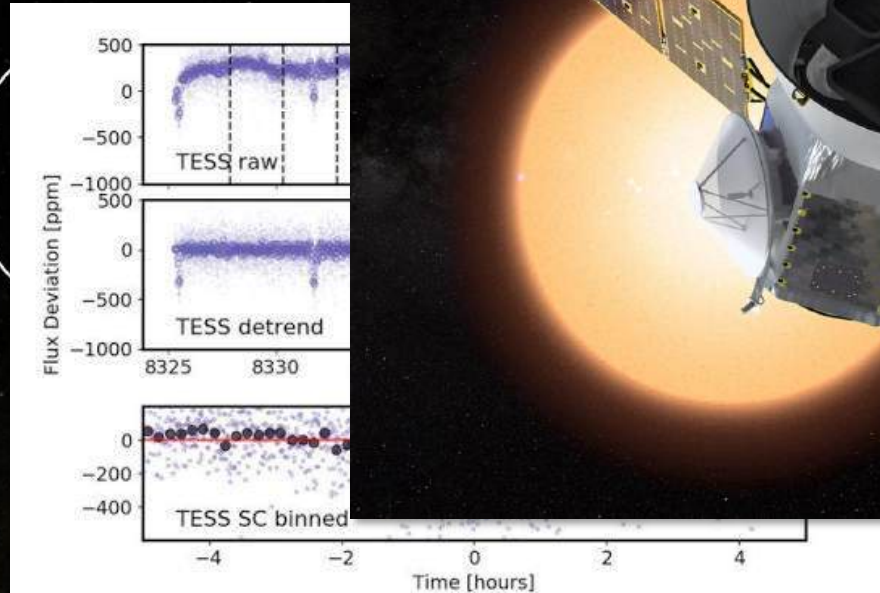
Artist conception, not to scale



FOTOMETRIA - EXOPLANÉTÁK



VIEW FROM ABOVE



Artist conception, not to scale

APP/BRIG

TIME

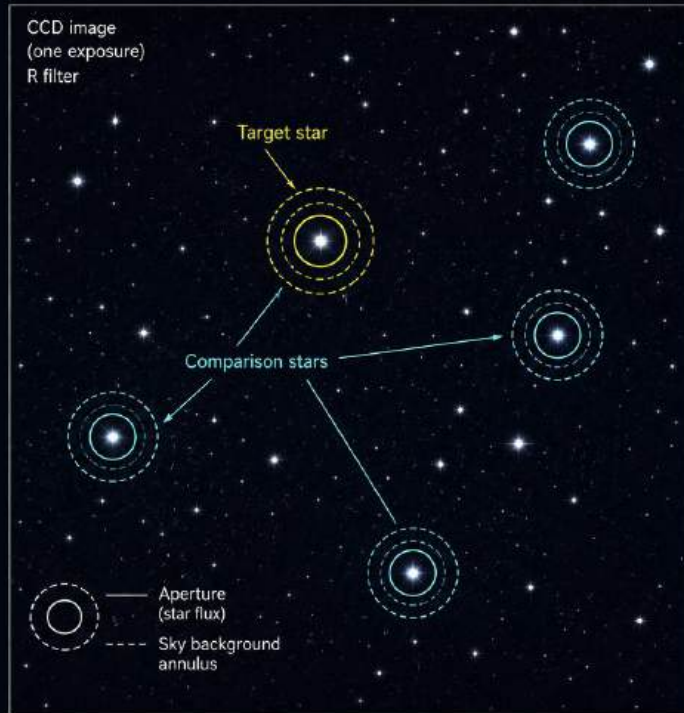


RELATÍVNA FOTOMETRIA



Relative (Differential) Photometry

Measuring variability of a target star relative to stable comparison stars



Aperture photometry is performed on the target and on several comparison stars to remove atmospheric and instrumental effects.

Method

1. Perform aperture photometry on the target star (yellow).
2. Perform aperture photometry on several comparison stars (cyan) assumed to have constant brightness.
3. Compute the ensemble comparison flux as the mean (or weighted mean) of the comparison stars.

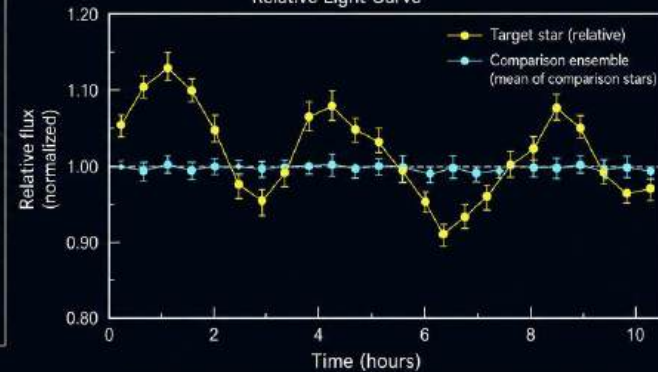
4. Compute the differential (relative) flux:

$$F_{\text{rel}}(t) = \frac{F_{\text{target}}(t)}{\langle F_{\text{comp}}(t) \rangle}$$

5. Convert to differential magnitude if desired:

$$\Delta m(t) = -2.5 \log_{10} F_{\text{rel}}(t)$$

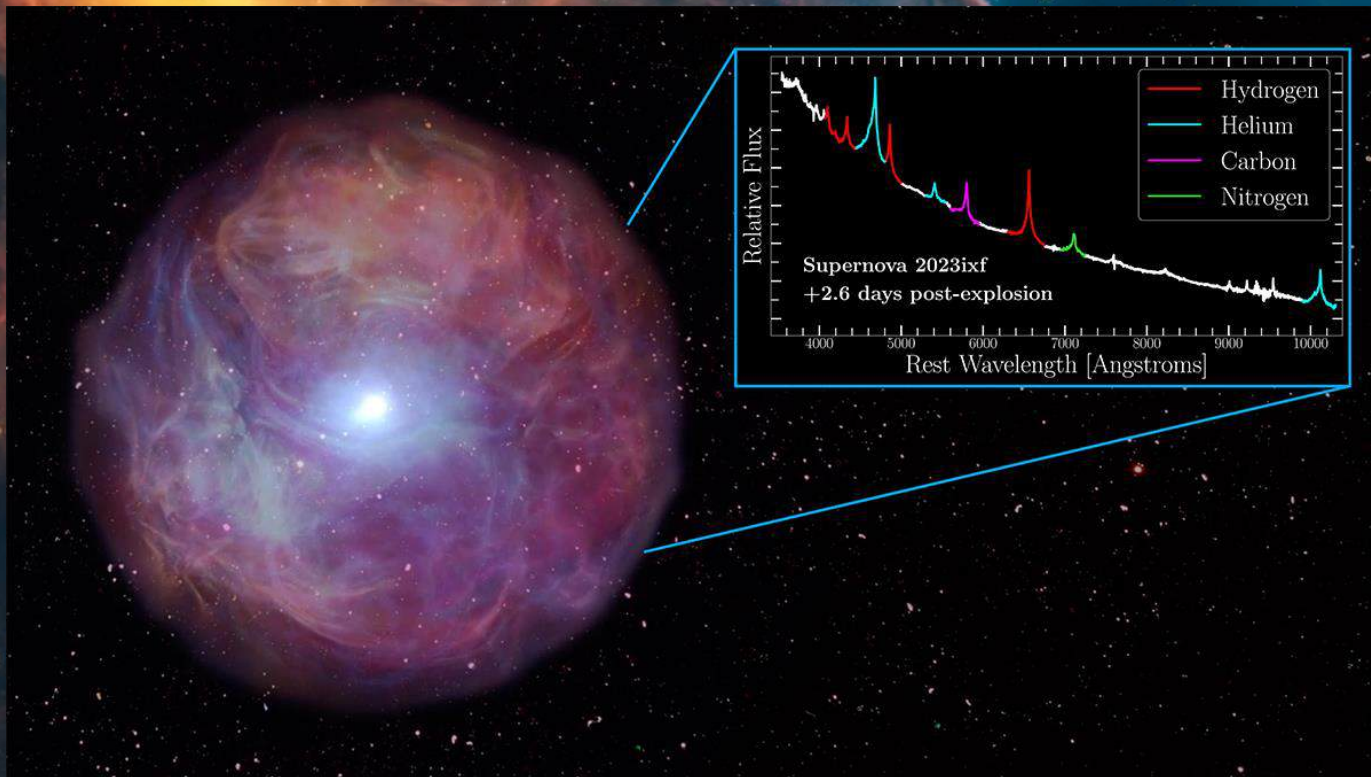
Relative Light Curve



The comparison ensemble is consistent with zero ($\Delta m = 0$).
The target star shows intrinsic variability.



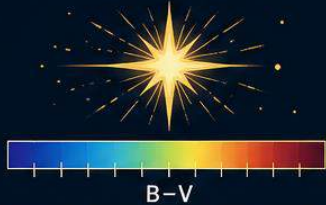
FOTOMETRIA - SUPERNOVA



PHOTOMETRY IN ASTRONOMY:

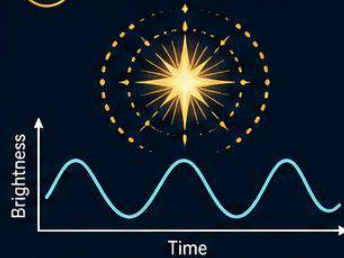
What Do We Study?

1 STARS & STELLAR POPULATIONS



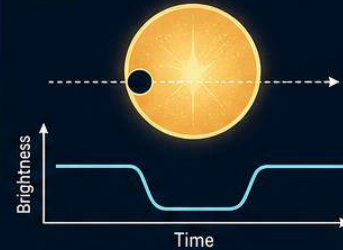
Determine temperature and spectral type from color indices ($B-V$), build Hertzsprung-Russell diagrams, and estimate cluster ages.

2 VARIABLE STARS



Cepheids and RR Lyrae as standard candles for measuring cosmic distances. Eclipsing binaries, novae and supernovae.

3 EXOPLANETS



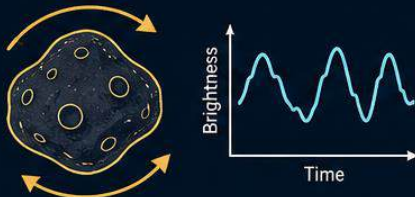
Transit photometry detects planets via brightness dips (Kepler, TESS, PLATO missions).

4 GALAXIES



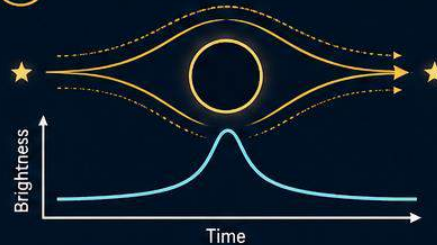
Photometric redshifts, morphological classification, and star formation rates from UV/IR flux.

5 SOLAR SYSTEM BODIES



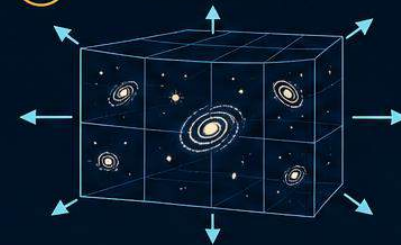
Light curves of asteroids and comets reveal rotation and shape. Photometry of moons and dwarf planets.

6 GRAVITATIONAL MICROLENSING



Detect dark objects and exoplanets via the temporary brightening of background stars caused by gravitational lensing.

7 COSMOLOGY



Type Ia supernovae as standard candles for measuring the expansion history of the Universe.

ASTROMETRIA — MERANIE POZÍCII HVIEZD



- Presné meranie polohy objektov na oblohe



ASTROMETRIA — MERANIE POZÍCII HVIEZD



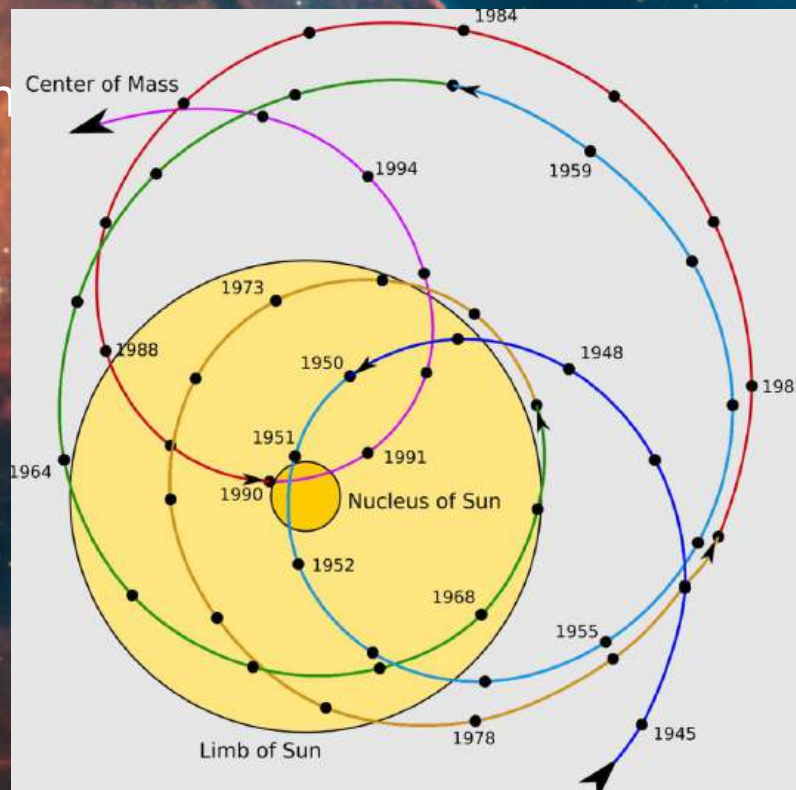
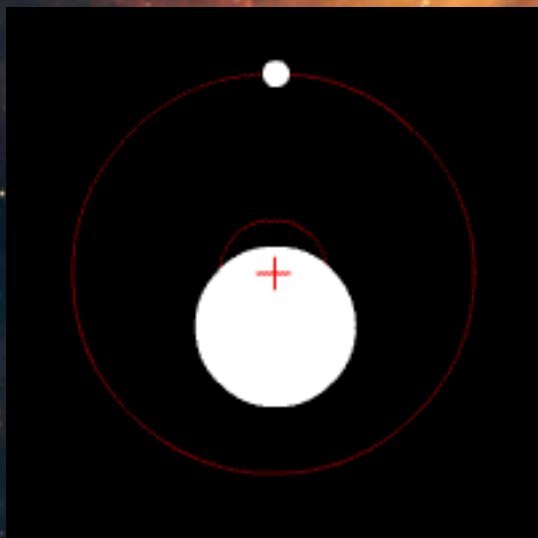
- Presné meranie polohy objektov na oblohe



ASTROMETRIA — MERANIE POZÍCII HVIEZD



- Presné meranie poloh

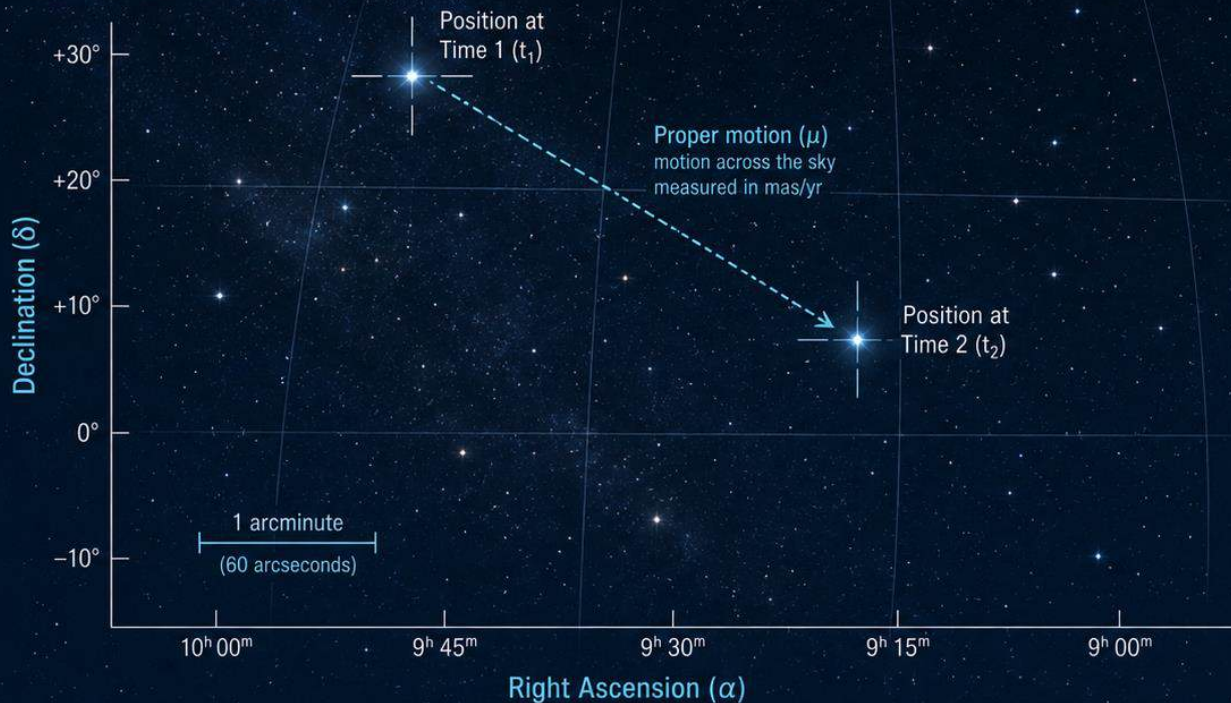




ASTROMETRY

MEASURING POSITIONS AND MOTIONS OF STARS

Astrometry is the precise measurement of the positions and motions of celestial objects on the sky, enabling studies of their distances, trajectories, and physical properties.



Proper motion (μ):

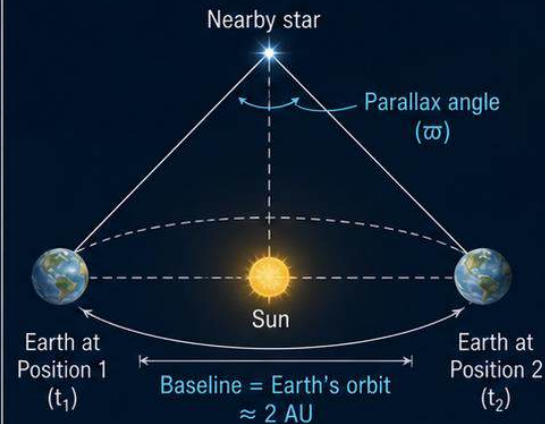
Angular motion of a star across the sky, measured in milliarcseconds per year (mas/yr).

Parallax (ϖ):

Apparent shift in a star's position due to Earth's orbit around the Sun; used to measure distance.

PARALLAX

Nearby stars appear to shift position when viewed from opposite sides of Earth's orbit.

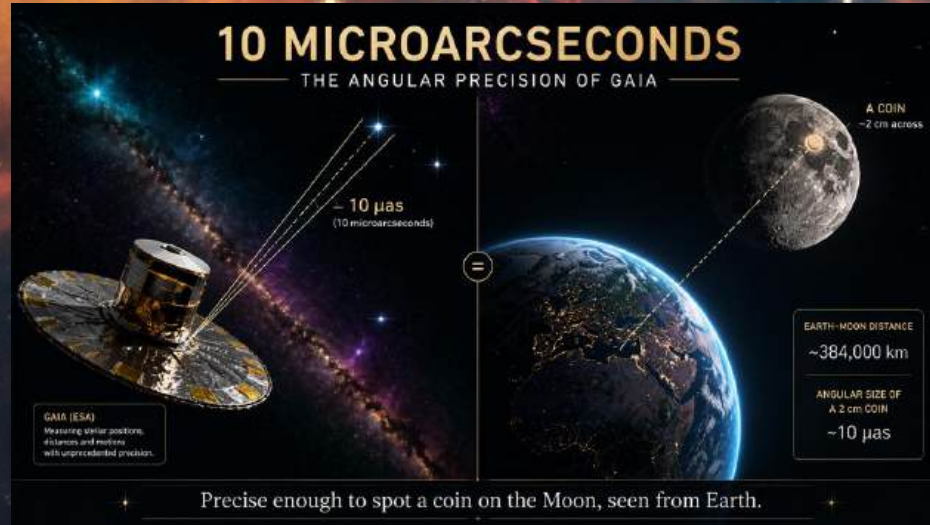


Astrometric observations yield positions, proper motions, and parallaxes, revealing distances and true motions of stars in 3D space.

PARALAXA - MISIA GAIA



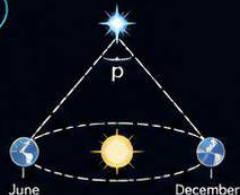
- Gaia merala uhly s presnosťou **10 mikroarcsekúnd** — to je ako rozoznať zo Zeme mincu na povrchu Mesiaca



ASTROMETRY:

Mapping the Universe in Motion

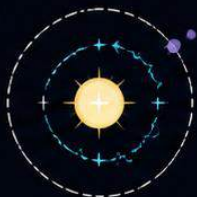
1



STELLAR DISTANCES & PARALLAX

Measure parallax angle (p) from Earth's orbit to determine distances to nearby stars.

3



EXOPLANET DETECTION

Detect unseen planets via the star's tiny astrometric wobble around the system barycenter.

5



STELLAR PROPER MOTION & HIGH-VELOCITY STARS

Measure stars' apparent motion across the sky; identify hypervelocity stars ejected from the Galactic center.

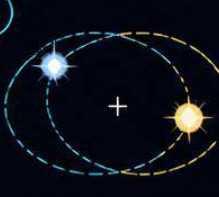
2



GALACTIC STRUCTURE & KINEMATICS

Map the Milky Way's spiral arms and stellar orbits using precise positions and velocities (Gaia mission).

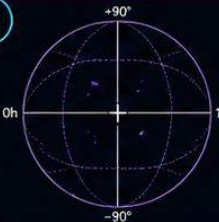
4



BINARY & MULTIPLE STAR SYSTEMS

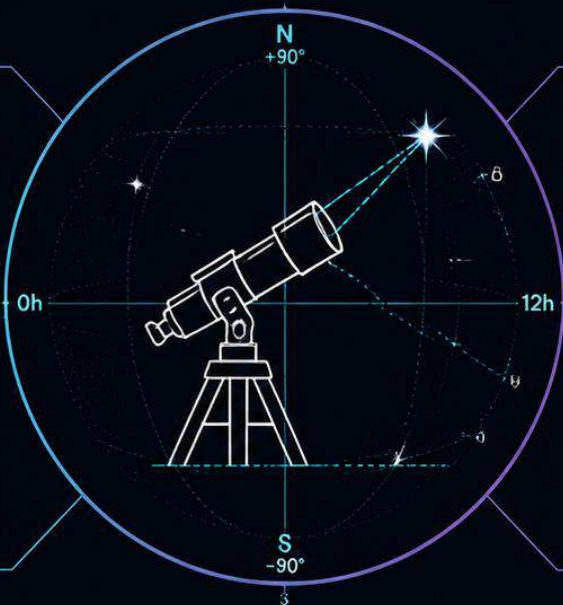
Track orbital motion of stellar companions to determine masses and orbital parameters.

6



REFERENCE FRAMES & FUNDAMENTAL PHYSICS

Use distant quasars to define inertial reference frames and test General Relativity via light bending.



Gaia

Mapping a billion stars with unprecedented precision.



Hipparcos

The foundation of modern astrometry.



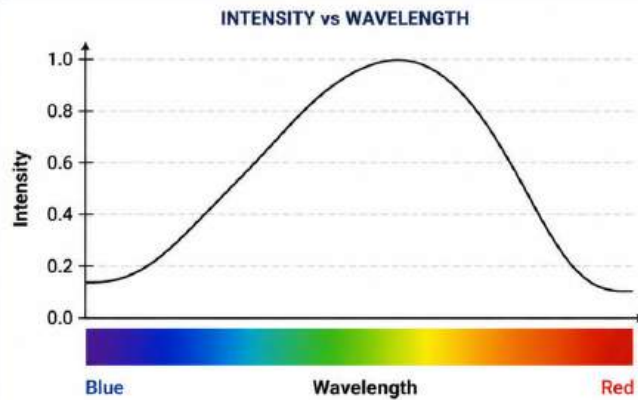
Astrometry: precise positions. Profound discoveries. Understanding the Universe in six dimensions.

SPEKTROSKOPIA



WHAT IS A SPECTRUM?

Light spread out by wavelength or frequency.



CONTINUOUS SPECTRUM



All wavelengths present smoothly.

EMISSION SPECTRUM



Bright lines at specific wavelengths.

ABSORPTION SPECTRUM



Dark lines where light is absorbed.



1. UNIQUE FINGERPRINTS

- Each chemical element has a unique set of spectral lines.
- These act like fingerprints, letting us identify what an object is made of.



2. SPECTRA REVEAL

- Temperature (shape and peak)
- Composition (which elements)
- Velocity (redshift/blueshift)



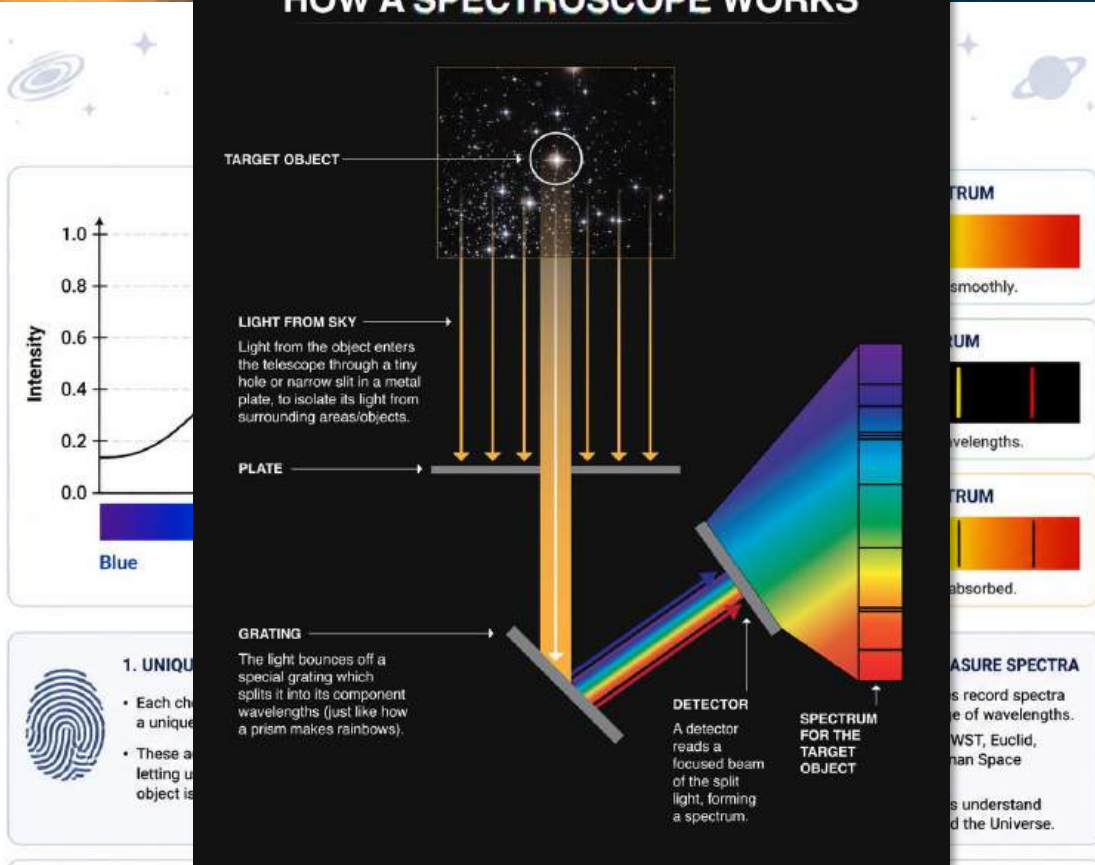
3. TELESCOPES MEASURE SPECTRA

- Modern telescopes record spectra across a vast range of wavelengths.
- Examples: Gaia, JWST, Euclid, Rubin (LSST), Roman Space Telescope.
- These data help us understand stars, galaxies, and the Universe.



KEY IDEA: A spectrum is light's story—carrying information about the physical nature of its source.

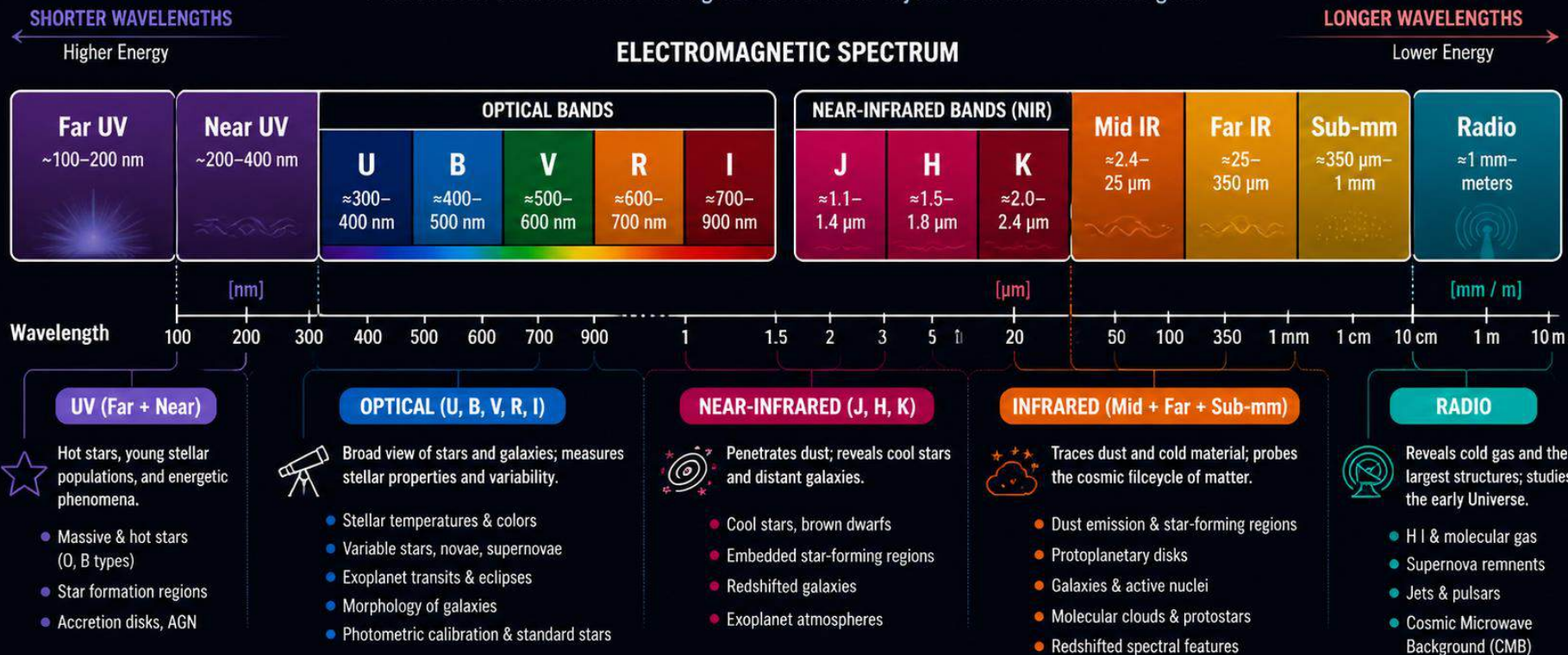






Astronomical Photometry Bands & Wavelength Ranges

Photometric bands measure the light from celestial objects at different wavelengths.



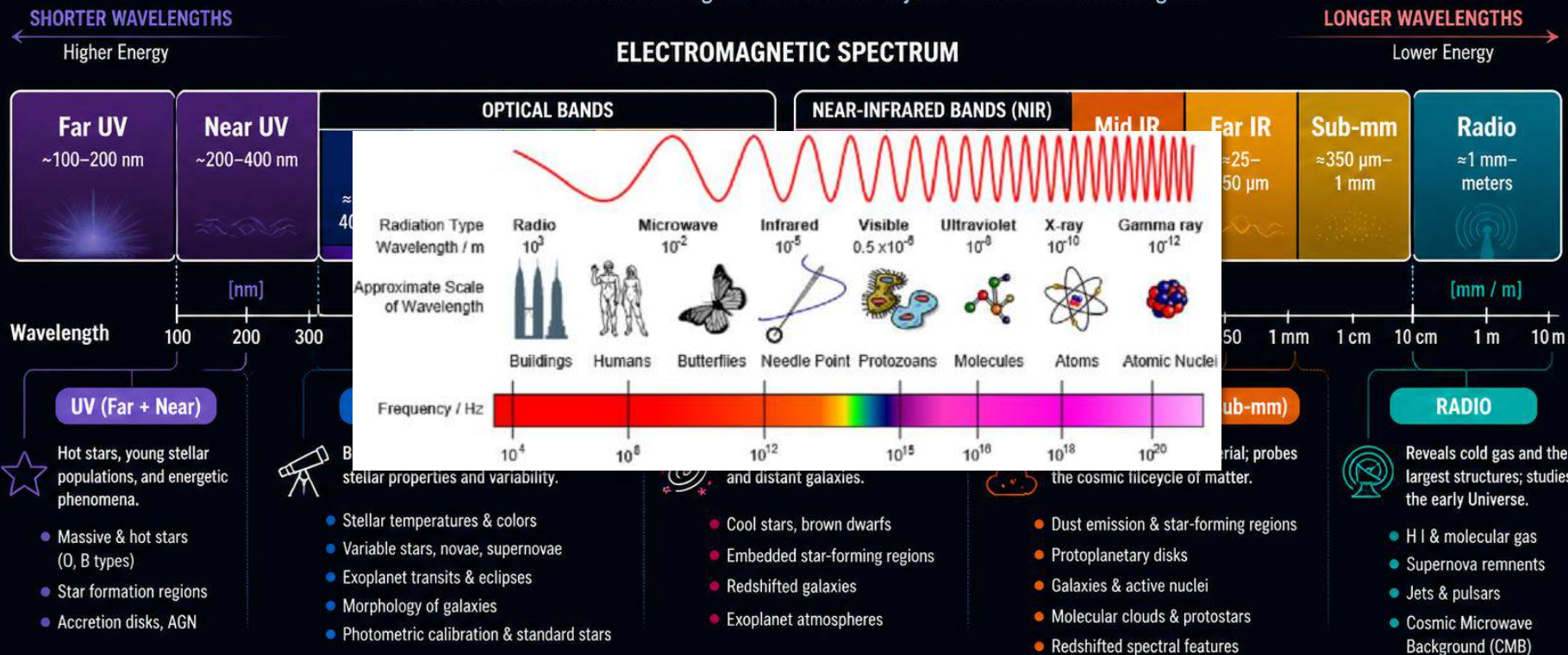
Photometric systems (e.g., Johnson–Cousins UBVRI, 2MASS JHK) define these bands with specific filter transmission curves. Exact ranges vary slightly by system and instrument.

Units: [nm] = nanometers (10^{-9} m) [μm] = micrometers (10^{-6} m) [mm / m] = millimeters / meters



Astronomical Photometry Bands & Wavelength Ranges

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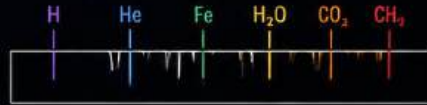
SPEKTROSKOPIA



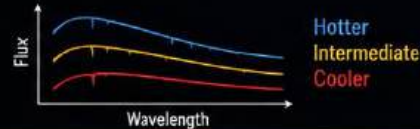
WHAT SPECTRA TELL US — KEY DIAGNOSTICS IN ASTRONOMICAL SPECTROSCOPY



1. Chemical Composition —
absorption/emission lines
identify elements & molecules
(H, He, Fe, H₂O, CO₂, CH₄)



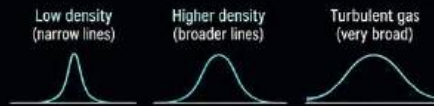
2. Temperature —
continuum shape & line ratios
→ effective temperature,
excitation



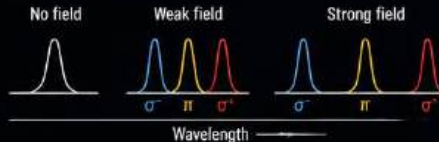
3. Velocity & Redshift —
Doppler shift of lines
→ radial velocity, expansion,
cosmological redshift



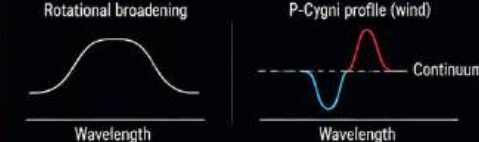
4. Density & Pressure —
line widths & ratios
→ gas density, pressure,
turbulence



5. Magnetic Fields —
Zeeman splitting
→ field strength



6. Rotation & Winds —
line broadening, P-Cygni profiles
→ rotation, outflows, accretion



UV
~10 – 400 nm



VISIBLE
~400 – 700 nm



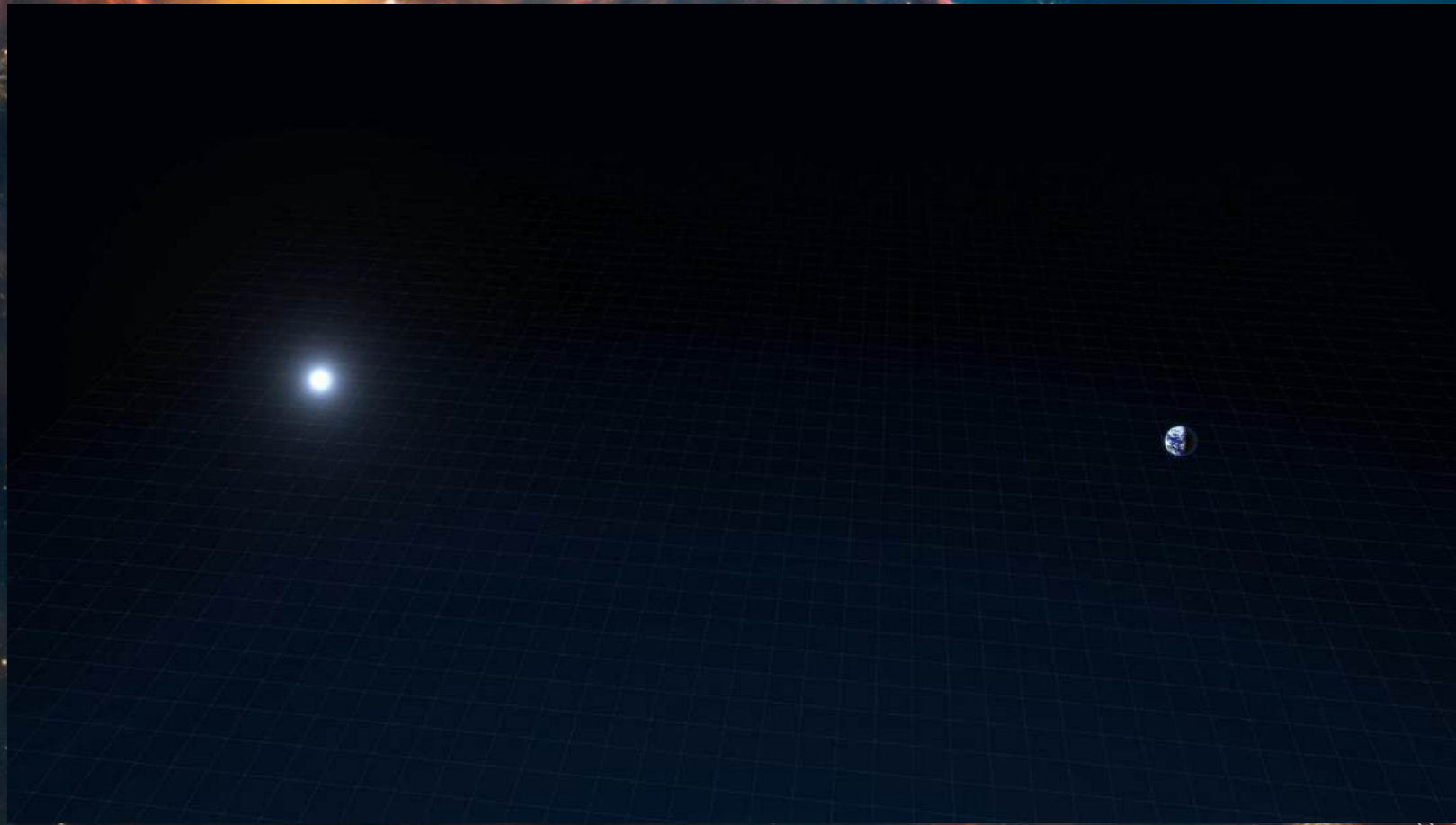
IR
~700 nm – 1 mm



RADIO
~1 mm – km



GRAVITAČNÉ ŠOŠOVKY



GRAVITAČNÉ ŠOŠOVKY



GRAVITATIONAL LENSING — HOW GRAVITY BENDS LIGHT

Mass curves spacetime. Light follows the curve.

1 SOURCE

Distant galaxy or quasar
Emits light that travels across cosmic distances.



2 LENS

Massive object
(galaxy cluster, black hole)
Its gravity bends spacetime and deflects light.



3 OBSERVER

Telescope on Earth
(JWST, Hubble, Euclid, Roman)
Detects the lensed light and reconstructs the image.



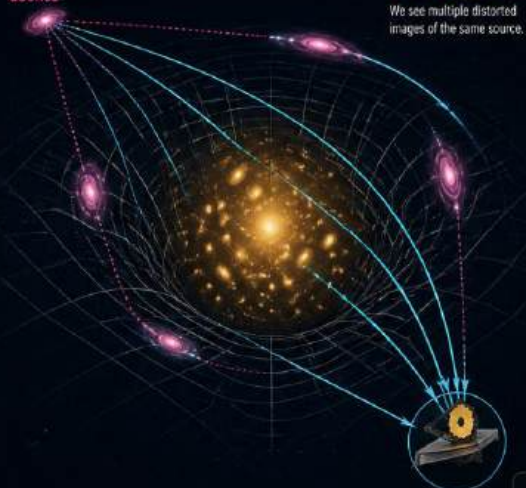
4 EINSTEIN RING

Perfect alignment of source, lens, and observer produces a circular ring.



BACKGROUND SOURCE

MULTIPLE IMAGES / ARCS
We see multiple distorted images of the same source.



OBSERVER

5 STRONG LENSING

Large deflections produce dramatic distortions: multiple arcs, giant arcs, and multiple images.



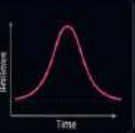
6 WEAK LENSING

Small distortions of many background galaxies are coherent and reveal the mass distribution.



7 MICROLENSING

Compact objects (stars, planets, black holes) cause temporary brightening of background stars or quasars.



DEFLECTION ANGLE

$$\alpha = \frac{4GM}{c^2 b}$$

G = gravitational constant

M = mass of the lens

c = speed of light

b = impact parameter (closest approach)

..... Unlensed light path (in straight space)

— Lensed light path (bent by gravity)

● Mass (lensing matter)

● Background source

WHAT WE LEARN



Mass distribution (including dark matter)
Map the invisible to understand structure.



Hubble constant (time delays)
Measure cosmic expansion with time delays between multiple images.



Exoplanets (microlensing)
Detect planets by their gravitational signature.

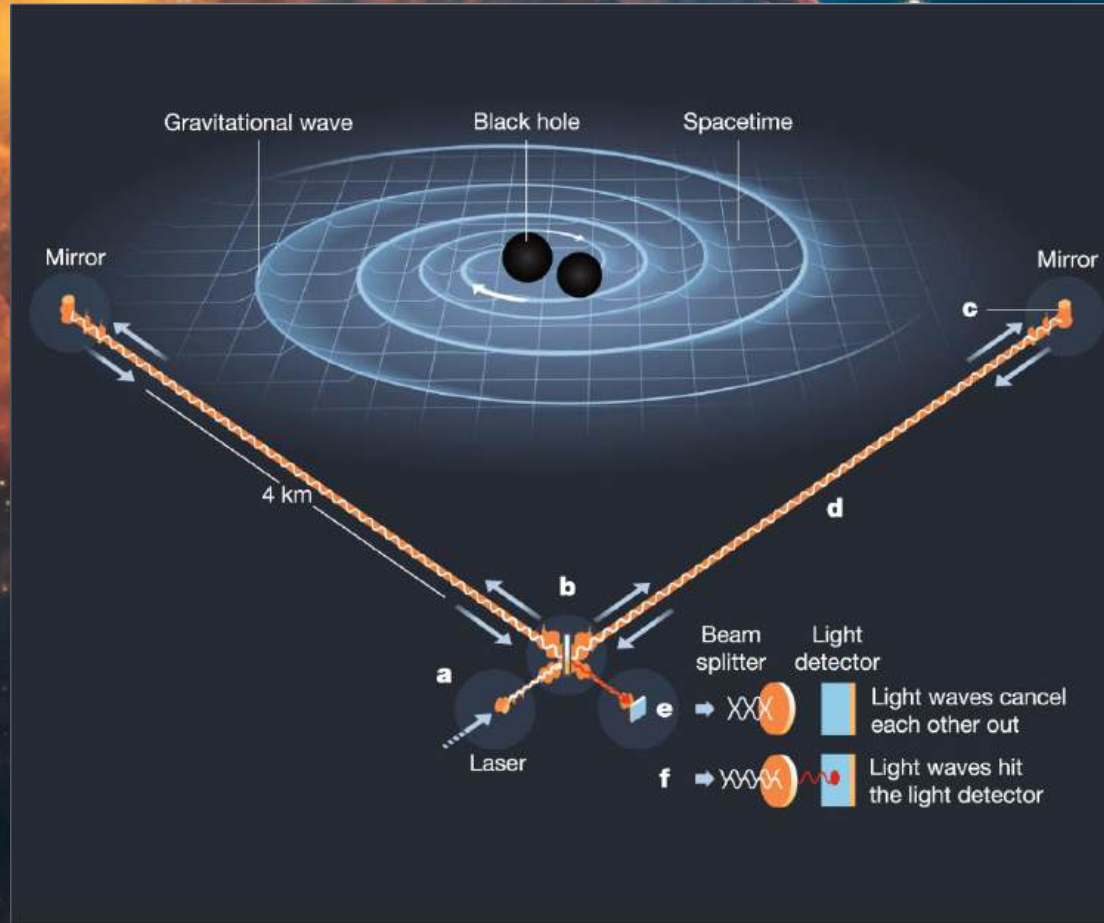


High- z galaxies (magnification)
Gravitational telescopes reveal the distant universe.

General Relativity predicts it. Observations confirm it. We use it to explore the universe.



GRAVITAČNÉ VLNY – BEYOND THE SCOPE





ČASŤ 2

TOP observatóriá 2026



TOP ĎALEKOHĽADY SVETA (A NIE LEN)



TOP ĎALEKOHĽADY SVETA (A NIE LEN)

- James Webb Space Telescope



TOP ĎALEKOHĽADY SVETA (A NIE LEN)

- James Webb Space Telescope
- ESA Gaia



TOP ĎALEKOHĽADY SVETA (A NIE LEN)

- James Webb Space Telescope
- ESA Gaia
- ESA Euclid



TOP ĎALEKOHĽADY SVETA (A NIE LEN)

- James Webb Space Telescope
- ESA Gaia
- ESA Euclid
- Vera Rubin Observatory



TOP ĎALEKOHĽADY SVETA (A NIE LEN)



- James Webb Space Telescope
- ESA Gaia
- ESA Euclid
- Vera Rubin Observatory
- Nancy Grace Roman Space Telescope





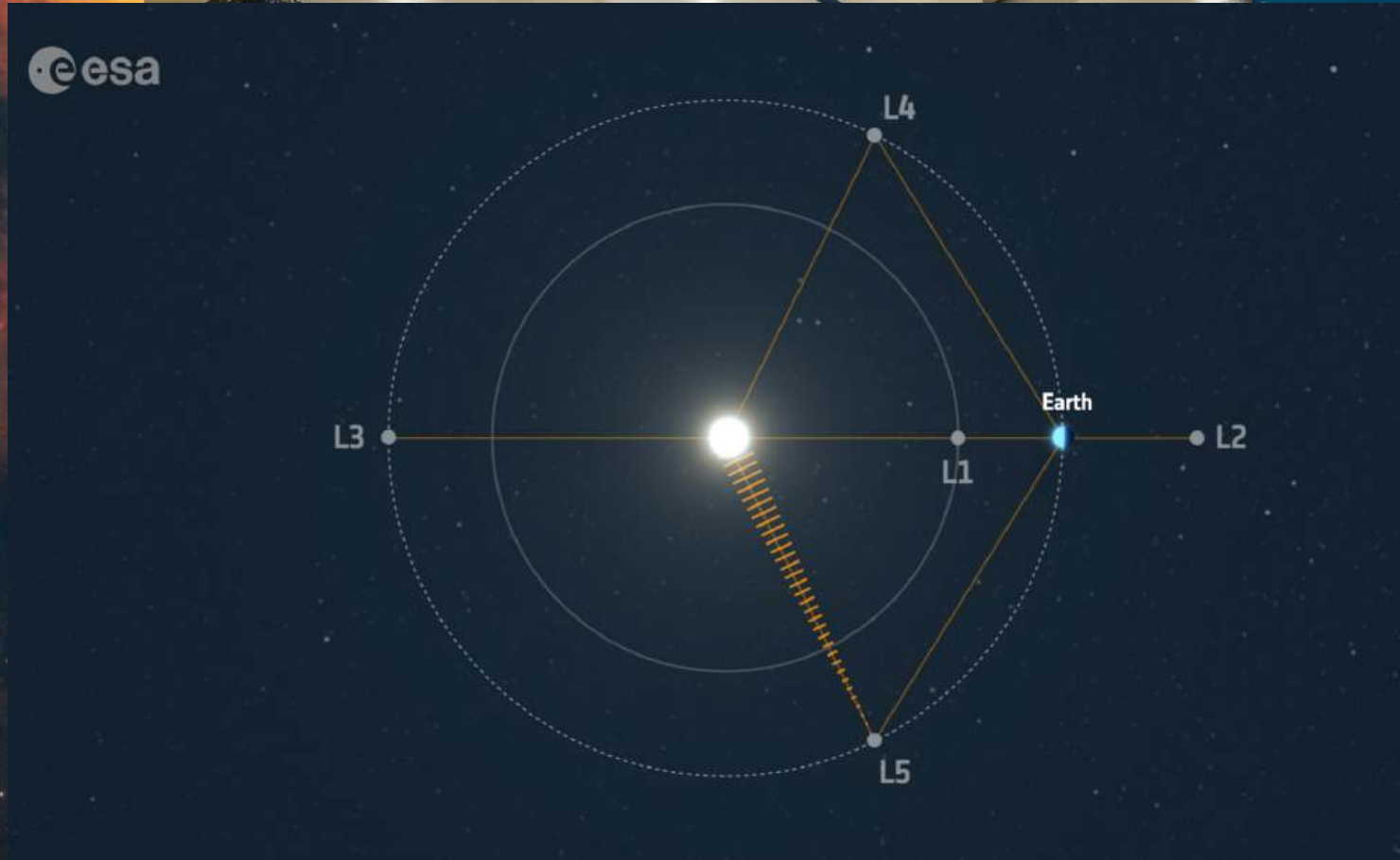
JAMES WEBB SPACE TELESCOPE



JAMES WEBB SPACE TELESCOPE



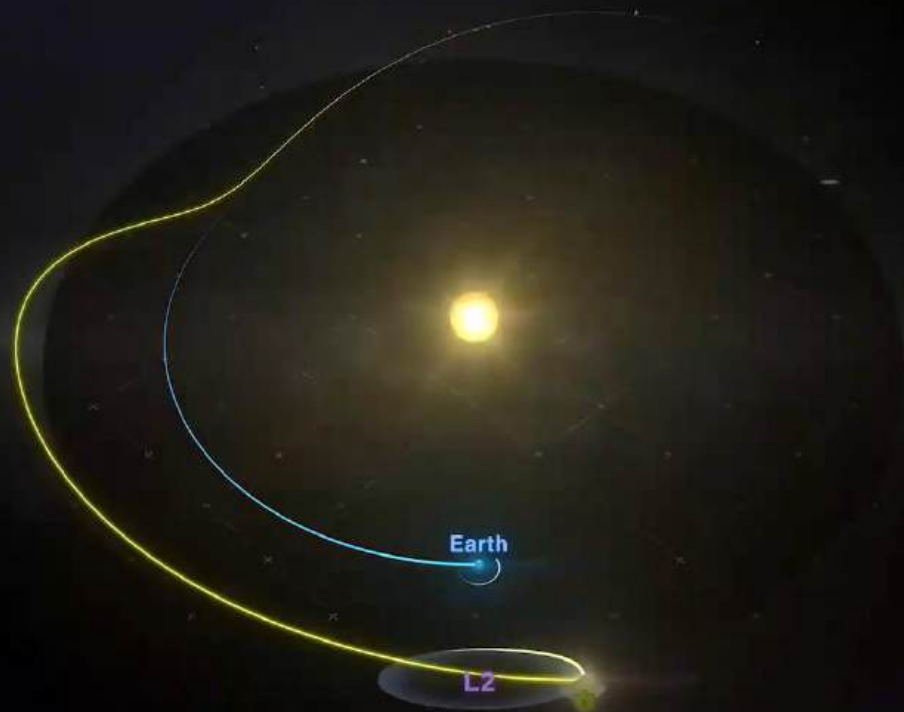
JAMES WEBB SPACE TELESCOPE



JAMES WEBB SPACE TELESCOPE







JWST

- 6,5m zrkadlo
- Infračervená oblasť



JWST

○ 6

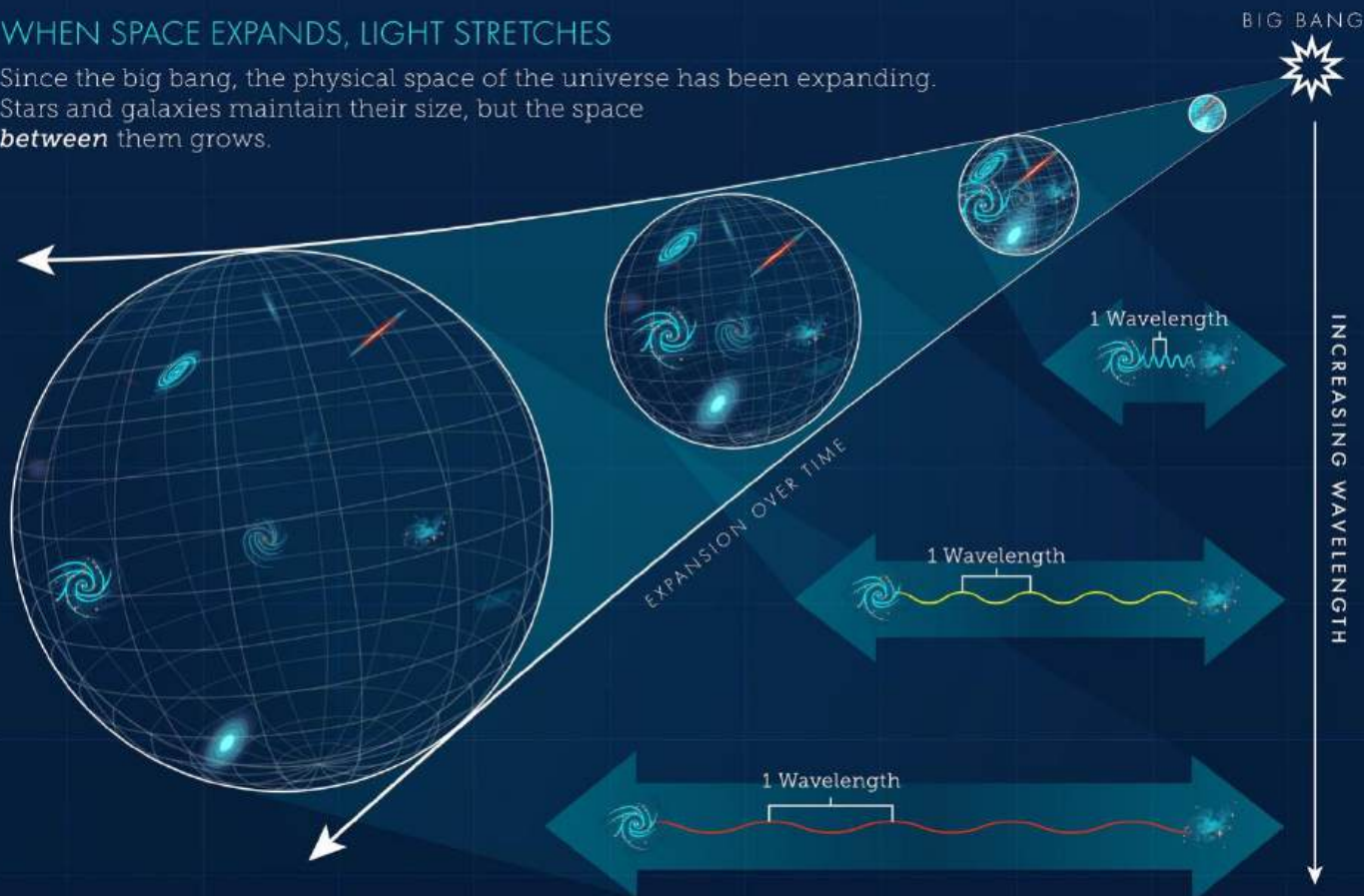
○ 11



WHAT IS COSMOLOGICAL REDSHIFT?

WHEN SPACE EXPANDS, LIGHT STRETCHES

Since the big bang, the physical space of the universe has been expanding. Stars and galaxies maintain their size, but the space *between* them grows.



As light travels through expanding space, it is stretched to longer wavelengths.



WHAT IS COSMOS

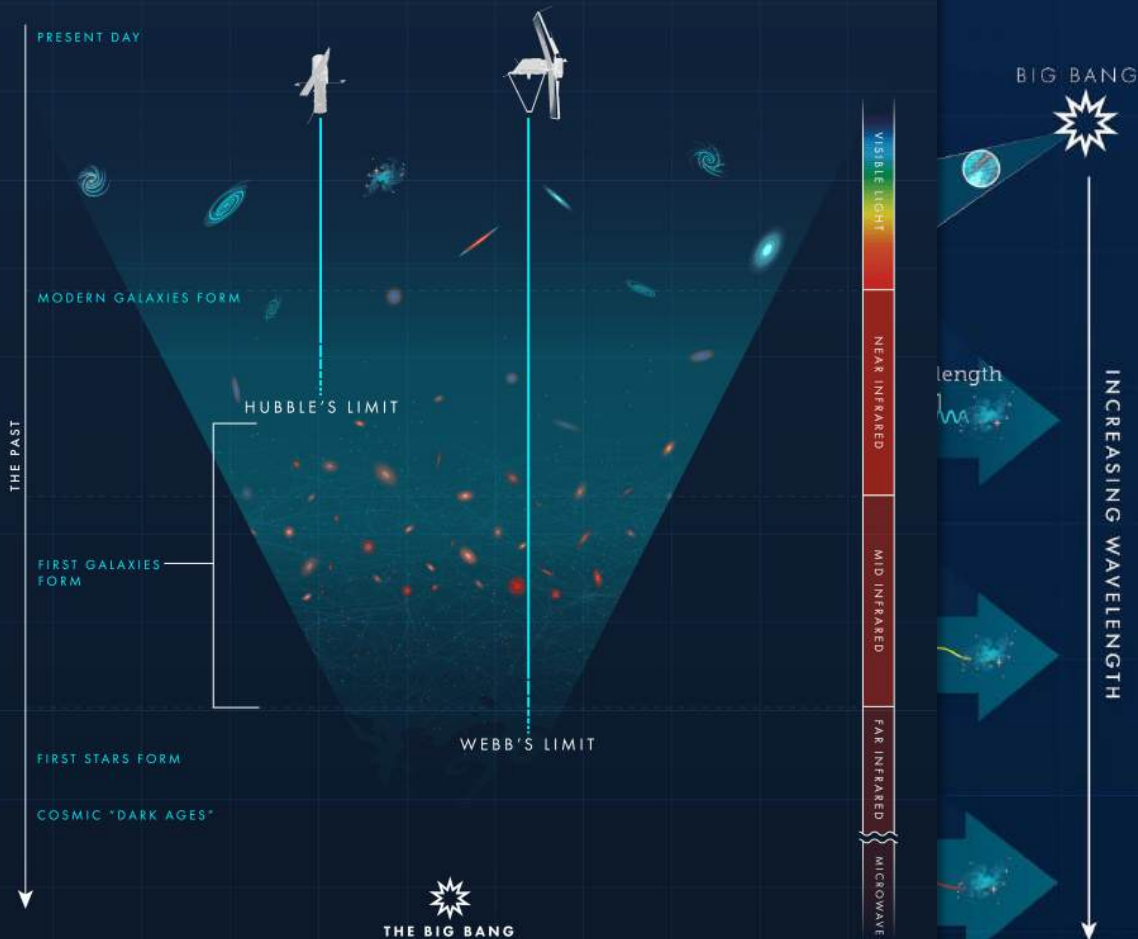
WHEN SPACE

Since the big bang
Stars and galaxies
between them



SEEING THE PAST

Telescopes with **infrared** detectors allow us to see the ancient light of the first galaxies, which has been redshifted over space and time.





A AKO WEBB POZORUJE?

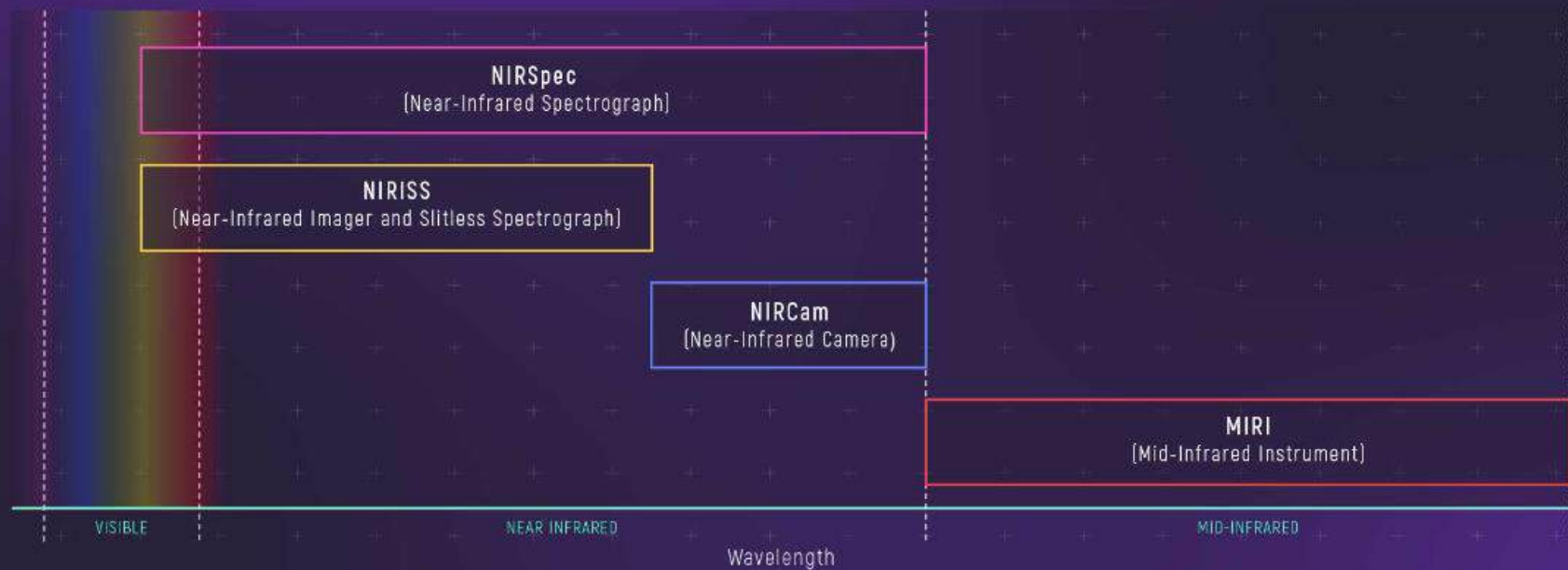
- NIRSpec - Near-Infrared Spectrograph
- FGS – Fine Guiding Sensor
- NIRISS - The Near Infrared Imager and Slitless Spectrograph
- NIRCам - Near-Infrared Camera
- MIRI - Mid-Infrared Instrument



A AKO WEBB POZORUJE?



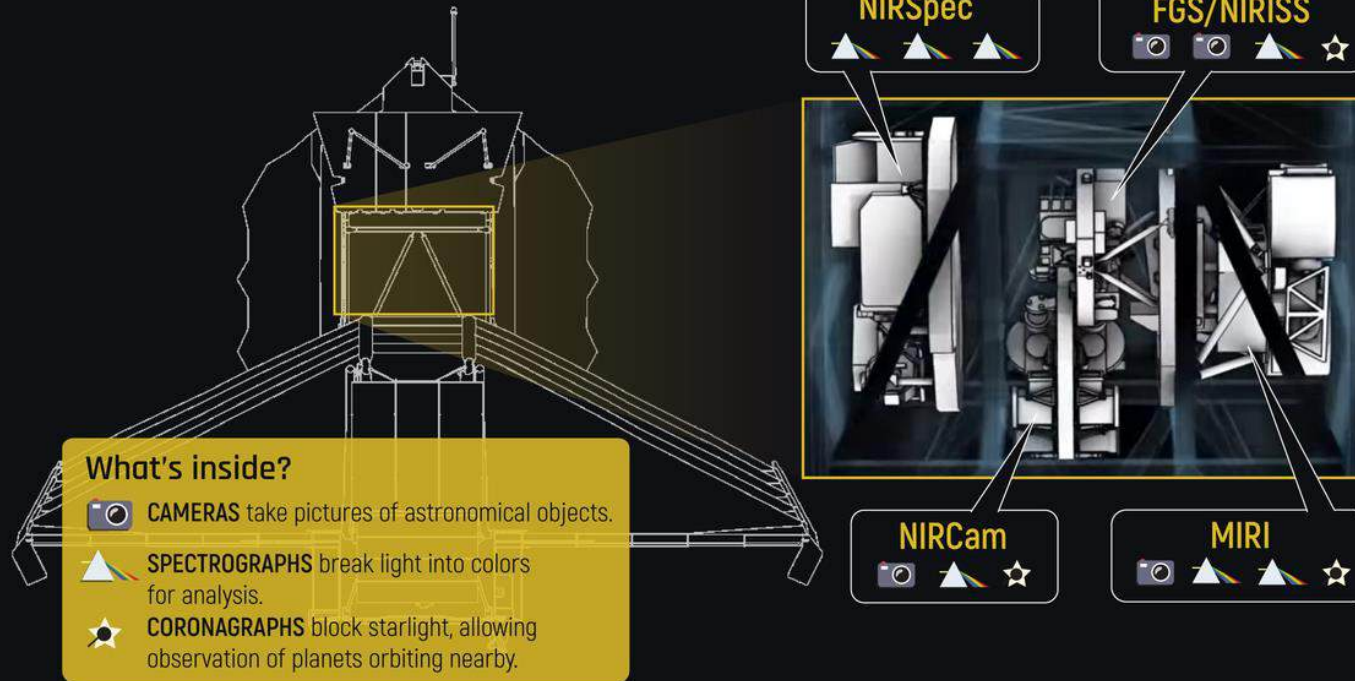
WAVELENGTH RANGE OF WEBB'S SPECTROSCOPIC INSTRUMENTS



A AKO WEBB POZORUJE?



Webb's Science Instruments



JADES deep field



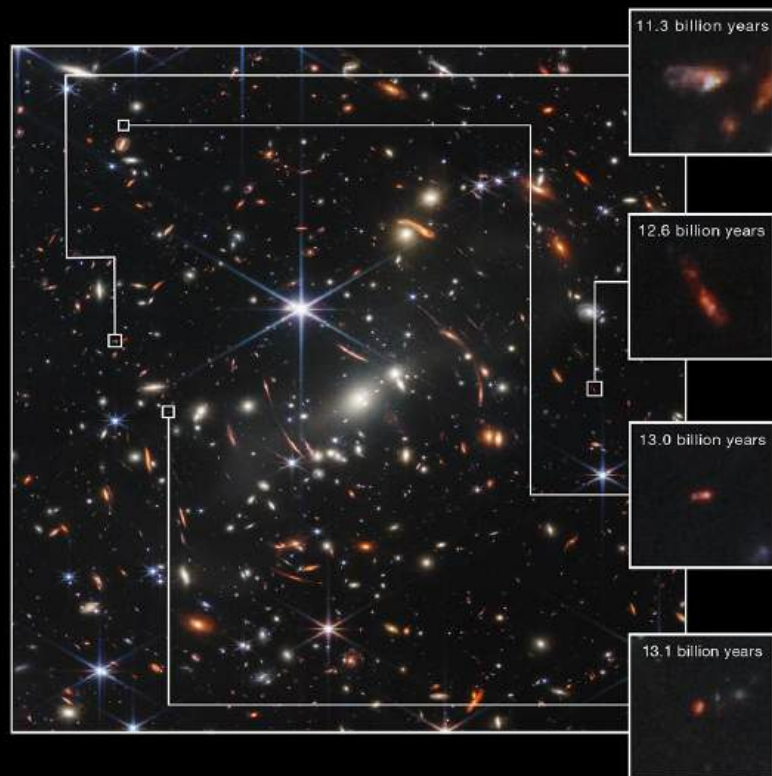
JADES deep field



GALAXY CLUSTER SMACS 0723

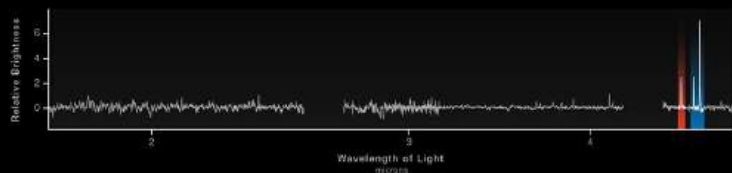
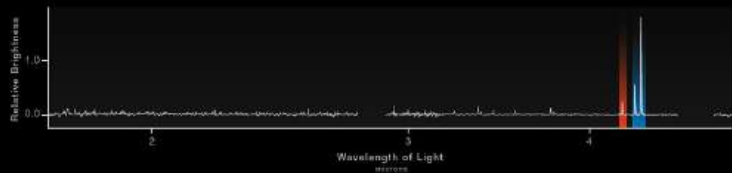
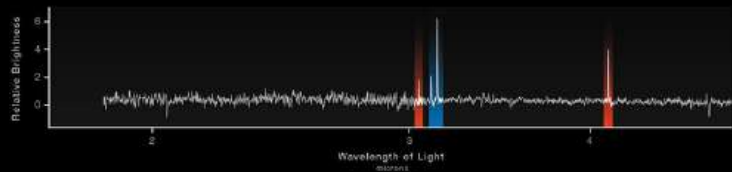
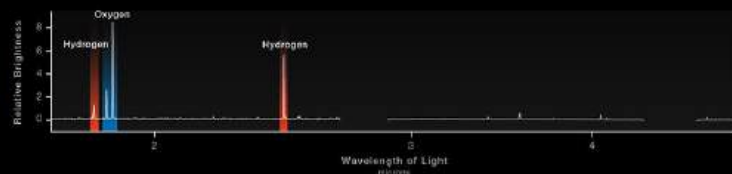
WEBB SPECTRA IDENTIFY GALAXIES IN THE VERY EARLY UNIVERSE

NIRCam Imaging

**WEBB**
SPACE TELESCOPE

NIRSpec Microshutter Array Spectroscopy

Lowest redshift



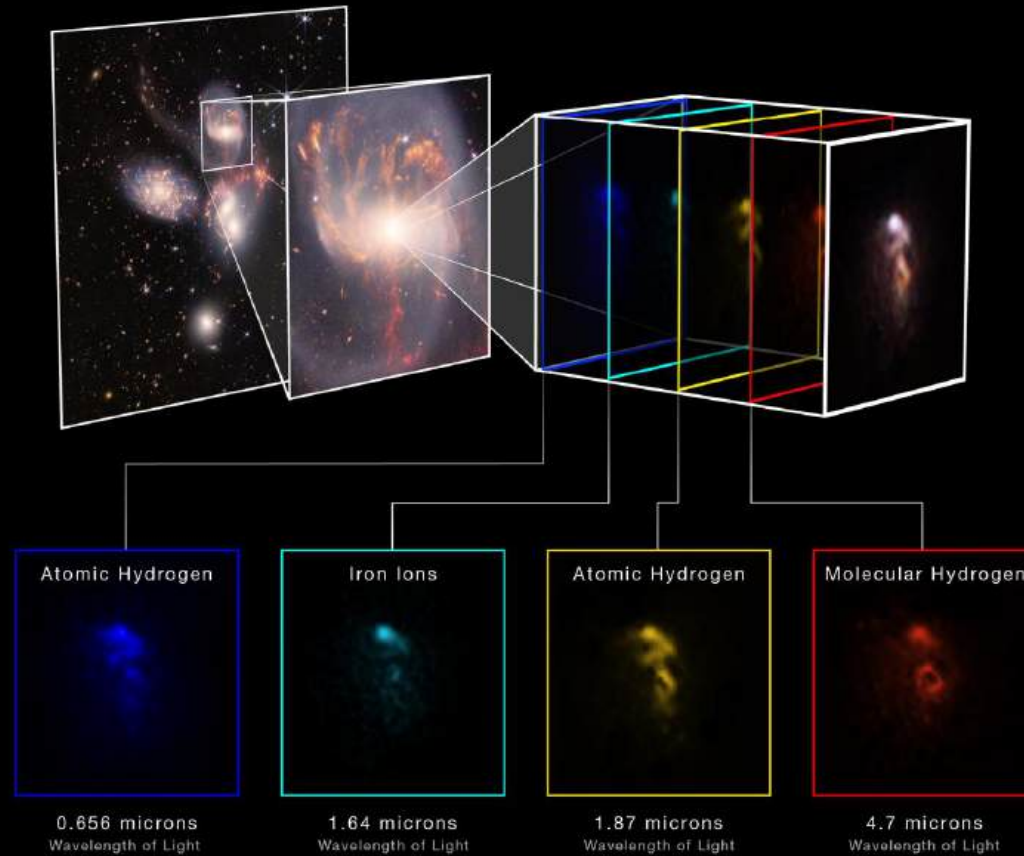
Highest redshift

INTERACTING GALAXIES | STEPHAN'S QUINTET

COMPOSITION OF GAS AROUND ACTIVE BLACK HOLE

NIRCam and MIRI Imaging

NIRSpec IFU Spectroscopy

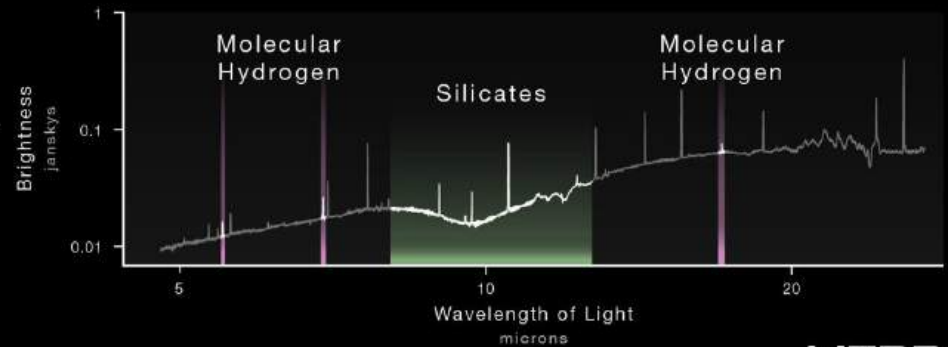
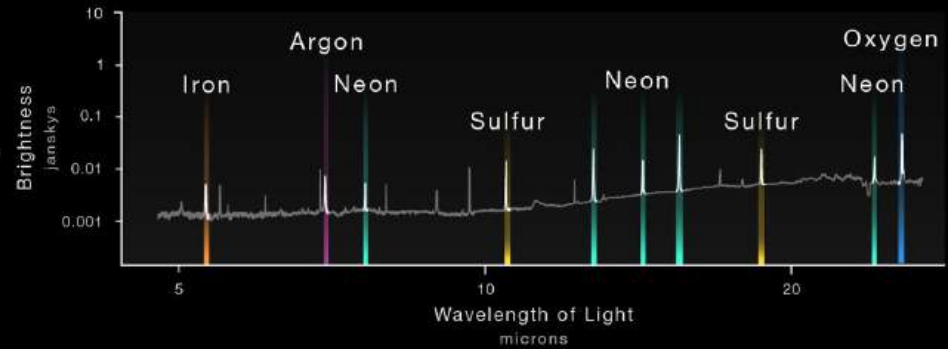
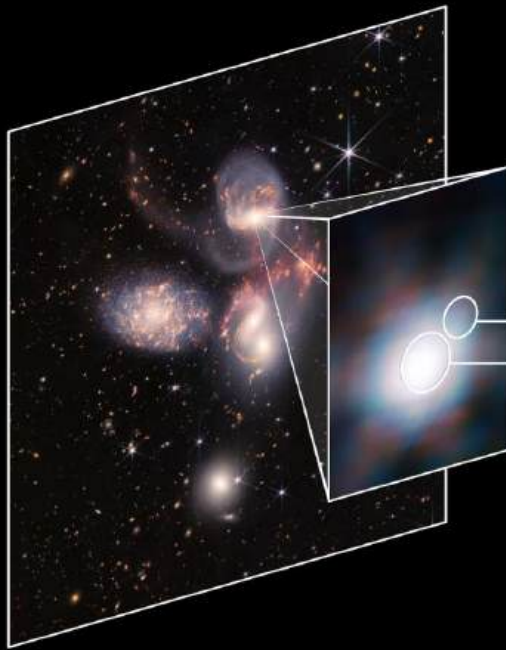
**WEBB**
SPACE TELESCOPE

INTERACTING GALAXIES | STEPHAN'S QUINTET

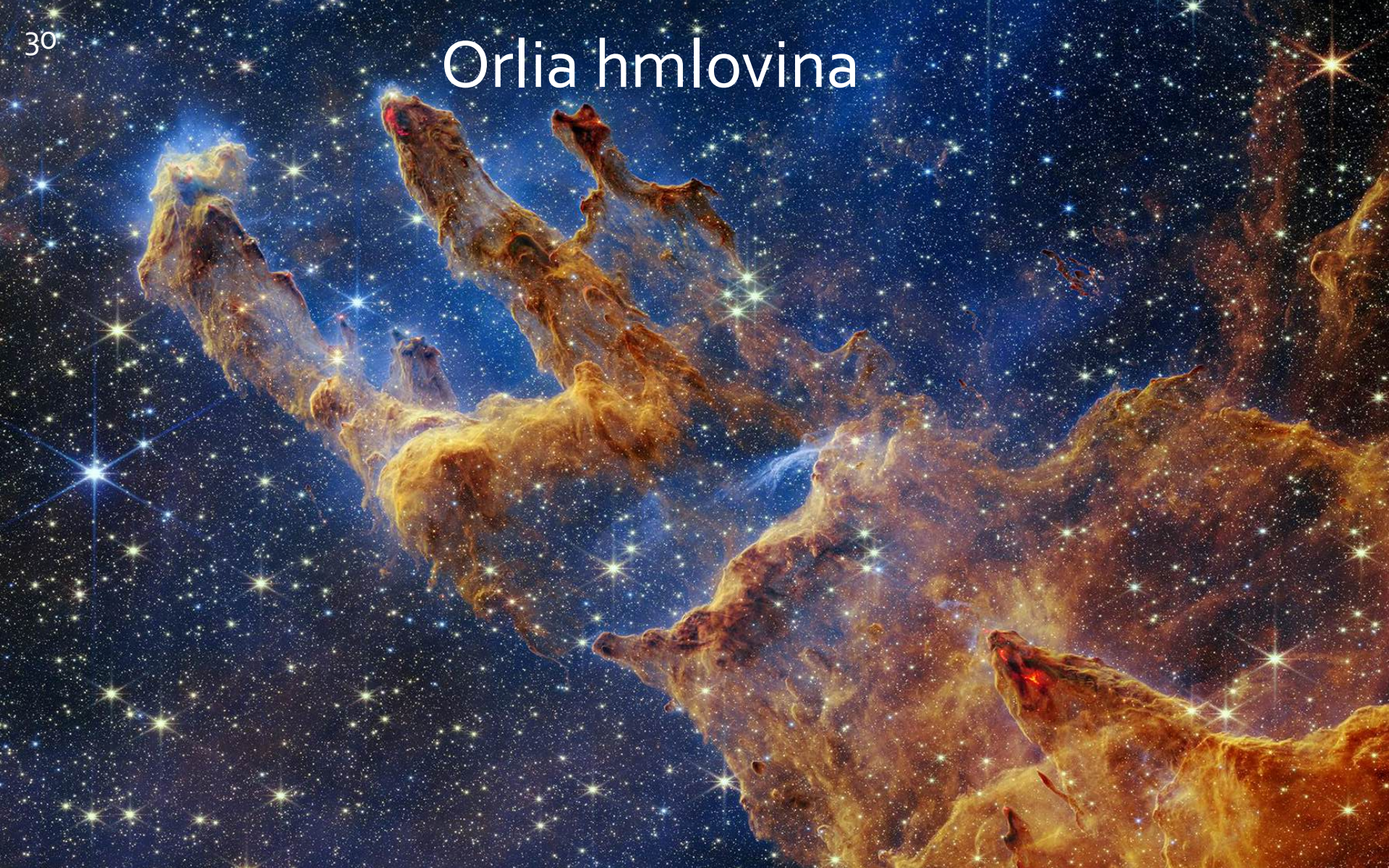
COMPOSITION OF GAS AROUND ACTIVE BLACK HOLE

NIRCam and MIRI Imaging

MIRI IFU Medium Resolution Spectroscopy

WEBB
SPACE TELESCOPE

Orla hmlovina



Orla hmlovina





južná prstencová hmlovina

NIRCam

NIRCam

MIRI

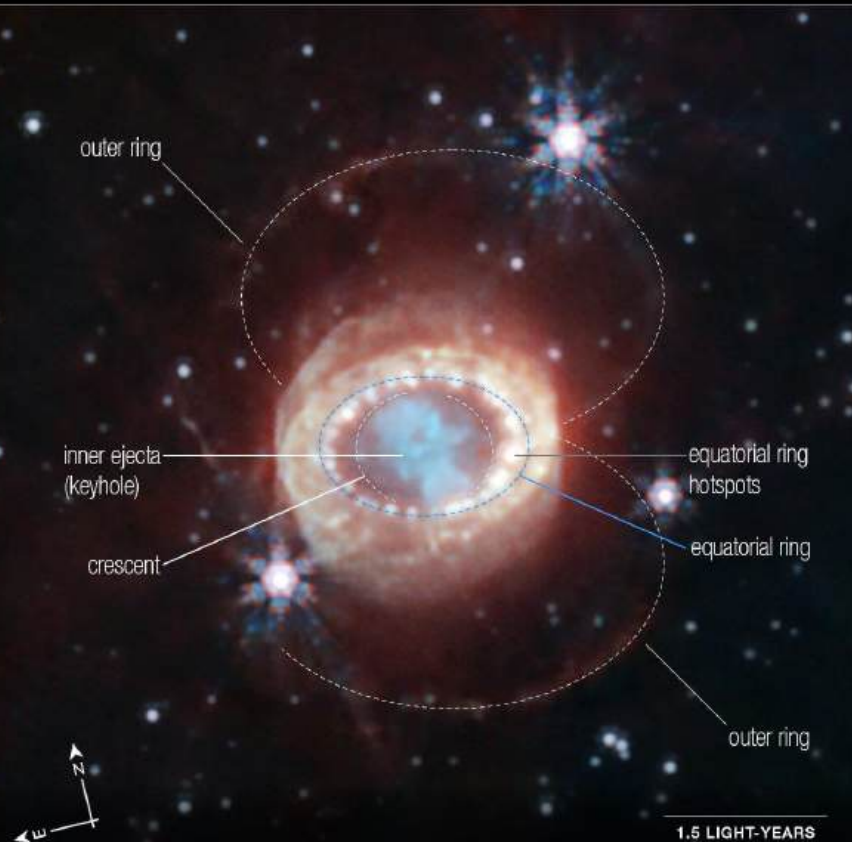




južná prstencová hmlo

NIRCam

JAMES WEBB SPACE TELESCOPE
SUPERNOVA 1987A



NIRCam Filters | F150W F164N F200W F322N F405N F444W

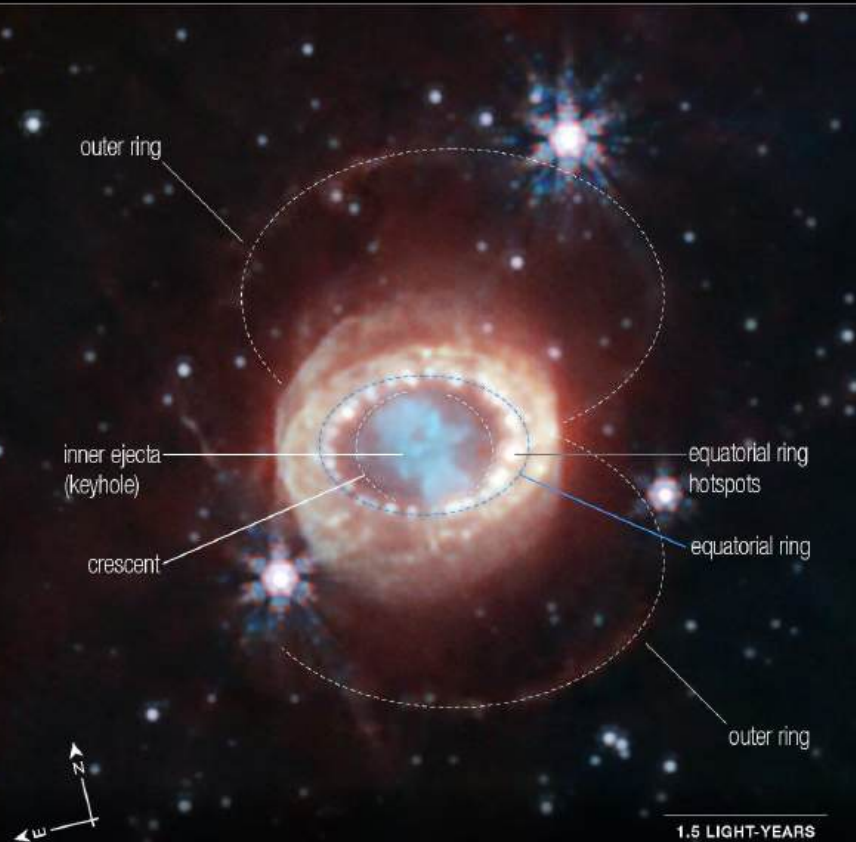




južná prstencová hmlo

NIRCam

JAMES WEBB SPACE TELESCOPE
SUPERNOVA 1987A



NIRCam Filters | F150W F164N F200W F322N F405N F444W

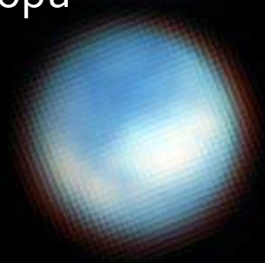


JUPITER A JEHO OKOLIE

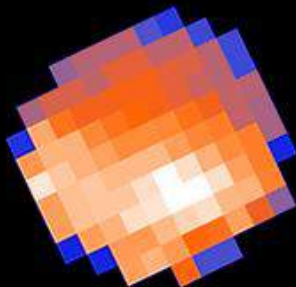
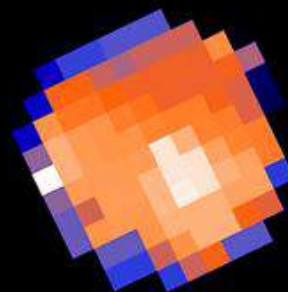
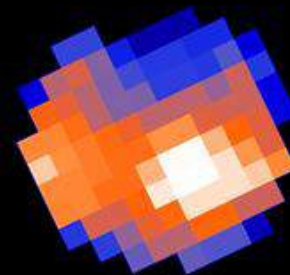


NIRCam

Europa



Europa (NIRCam)

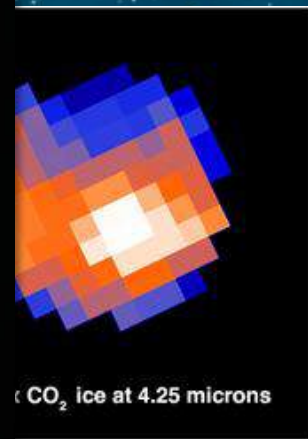
Crystalline CO₂ ice at 2.7 micronsCrystalline CO₂ ice at 4.27 micronsComplex CO₂ ice at 4.25 microns

JUPIT

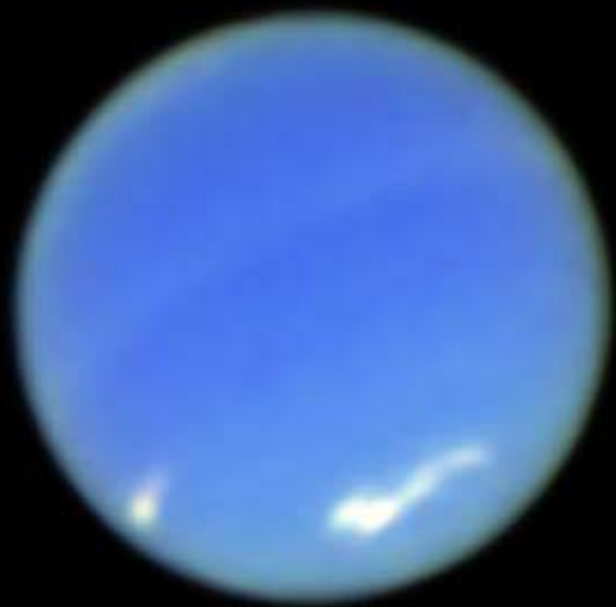
NIRCam

Europa

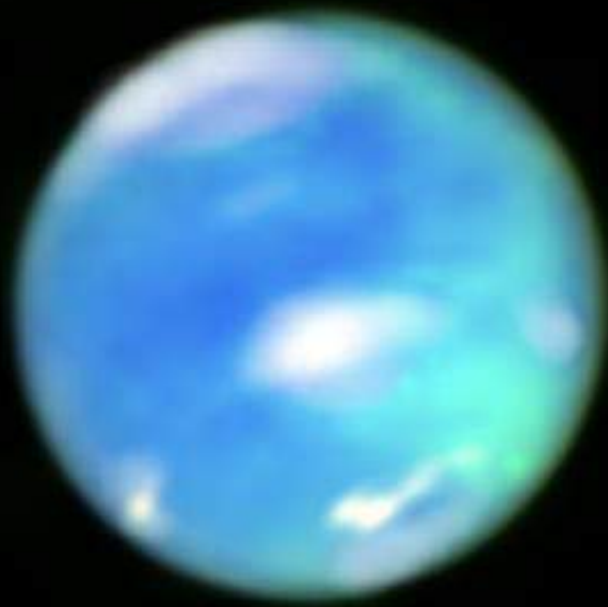
Europa (NIRCam







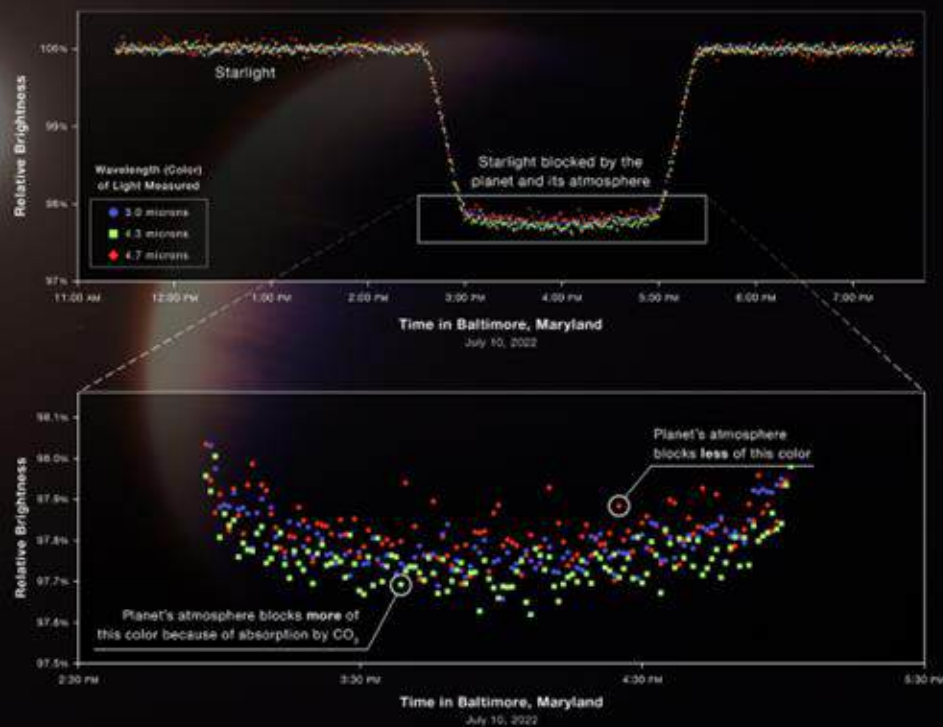
Hubble



Hubble + Webb

HOT GAS GIANT EXOPLANET WASP-39 b TRANSIT LIGHT CURVE

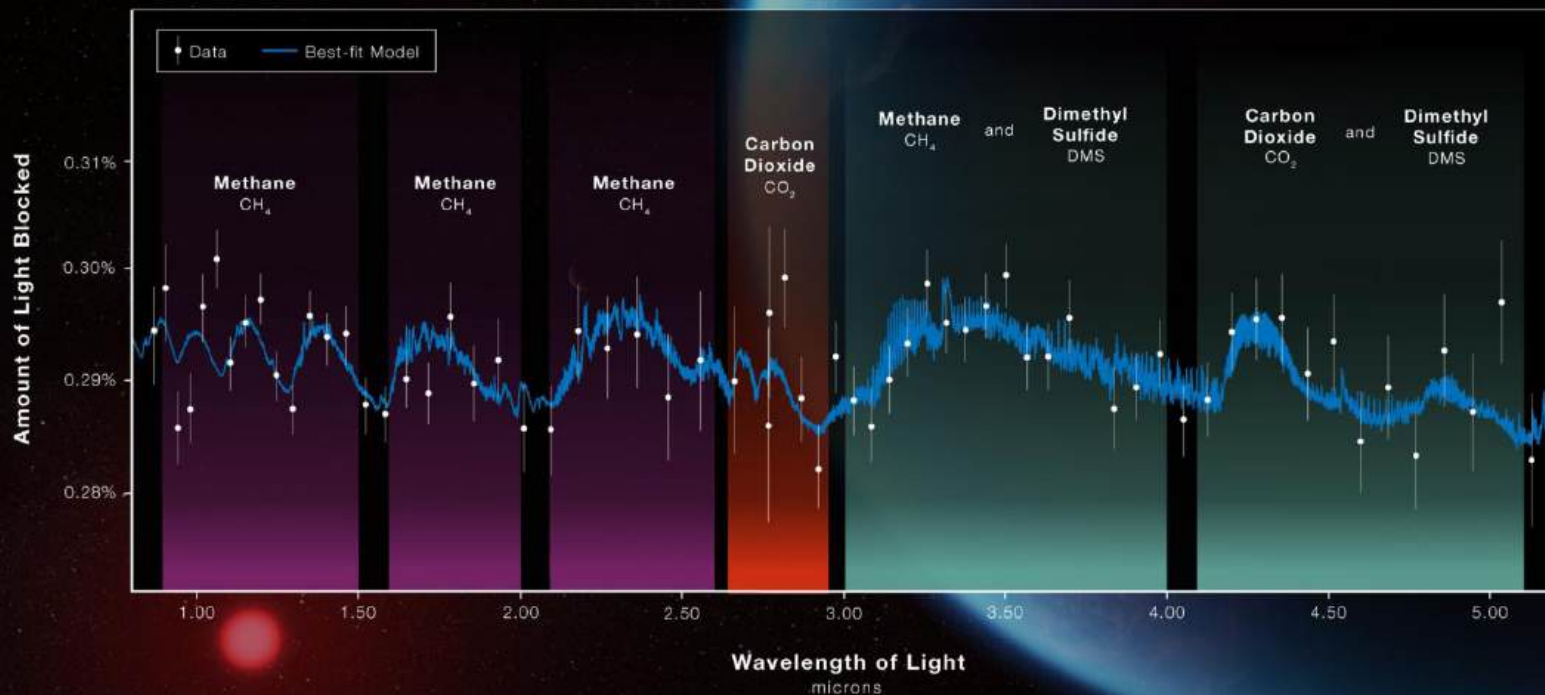
NIRSpec | Bright Object Time-Series Spectroscopy

**WEBB**
SPACE TELESCOPE

EXOPLANET K2-18 b

ATMOSPHERE COMPOSITION

NIRISS and NIRSpec (G395H)



WEBB
SPACE TELESCOPE

GAS-GIANT EXOPLANET WASP-107 b

TRANSMISSION SPECTRUM

Hubble WFC3

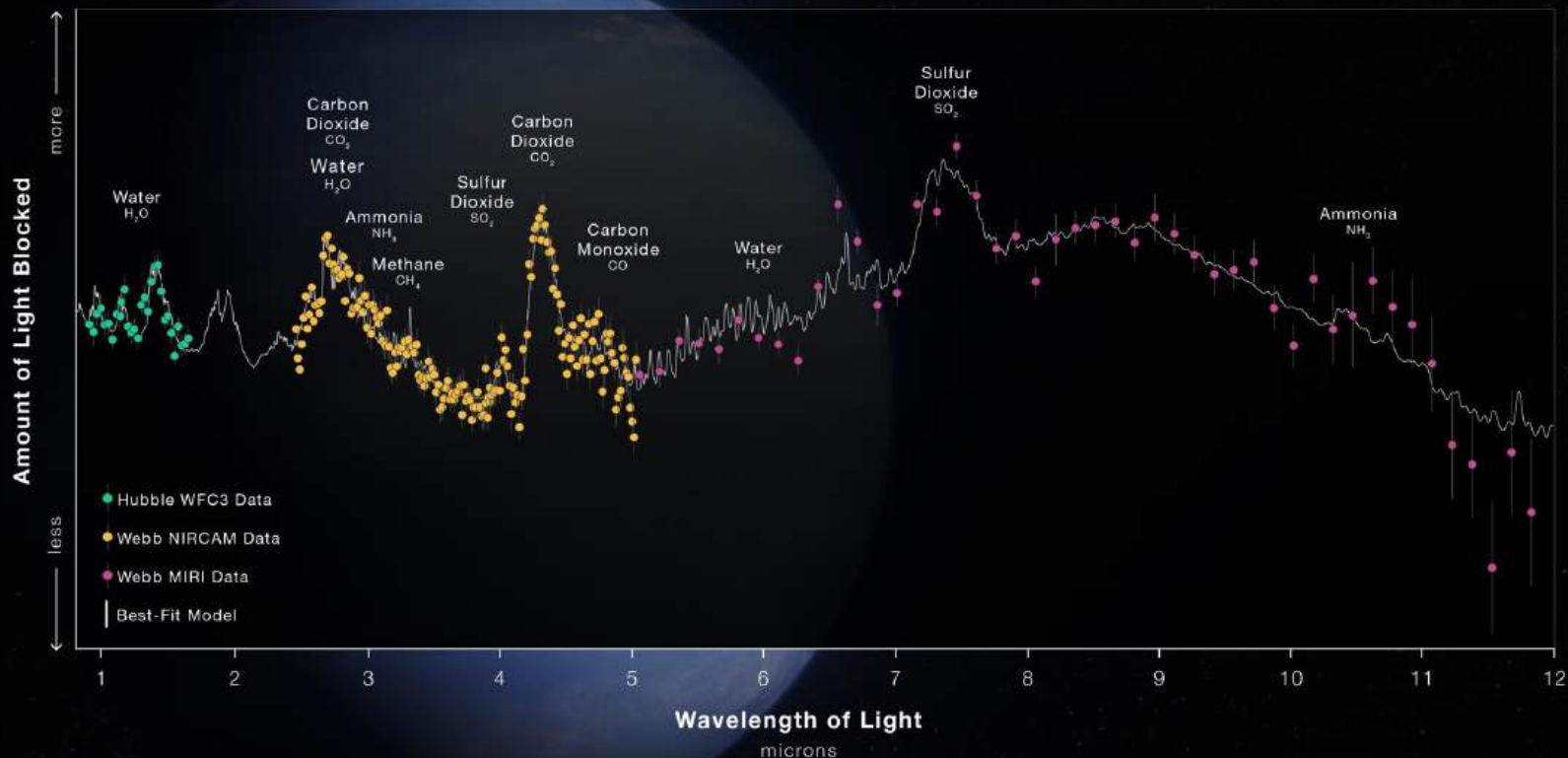
Grism Spectroscopy

Webb NIRCarn

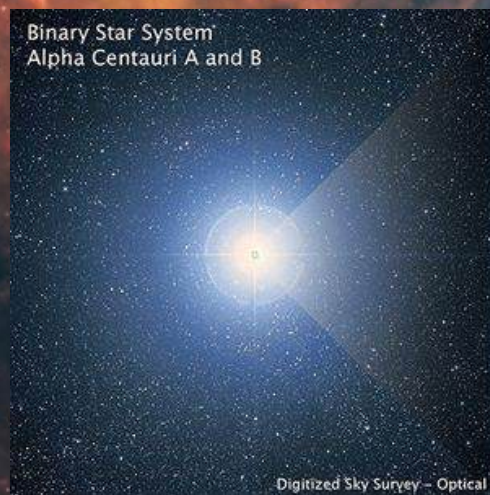
Grism Spectroscopy

Webb MIRI

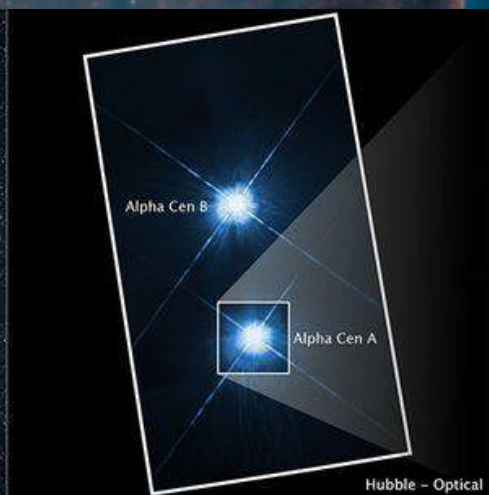
Low-Resolution Spectroscopy



Binary Star System Alpha Centauri A and B



Digitized Sky Survey – Optical



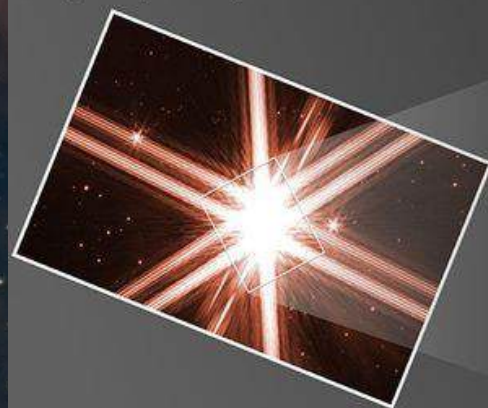
Hubble – Optical



Webb – Infrared (starlight subtracted)

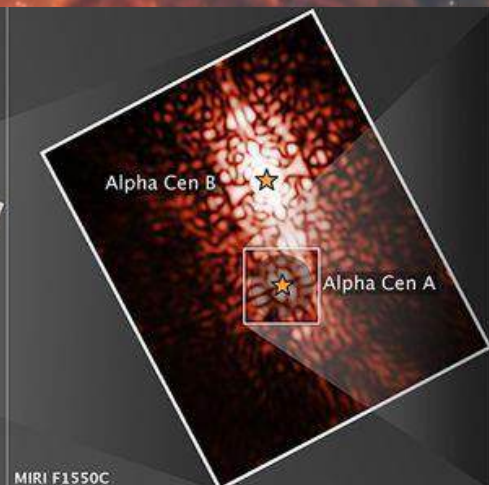


Binary Star System Alpha Centauri A and B



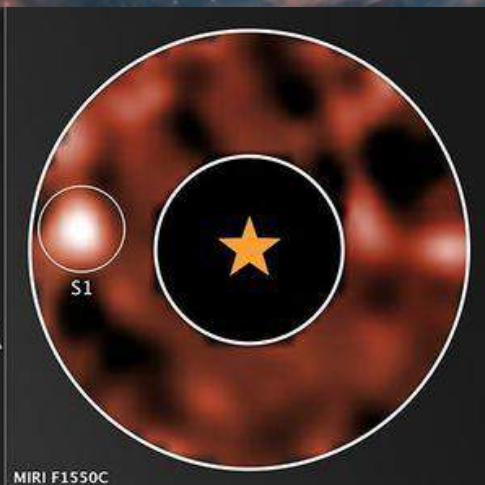
MIRI F1000W

Webb's first image of the binary system, before a light-blocking mask was put in place.



MIRI F1550C

A zoomed-in version with the mask placed over Alpha Centauri A, blocking more than 99 percent of the star's light.



MIRI F1550C

Remaining starlight from both stars subtracted to reveal candidate planet.



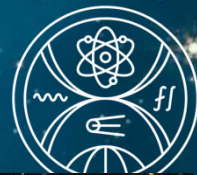
MEGA JET – 8LY



BULLET CLUSTER



BULLET CLUSTER

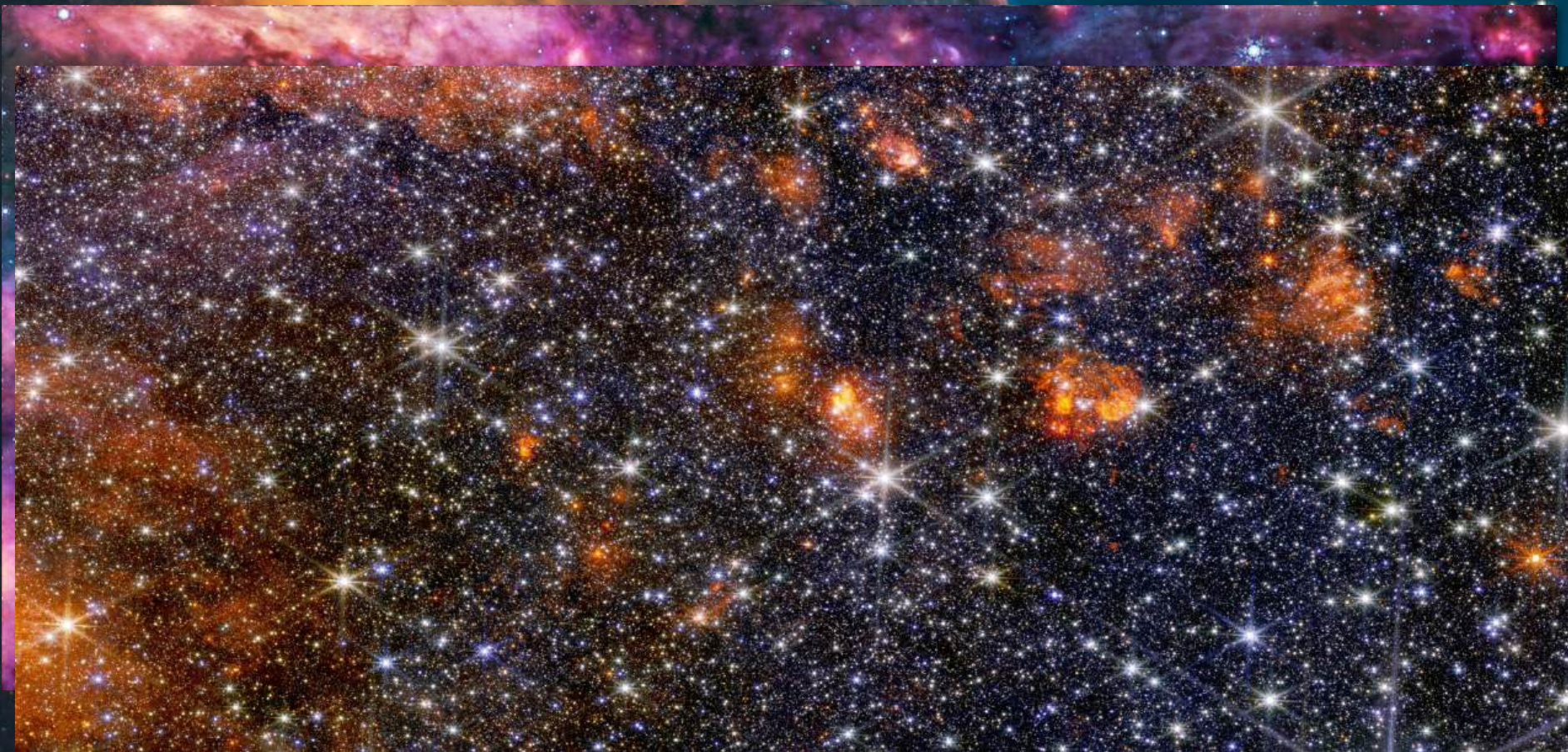




Sombrero Galaxy (NIRCam/MIRI)



SAGITTARIUS B₂ - HVIEZDOTVORBA

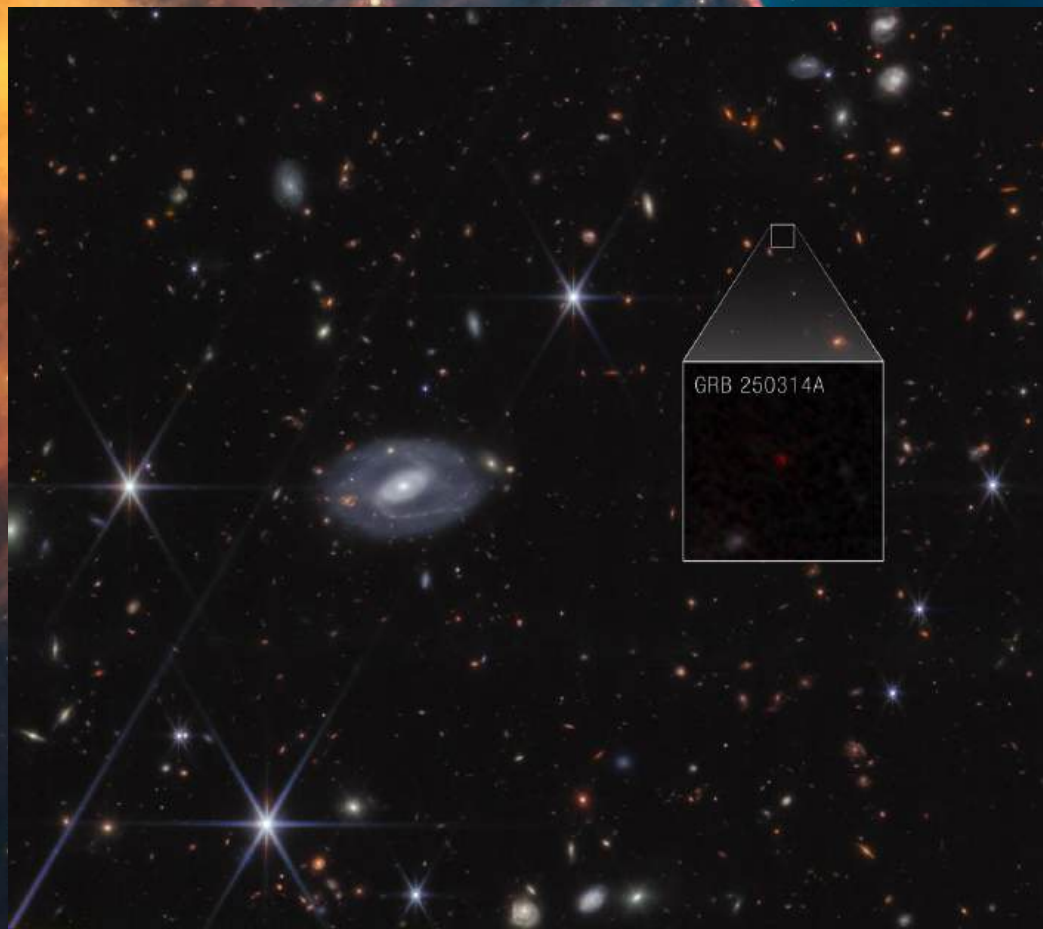
SAGITTARIUS B₂ - HVIEZDOTVORBA



Cranium Nebula



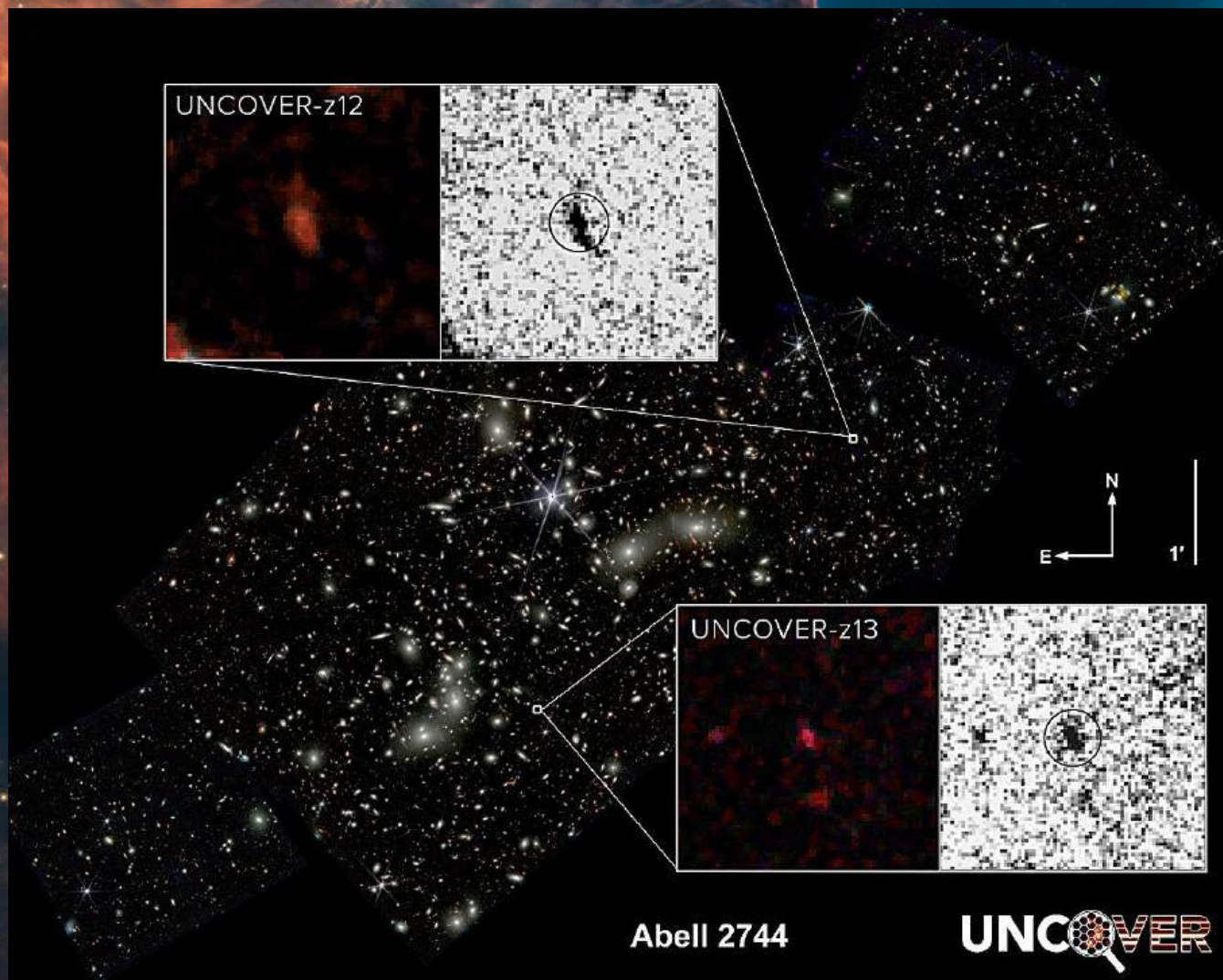
NASJTARŠIA SUPERNOVA



ČIERNE DIERY BEZ GALAXIE

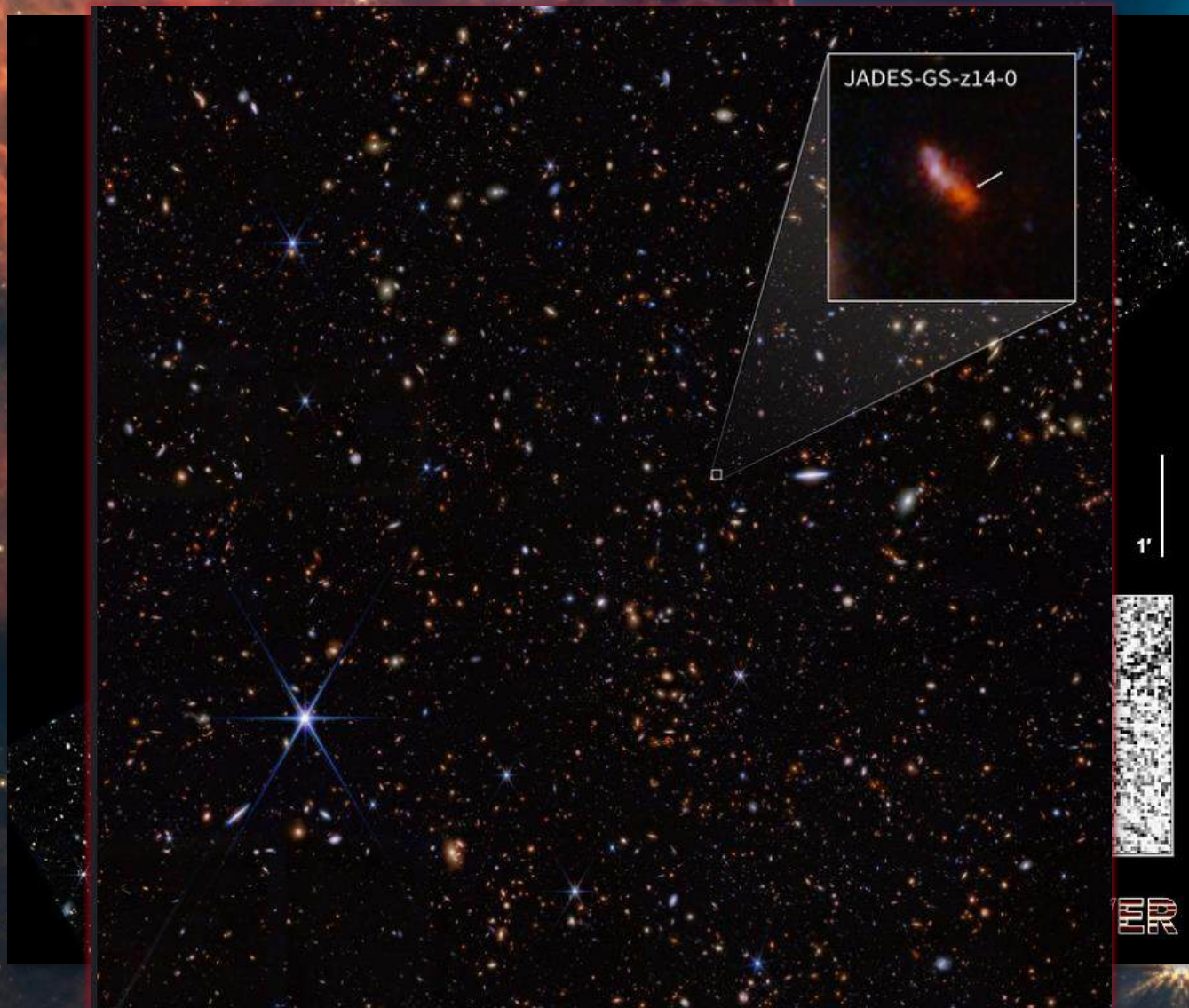


NA



41

NA



JADES-GS-z14-0



NA

JADES-GS-z14-0



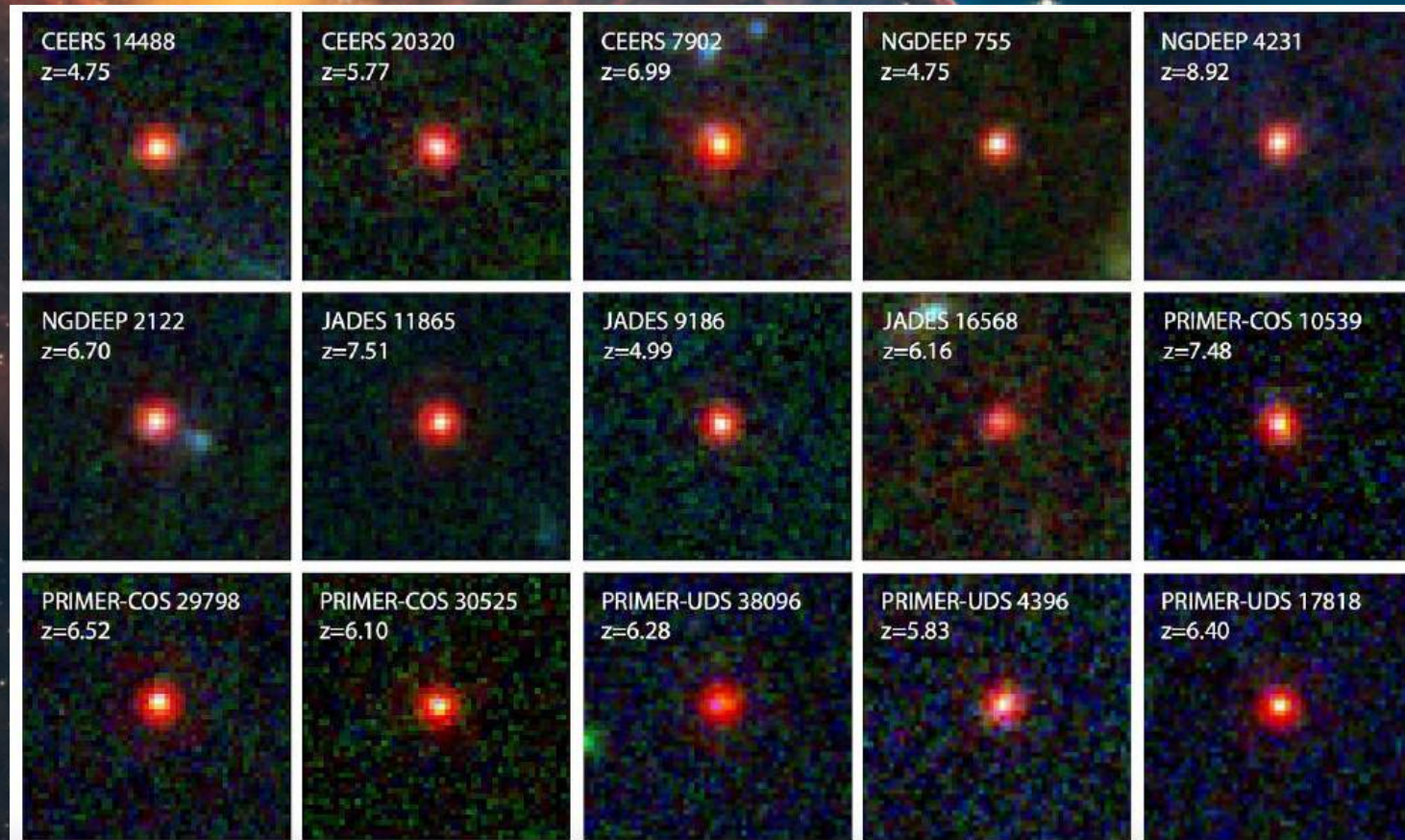
MoM-z14



ER



LITTLE RED DOTS



1st anniversary



2nd anniversary



3rd anniversary



3rd anniversary

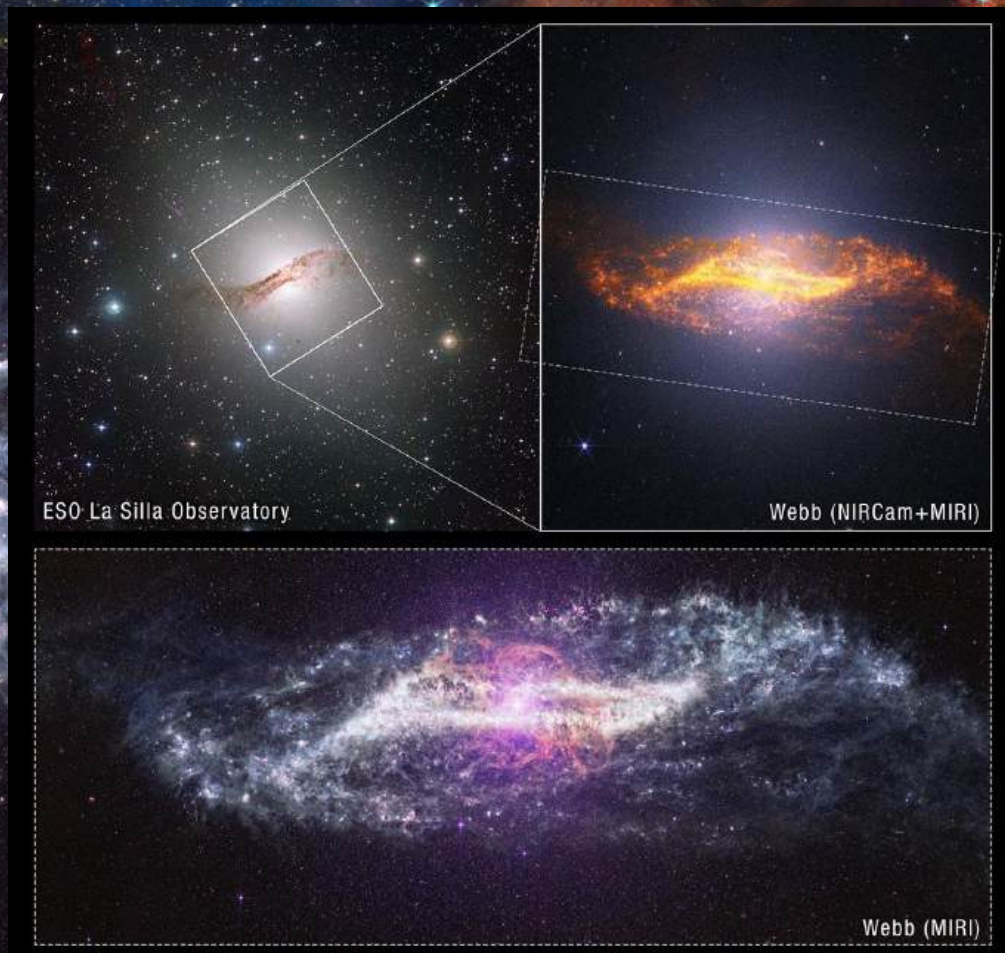


4th anniversary



3rd anniversary

4th anniversary





ESA GAIA



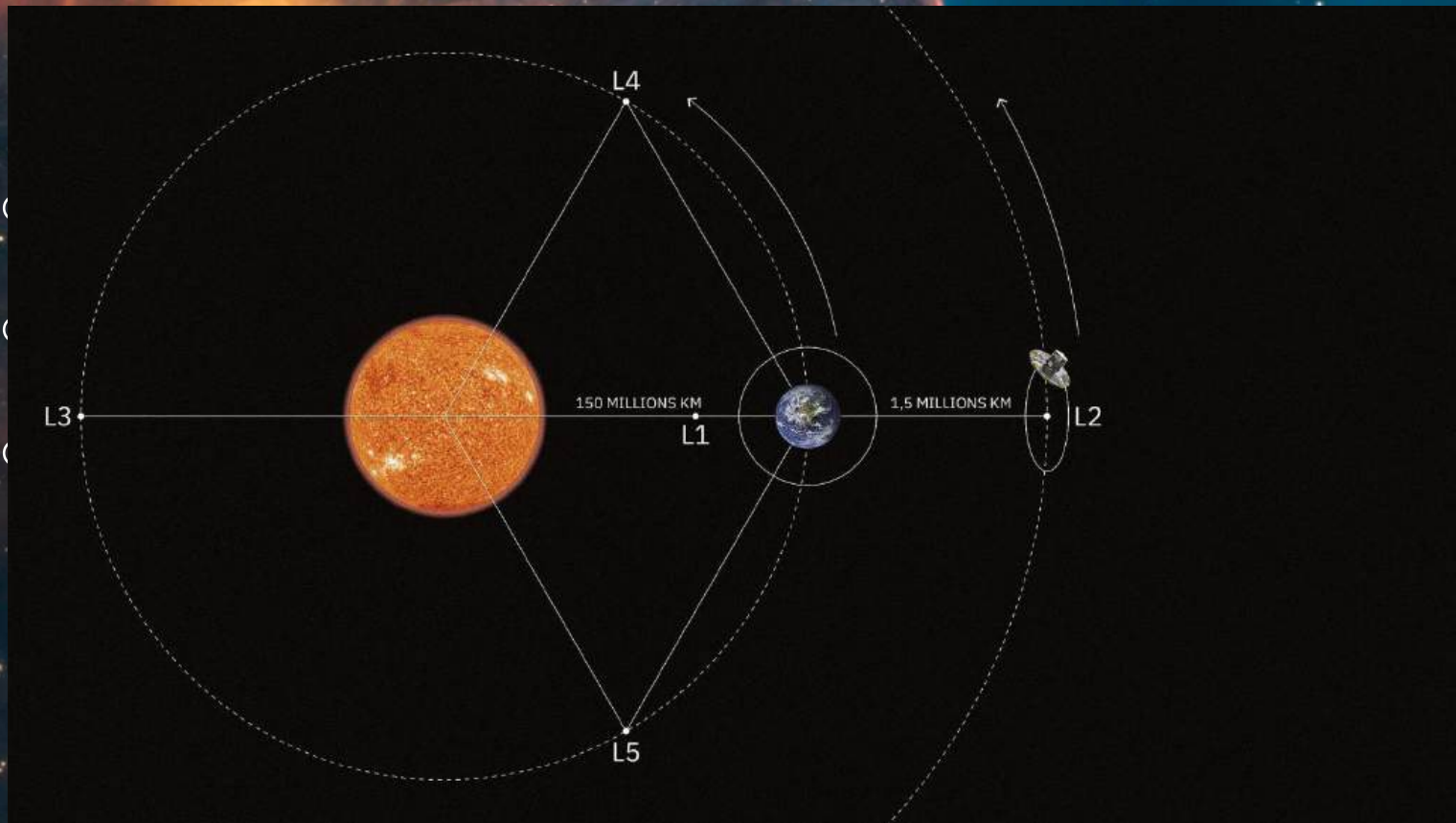
GAIA – GALAKTICKÁ REVOLÚCIA



- December 2013 – 2025 (6 rokov dlhšie ako sa plánovalo)
- Misia ESA v L2
- Astrometrická, fotometrická a spektroskopická misia



GAIA – GALAKTICKÁ REVOLÚCIA



GAIA – GALAKTICKÁ REVOLÚCIA



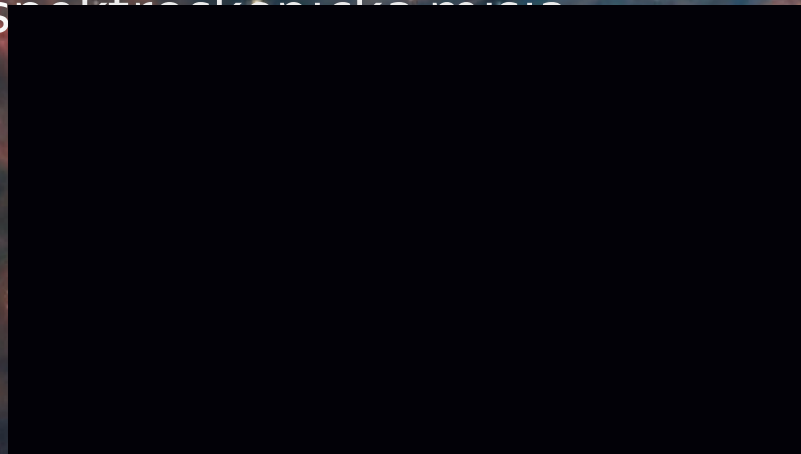
- December 2013 – 2025 (6 rokov dlhšie ako sa plánovalo)
- Misia ESA v L2
- Astrometrická, fotometrická a spektroskopická misia



GAIA – GALAKTICKÁ REVOLÚCIA



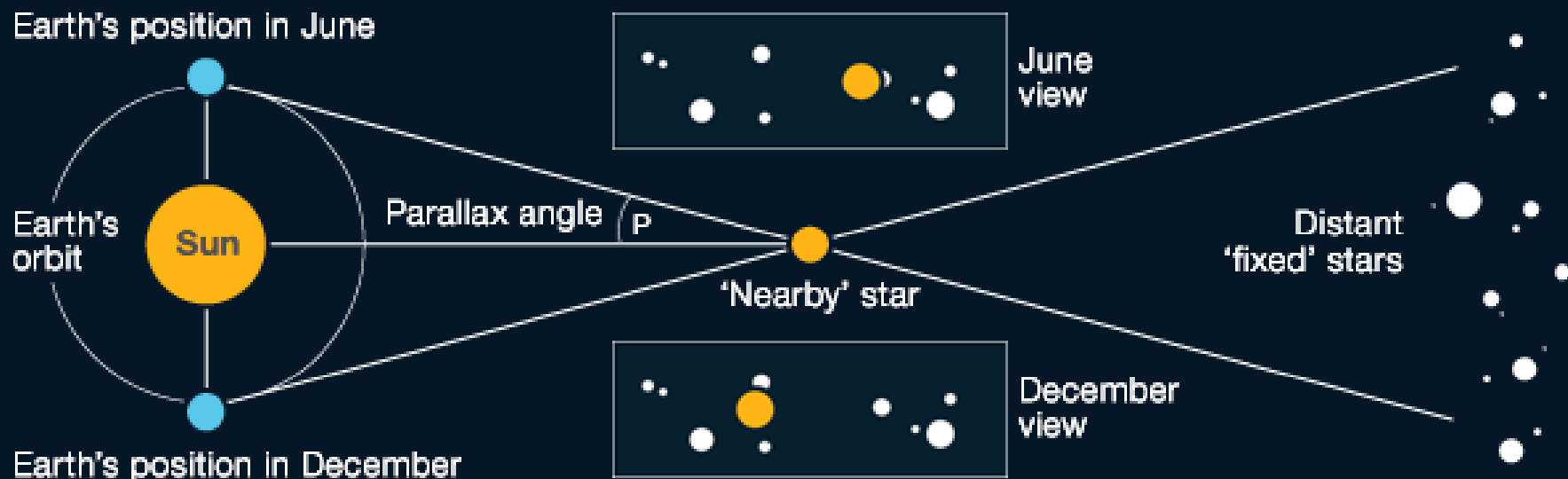
- December 2013 – 2025 (6 rokov dlhšie ako sa plánovalo)
- Misia ESA v L2
- Astrometrická, fotometrická a spektroskopická misia



GAIA – GALAKTICKÁ REVOLÚCIA



Stellar parallax: Measuring the distance of stars



Source: University of Virginia





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ASTRONÓFIE A
ASTROFYZIKY
Fakulta vied
Univerzita Komenského
Bratislava

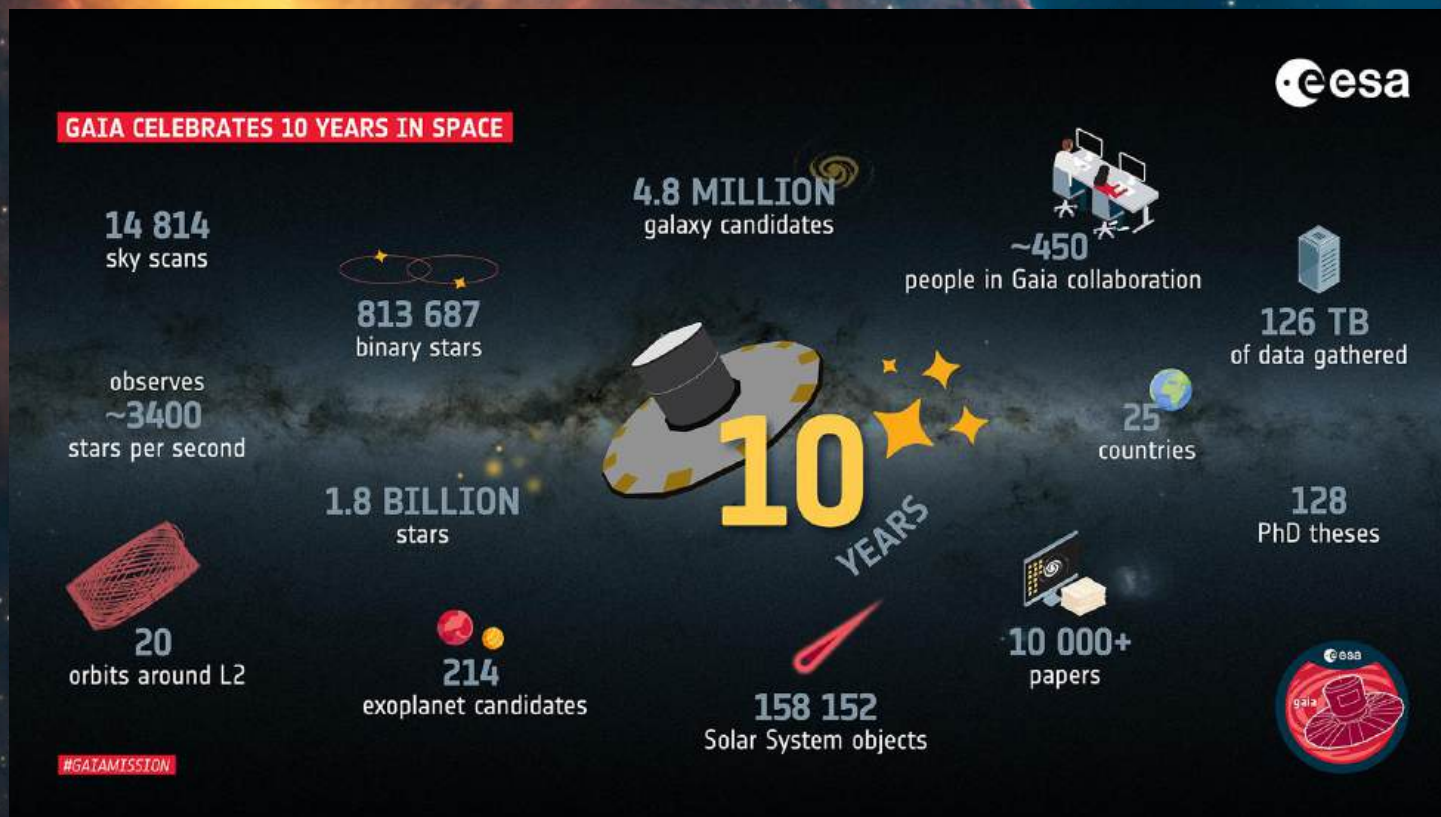


Gaia – Galactic Revolution



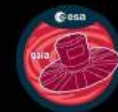
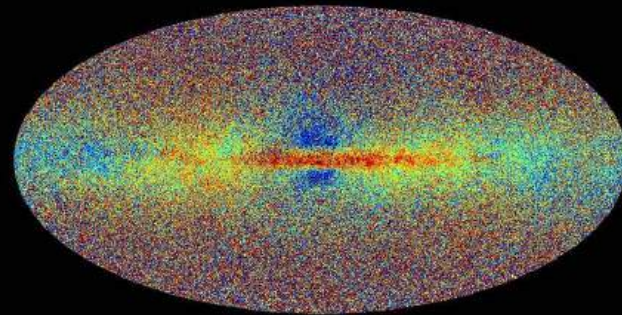
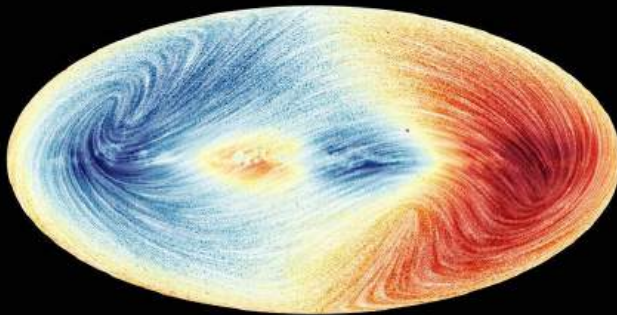
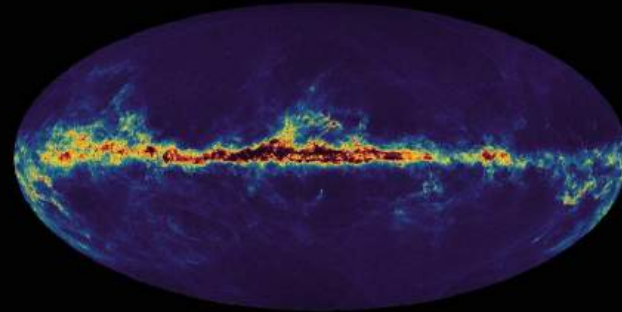
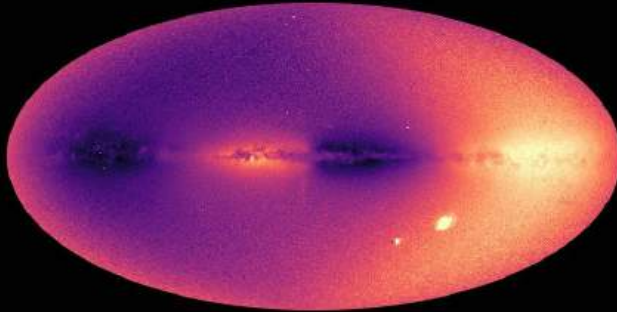


GAIA – GALAKTICKÁ REVOLÚCIA





GAIA: EXPLORING THE MULTI-DIMENSIONAL MILKY WAY



GALAKTICKÉ PREDMESTIA

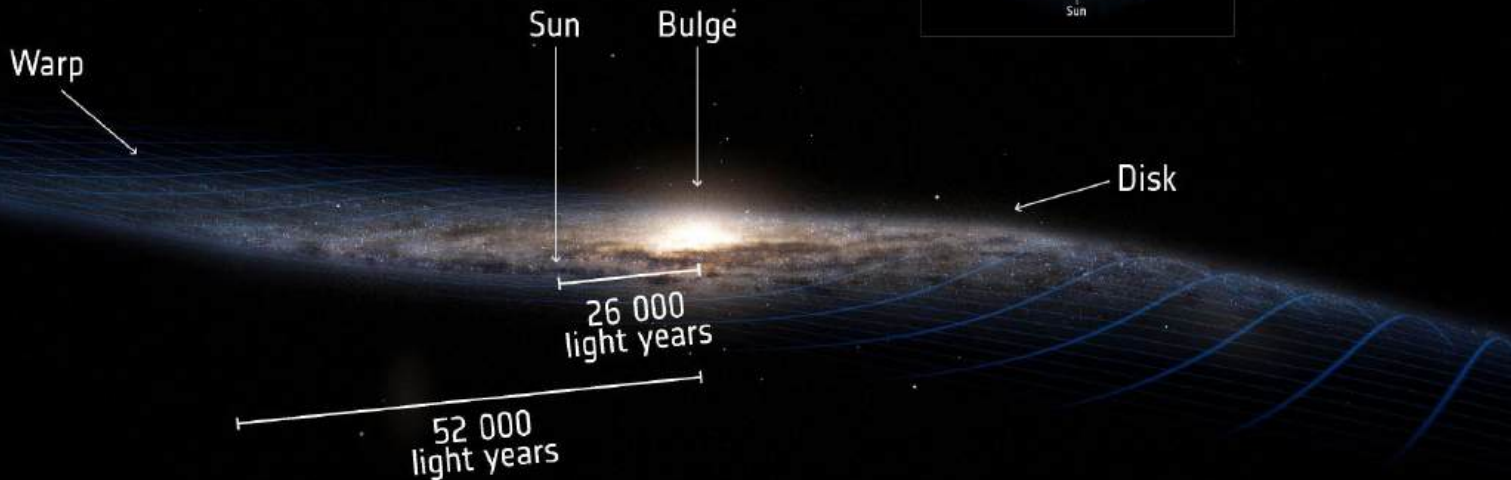
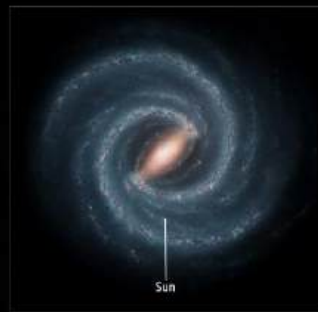
- Galaktický warp

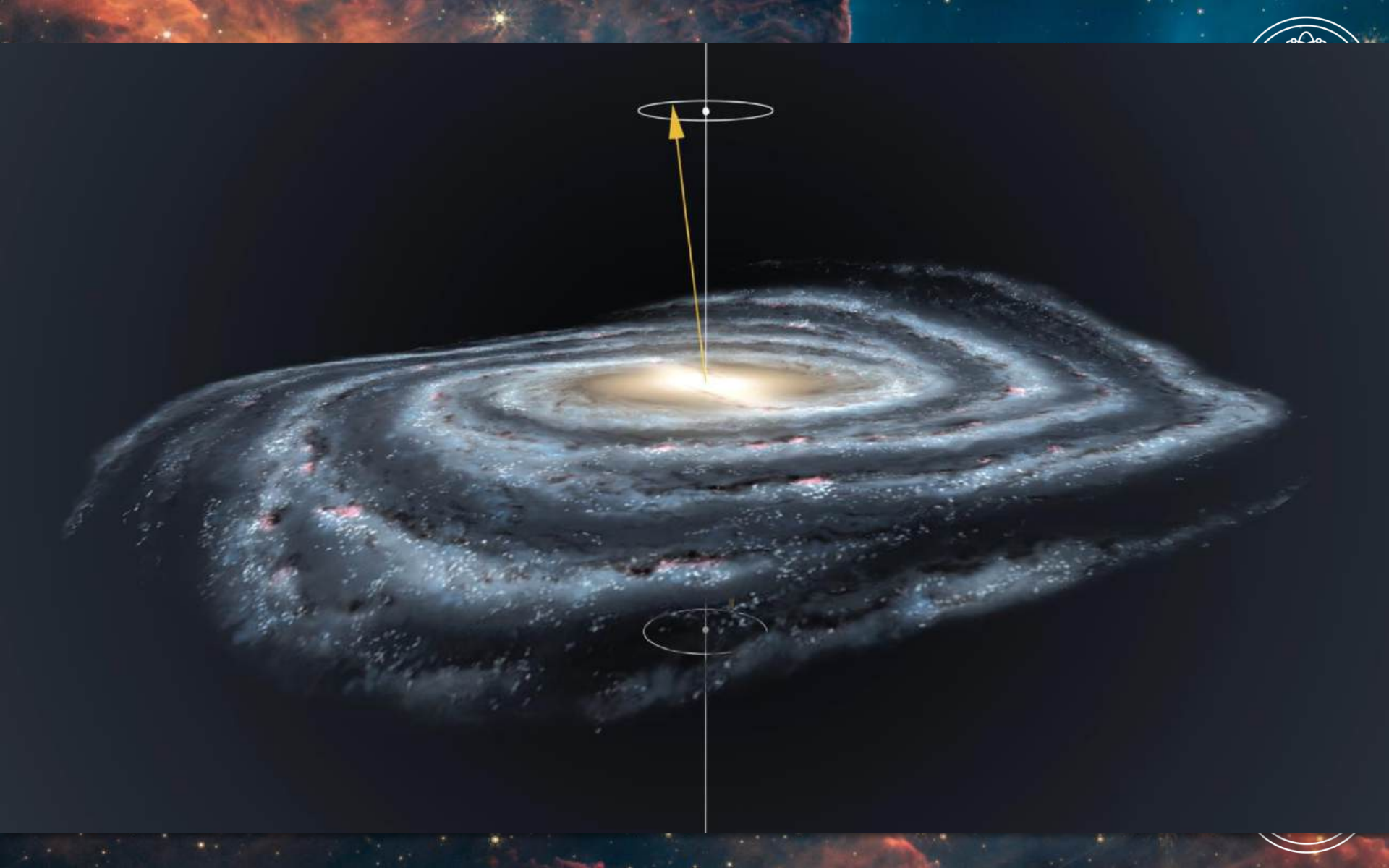


DIVISION OF
ASTRONOMY AND ASTROPHYSICS
Faculty of Mathematics, Physics and Informatics
Comenius University
Bratislava



THE MILKY WAY'S WARP









GALAKTICKÉ PREDMESTIA

- Galaktický warp
- Galaktický flare – zhrubnutie disku



GALAKTICKÉ PREDMESTIA



- Galaktický warp
- Galaktický flare – zhrubnutie disku

$$\rho(R, z) = \rho_0(R) \times \exp(-R/h_r) \times \exp(-|z|/h_z)$$



GALAKTICKÉ PREDMESTIA



- Galaktický warp
- Galaktický flare – zhrubnutie disku

$$\rho(R, z) = \rho_0(R) \times \exp(-R/h_r) \times \exp(-|z|/h_z)$$

- Zvyšovanie škálovacej hrúbky (h_z) disku so zvyšujúcou sa vzdialenosťou od stredu Galaxie
- Pozorované v plynných vodíkových aj hviezdnych populáciách



GALAKTICKÉ PREDMESTIA

- Galaktický warp – skútenie
-
- Galaktický flare – zhrubnutie disku
- Hviezdne prúdy



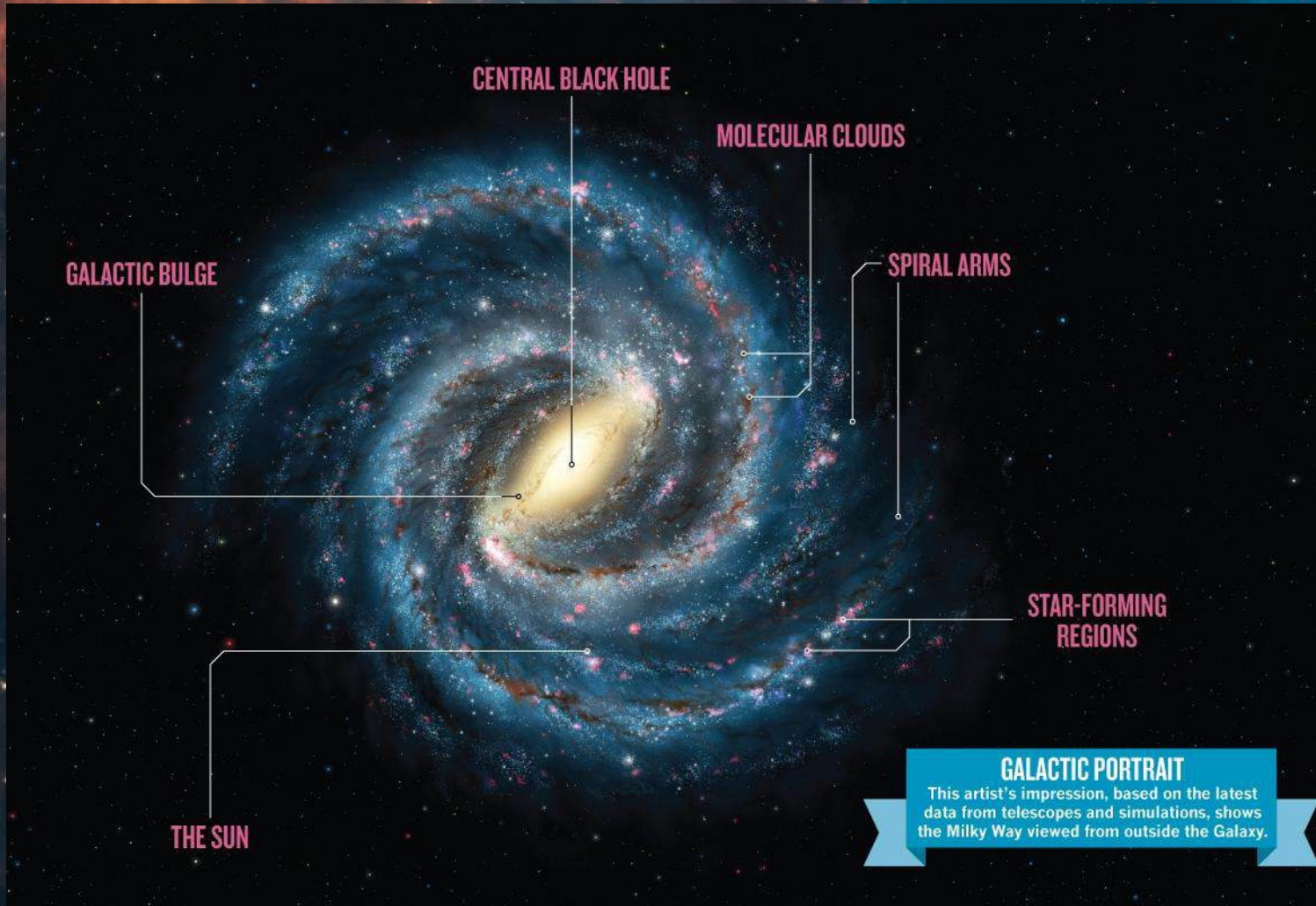
GALAKTICKÉ PREDMESTIA

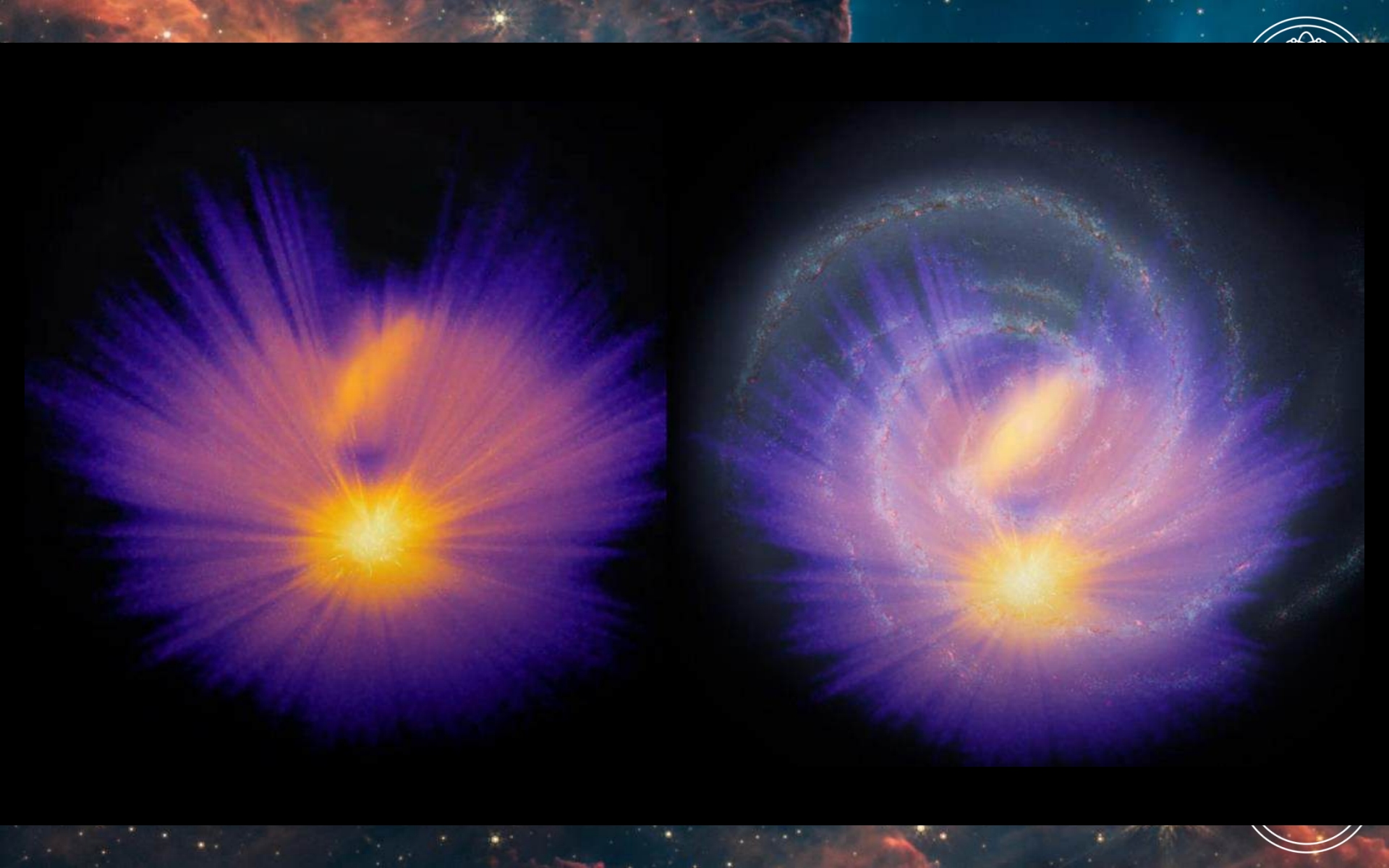


GALAKTICKÉ PREDMESTIA



©Trevor Johnson





GAIA A NEDÁVNE GALAKTICKÉ SPLYNUTIE



- Známe galaktické splynutie Gaia-Sausage-Enceladus – 8 až 11 Gyr
- Gaia DR3 naznačujú nedávnu veľkú zrážku 3 Gyr
- Virgo Radial Merger



GAIA A NEDÁVNE GALAKTICKÉ SPLYNUTIE



esa

9.8 billion years ago
1.3 billion years before collision

8



©Trevor Gibson



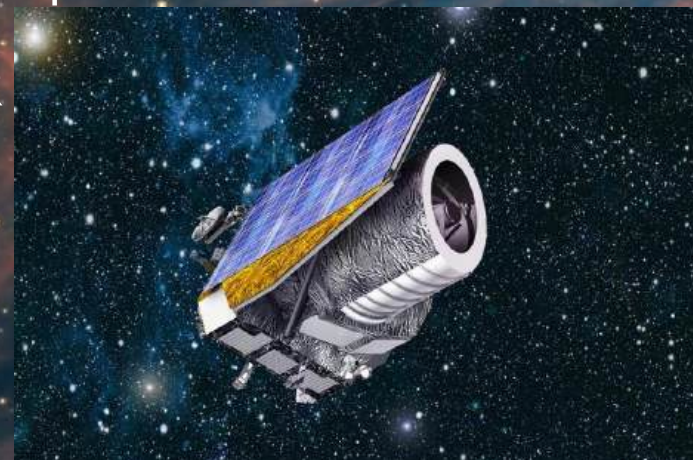
ESA EUCLID



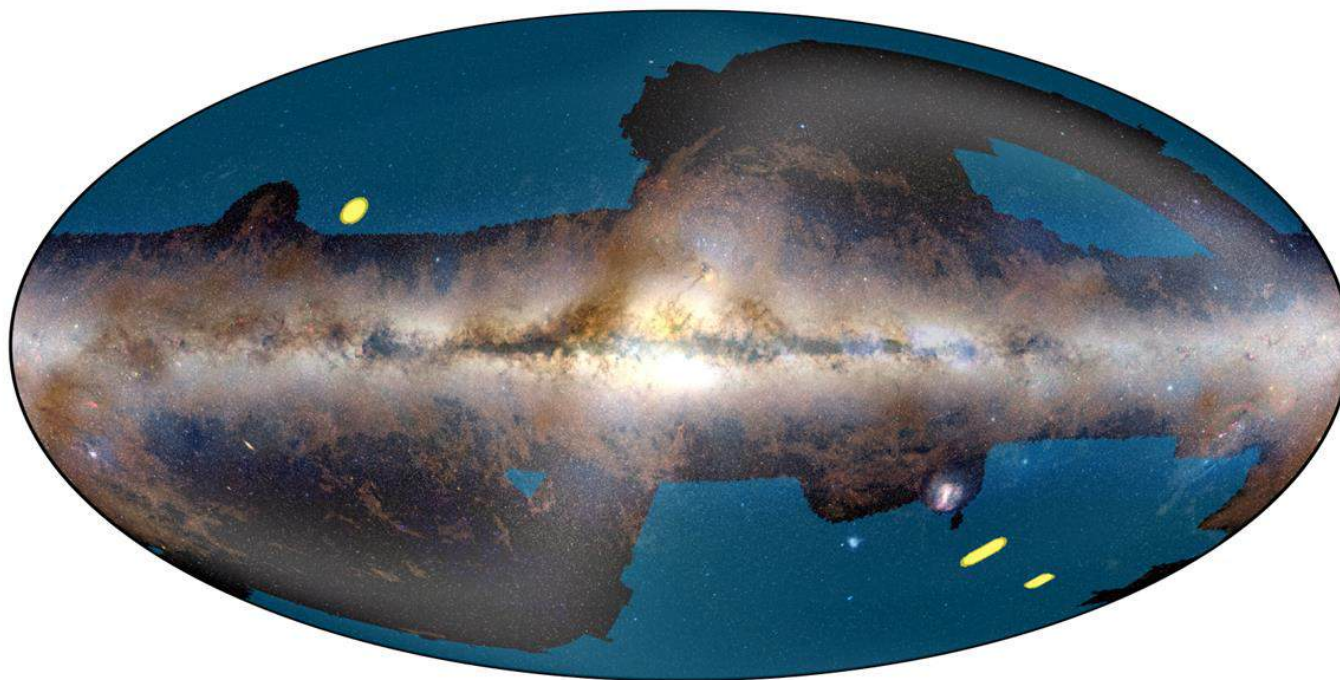
EUCLID – PRIESKUM HLBOKÉHO VESMÍRU



- 2023, prvé údaje 2025
- Misia ESA v L2
- 600 Mpx fotoaparát, 0,5 stupňa x 0,5 stupňa
- Priemer 1,2 m v optickom a blízkom IR
- Fotometrické a spektroskopické



EUCLID – PRIESKUM HL BOKÉHO VESMÍRU



The Euclid Wide Survey and the Euclid Deep Survey

■ Euclid Wide Survey : 13,345 deg² in 6 years, avoiding the galactic plane (stars, dust) and the ecliptic plane (zodiacal light)

■ Euclid Deep Fields : North=20 deg² (top left), Fornax=10 deg² (bottom right), South=23 deg² [+ extended coverage]

⇒ The Euclid survey focuses on the two galactic caps to explore the extragalactic sky (total coverage = 2200 deg²/year) Background: Euclid Consortium / Planck Collaboration / A. Mellinger



EUCLID =

MIRU



The Euclid Wide

Euclid Wide Sur

Euclid Deep Fi

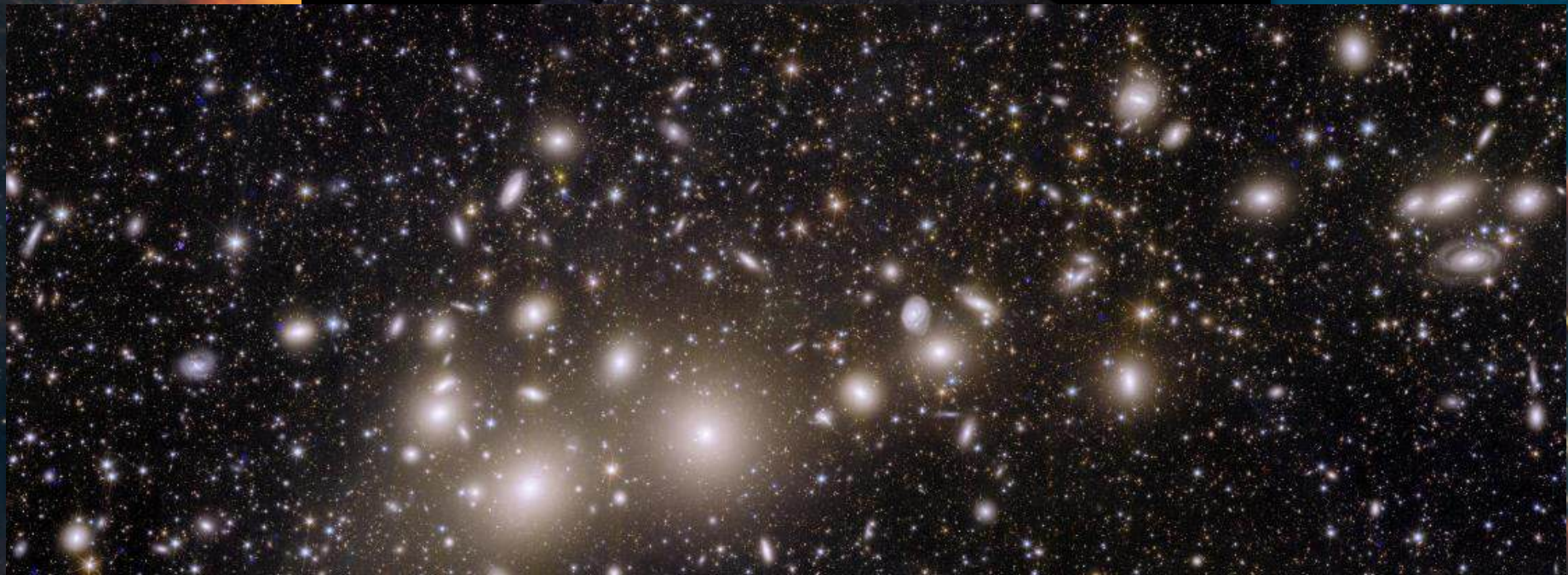
⇒ The Euclid sur



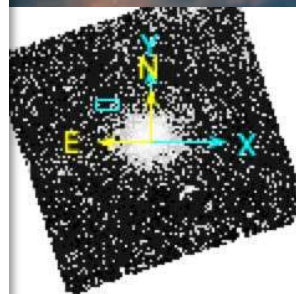
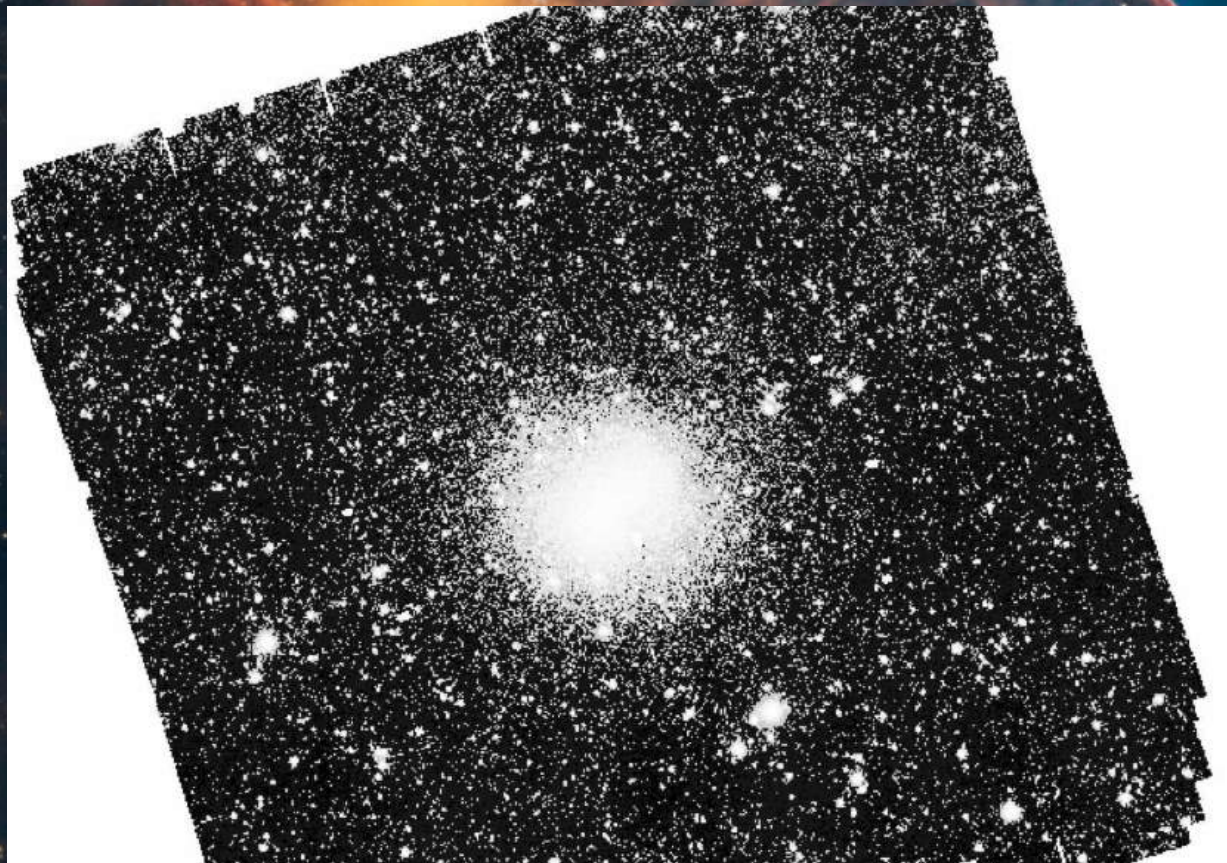
Euclid Collaboration / A. Mellinger

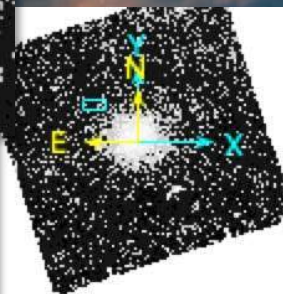
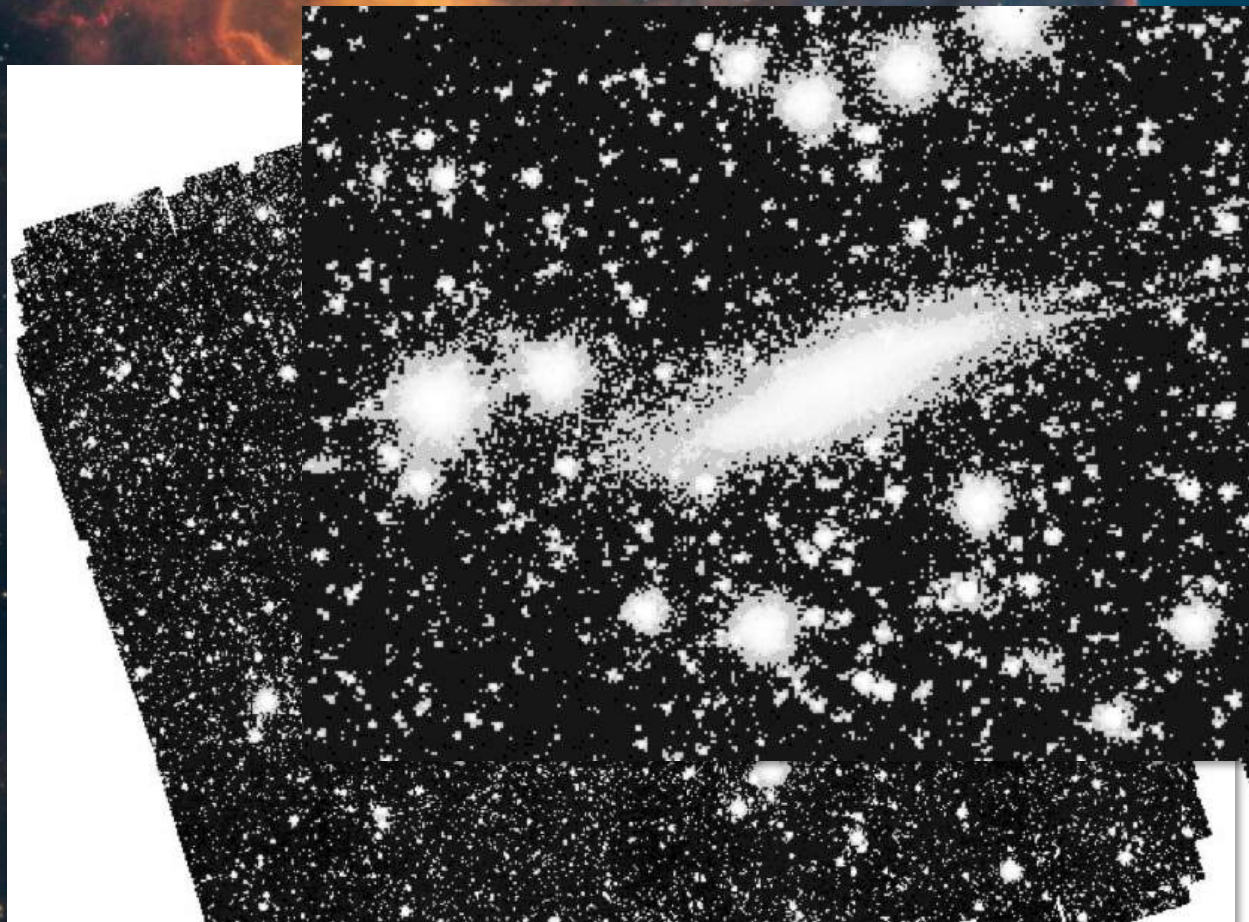


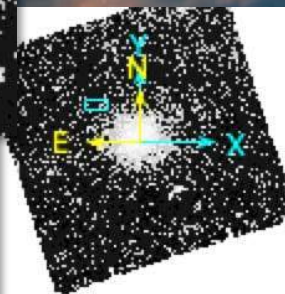
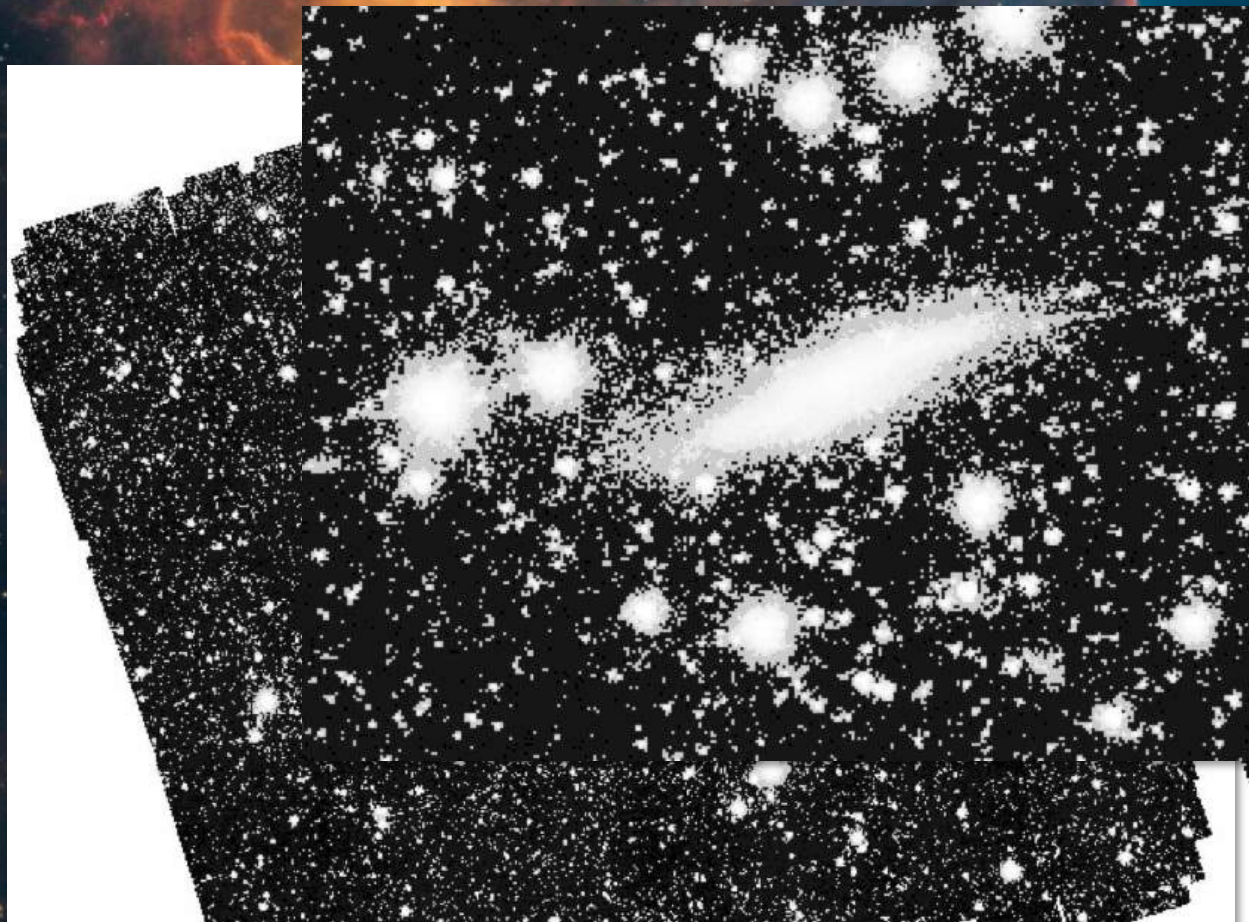


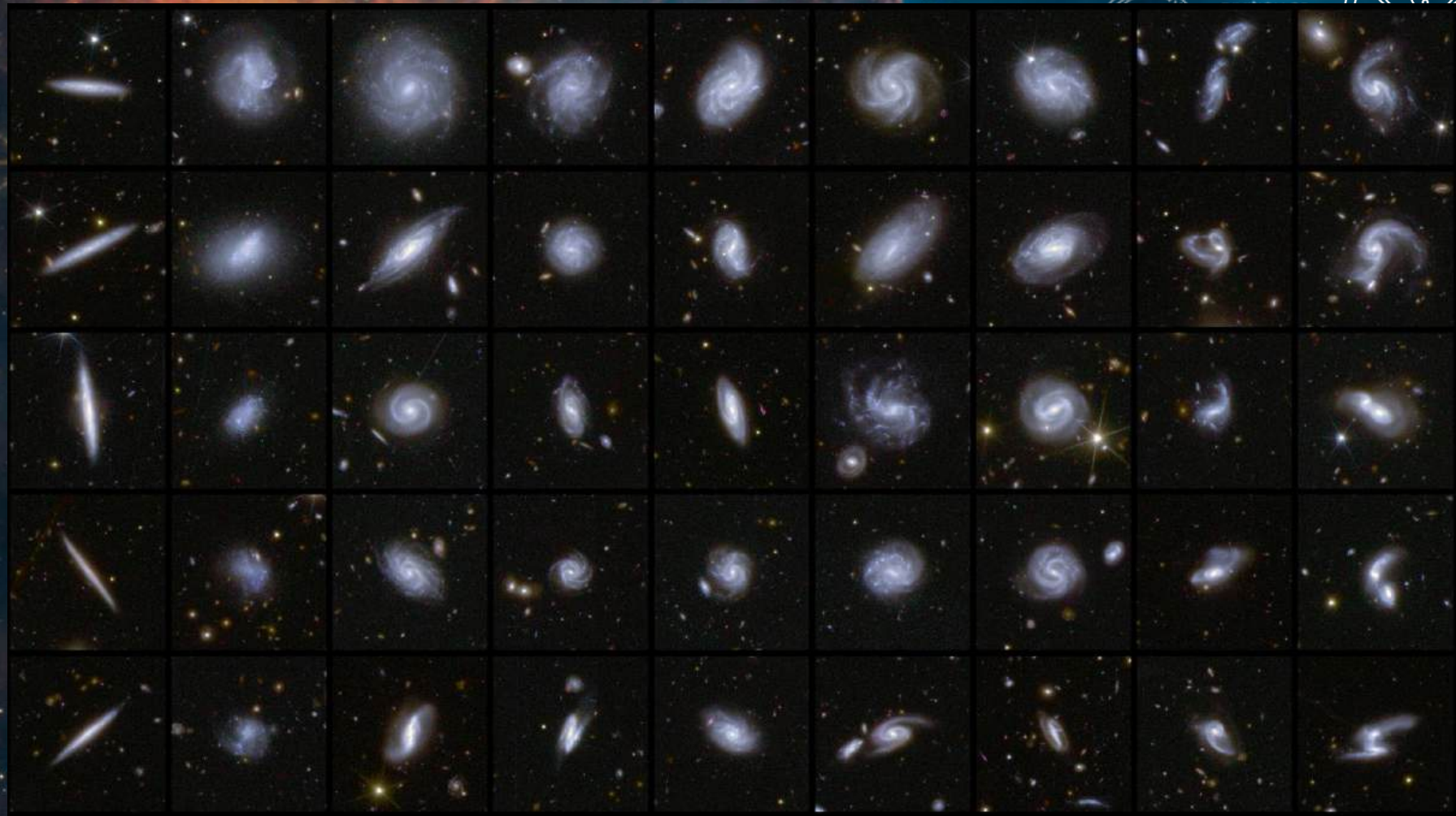


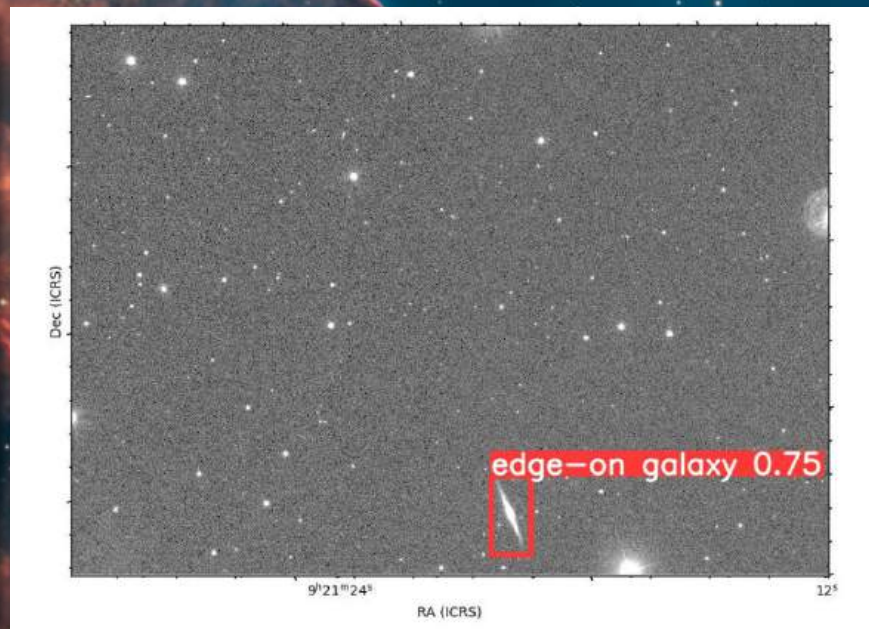


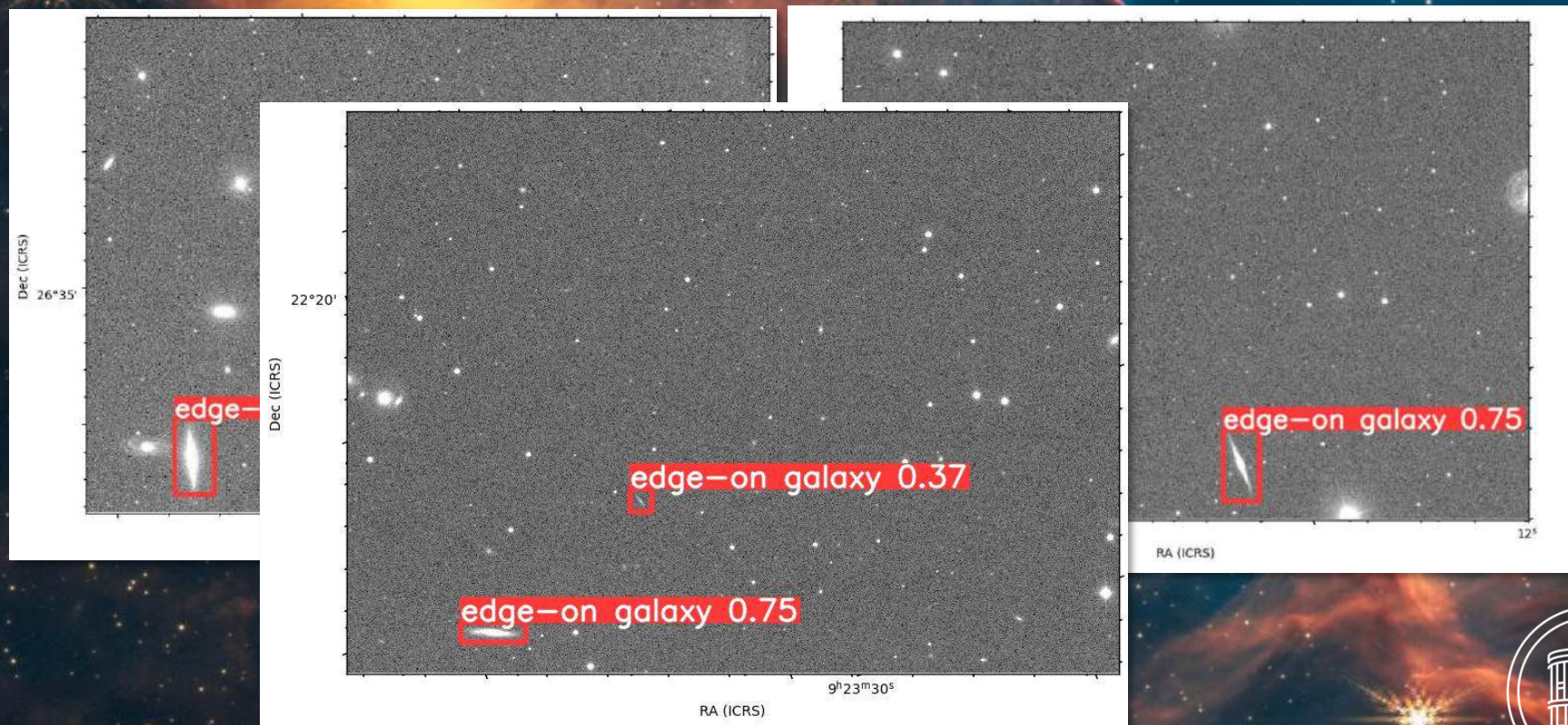


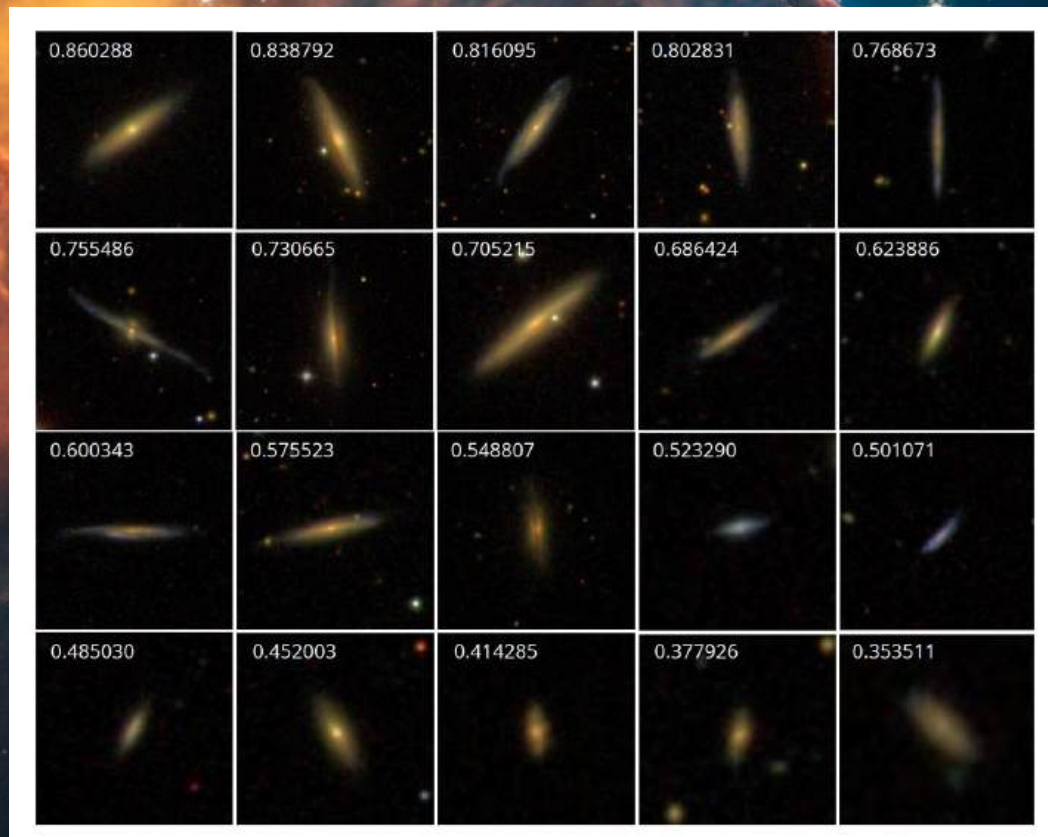


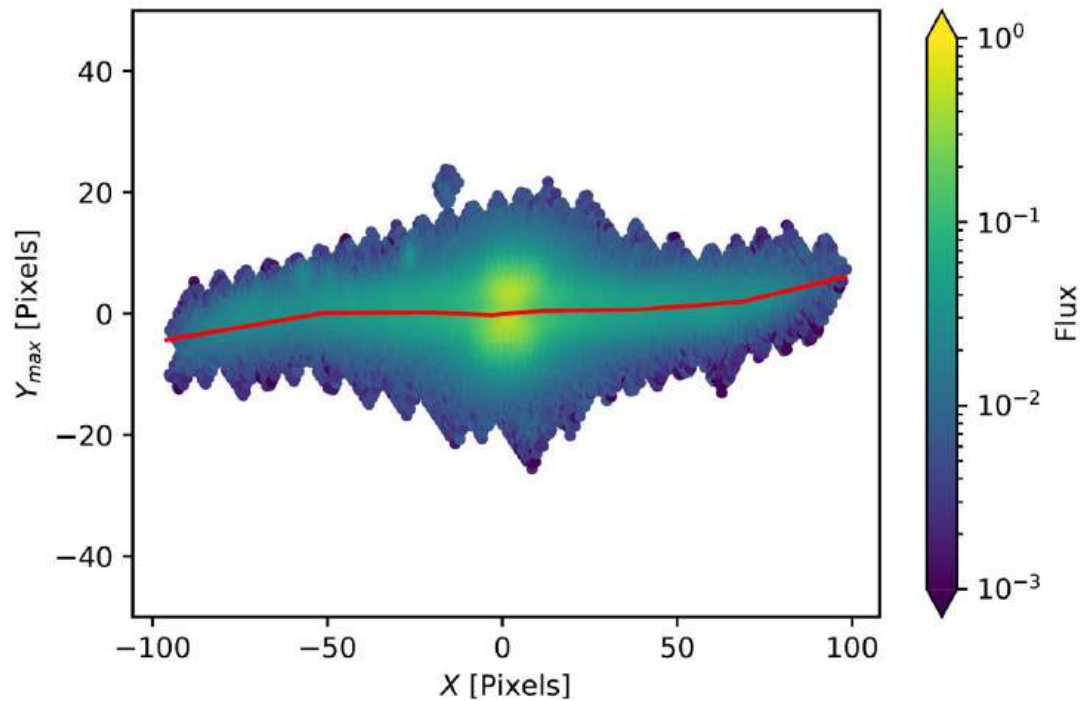
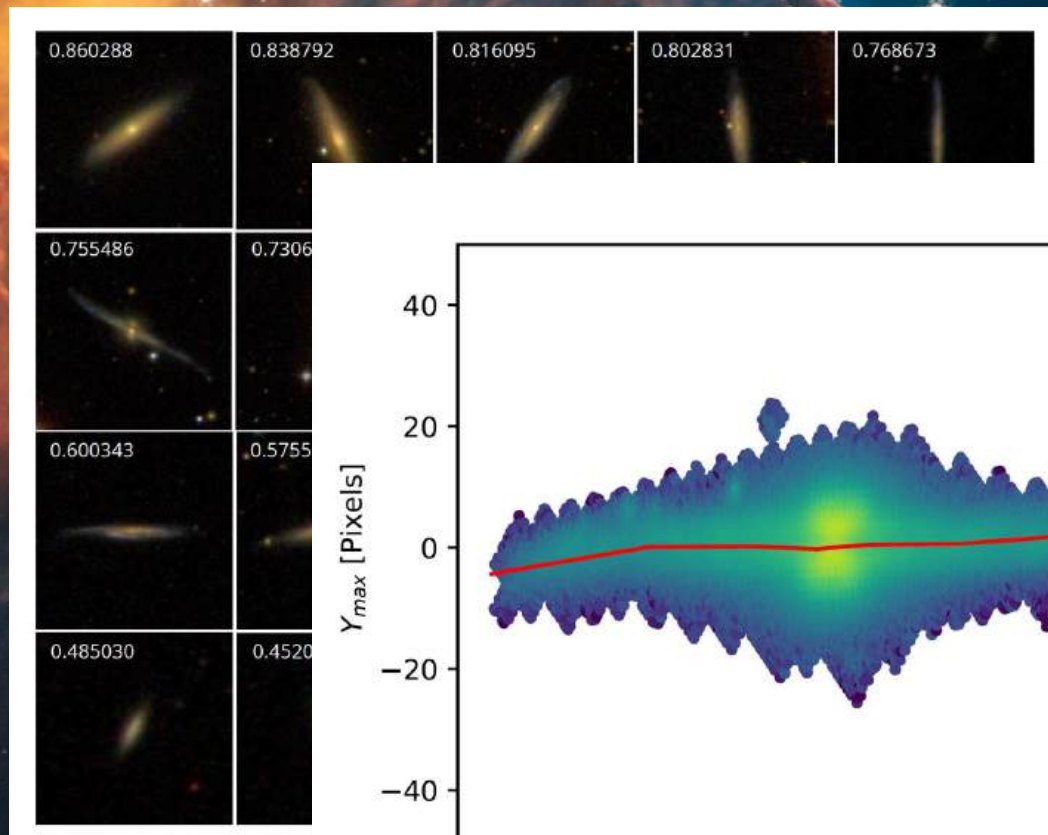






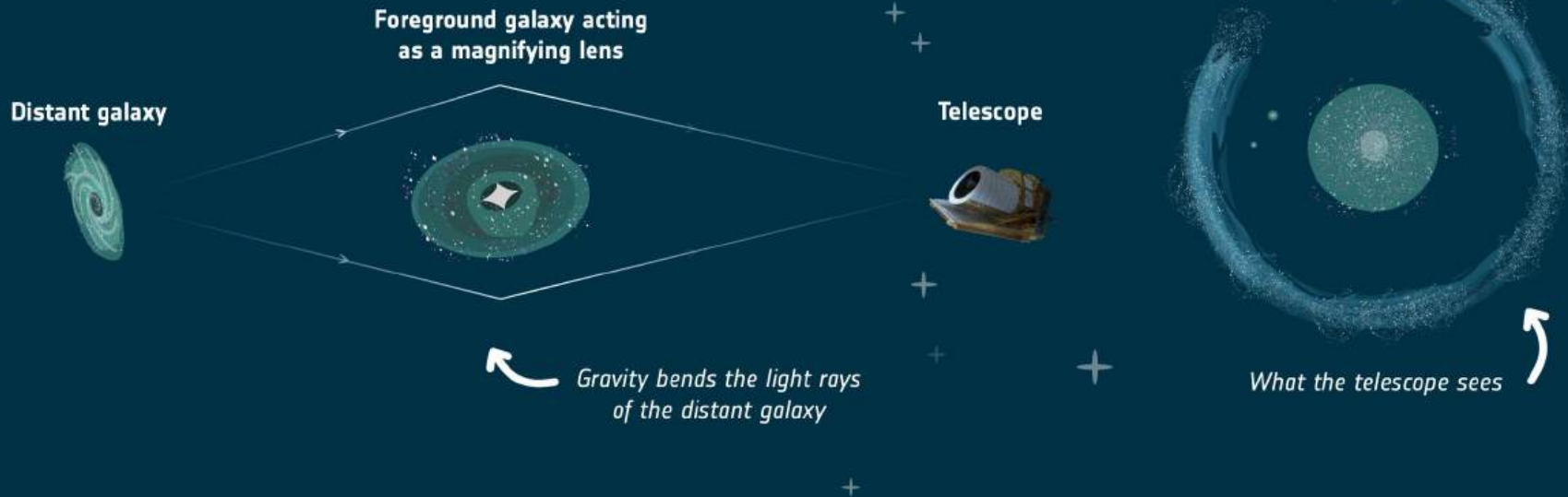






EINSTEIN RING – EXPLAINED

When we observe a distant galaxy with our telescope, its light may encounter another galaxy on its way to us. The foreground galaxy acts like a magnifying lens, bending the travelling light rays due to its gravity. This is called gravitational lensing. If the background galaxy, the lensing galaxy, and the telescope are perfectly aligned, the image appears as a ring – called an Einstein ring.





sa

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When we
way to us
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telescope

Distant g



RX J0911+0551



WGD J0405-3308



HS 0810+2554



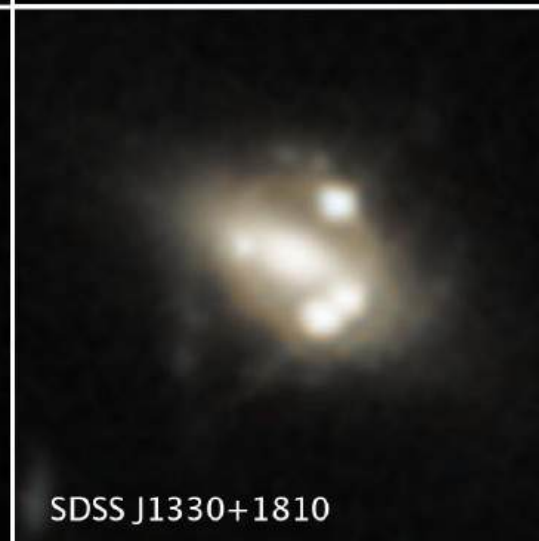
PS J1606-2333



WFI 2033-4723



SDSS J1330+1810





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RX J0911+0551

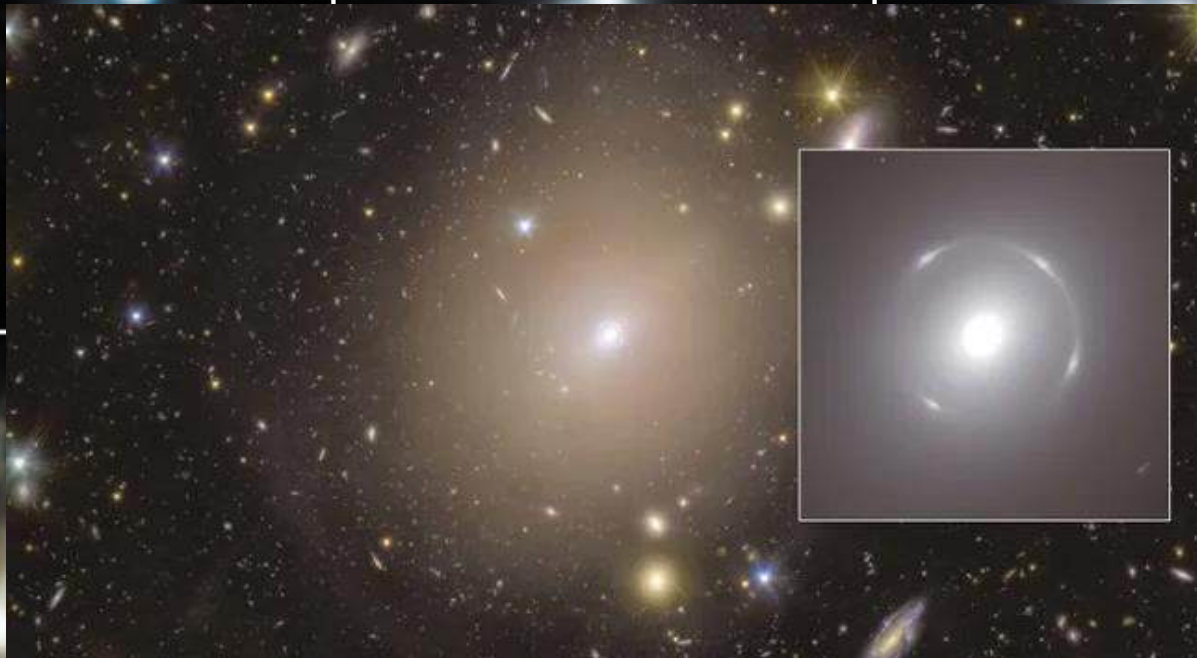
WGD J0405-3308

HS 0810+2554

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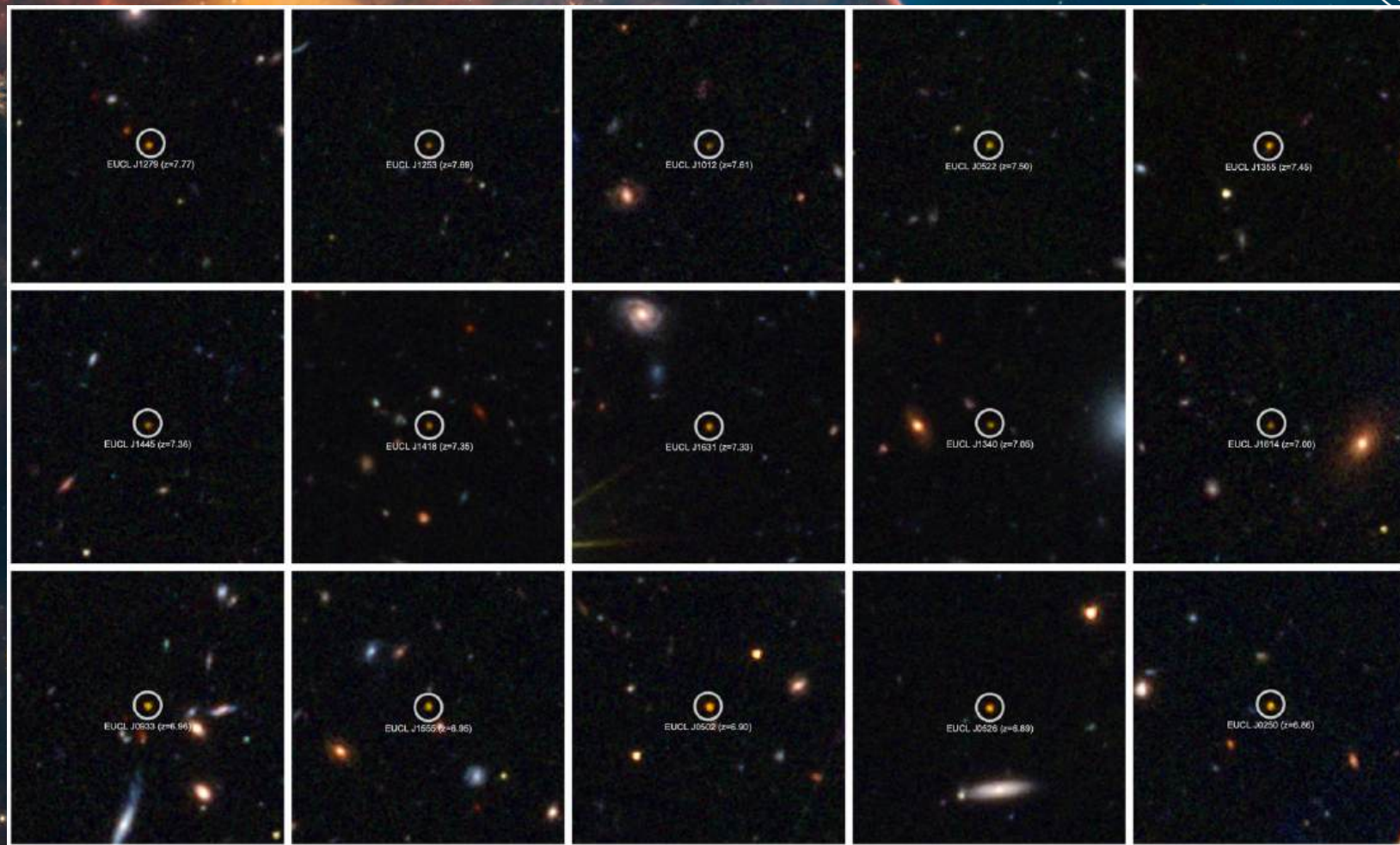


PS J1606-2333

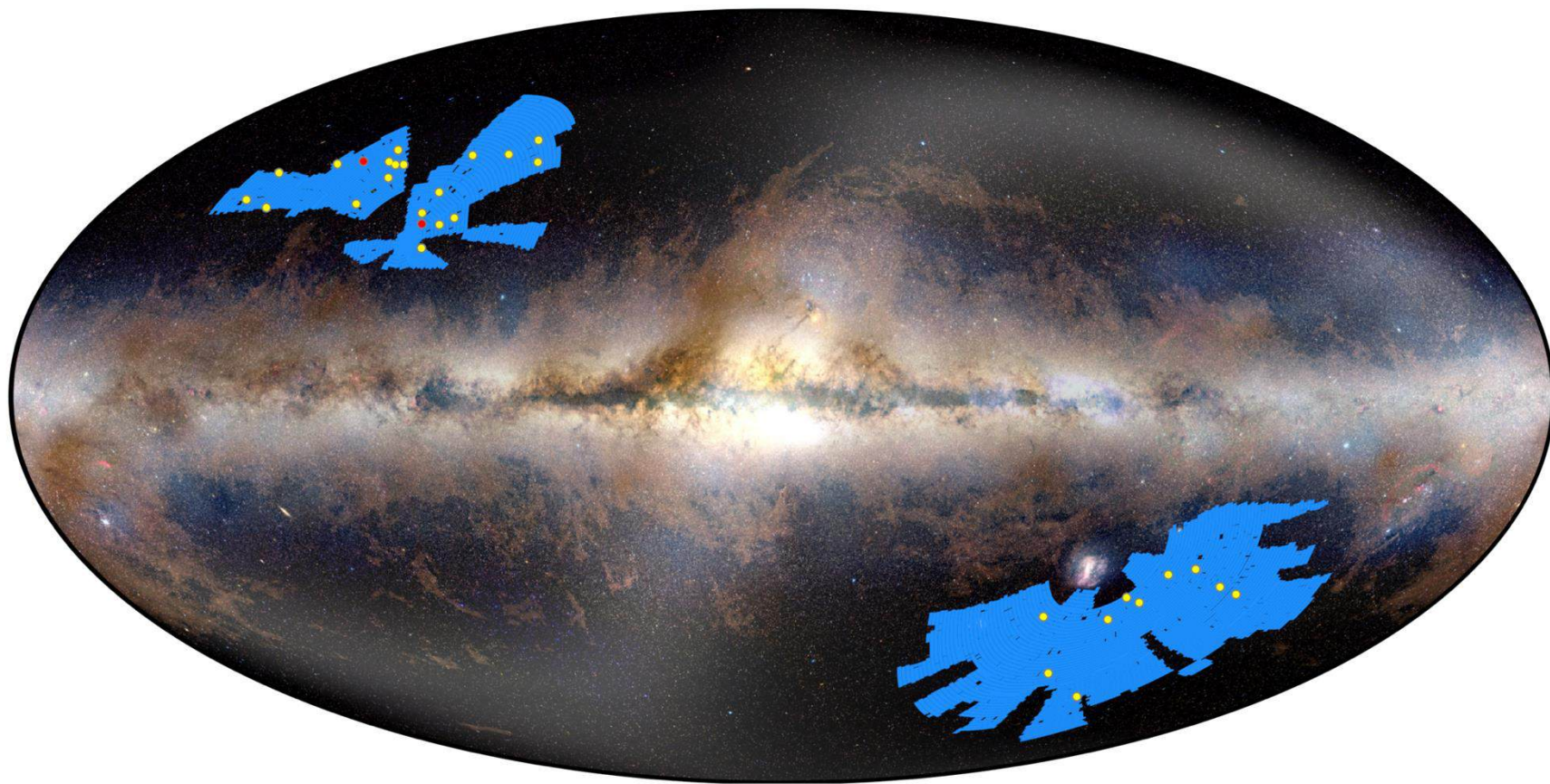
WFI 2033-4723

SDSS J1330+1810

NAJVZDIALENEJŠIE KVAZARY



NAJVZDIALENEJŠIE KVAZARY



WHAT EUCLID WILL MEASURE: WEAK LENSING

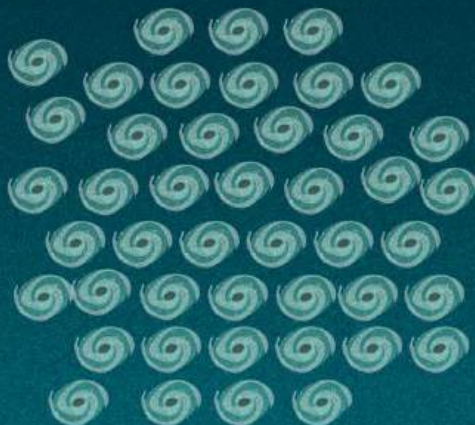
A concentration of matter along the line of sight can act like a magnifying glass, bending and distorting light from galaxies and clusters behind it. This effect is called gravitational lensing. Scientists distinguish between strong gravitational lensing, when the distortions are very apparent, like in the case of Einstein rings, arcs, and multiple images, and weak gravitational lensing, when the distortions of background sources are much smaller. In this case, distortions (of a few percent) can only be detected by analysing large numbers of sources in a statistical way.

Euclid will measure the distorted shapes of billions of galaxies over 10 billion years of cosmic history, providing a 3D view of the dark matter distribution in our Universe. The map of the distribution of galaxies over cosmic time will also teach us about dark energy, which affects the spatial evolution of the large-scale structure.

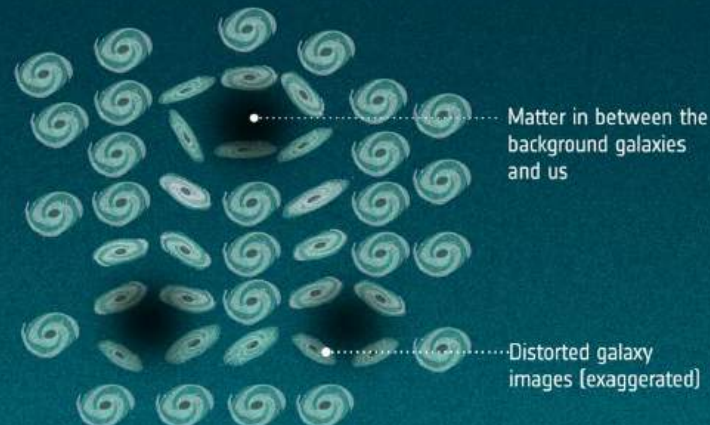
Strong lensing



Unlensed sources



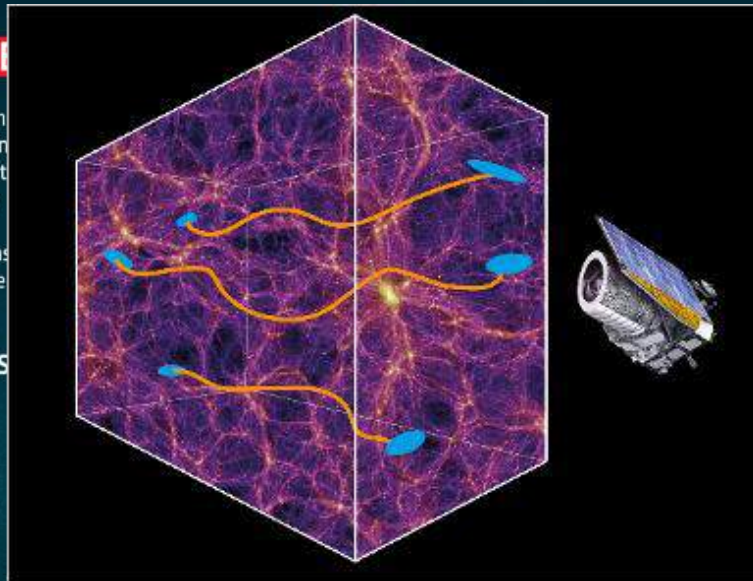
Weak lensing



WHAT IS

A concentration of dark matter. Scientists distinguish between strong gravitational lensing, when the effect is obvious, and weak gravitational lensing, when the effect is subtle and must be detected in a statistical way.

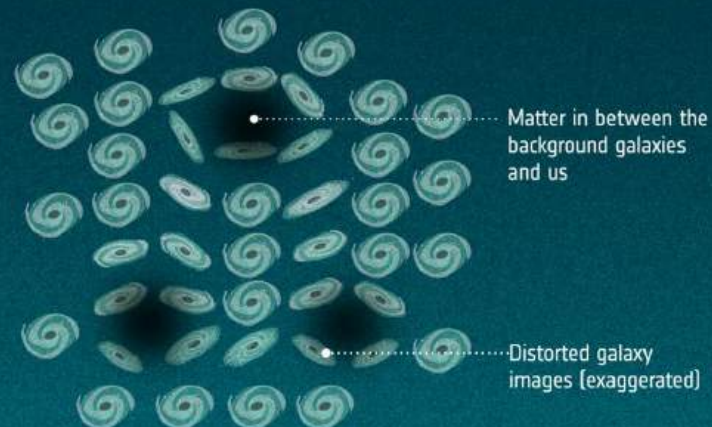
Euclid will measure the dark matter distribution in our Universe. The map of the dark matter distribution will provide a 3D view of the dark matter distribution in our Universe. This map will affect the spatial evolution of the large-scale structure.



ing light from galaxies and clusters behind it. This effect is called gravitational lensing. Like in the case of Einstein rings, arcs, and multiple images, and weak gravitational lensing (a few percent) can only be detected by analysing large numbers of sources in a statistical way.

y, providing a 3D view of the dark matter distribution in our Universe. This map will affect the spatial evolution of the large-scale structure.

Weak lensing

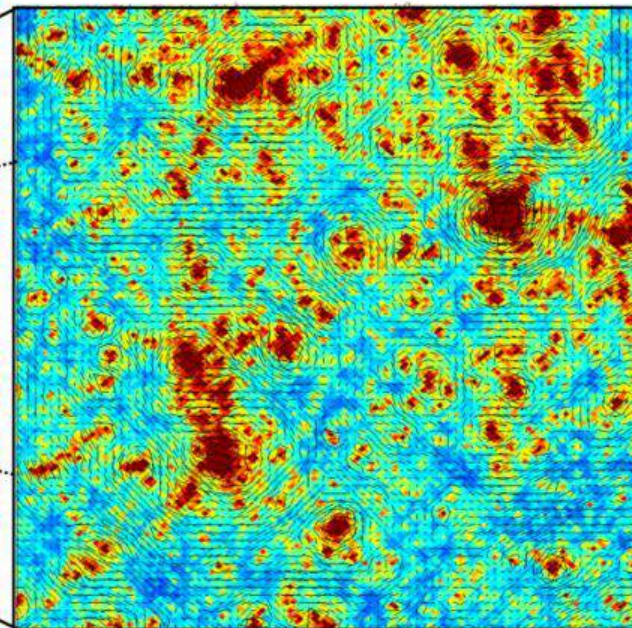
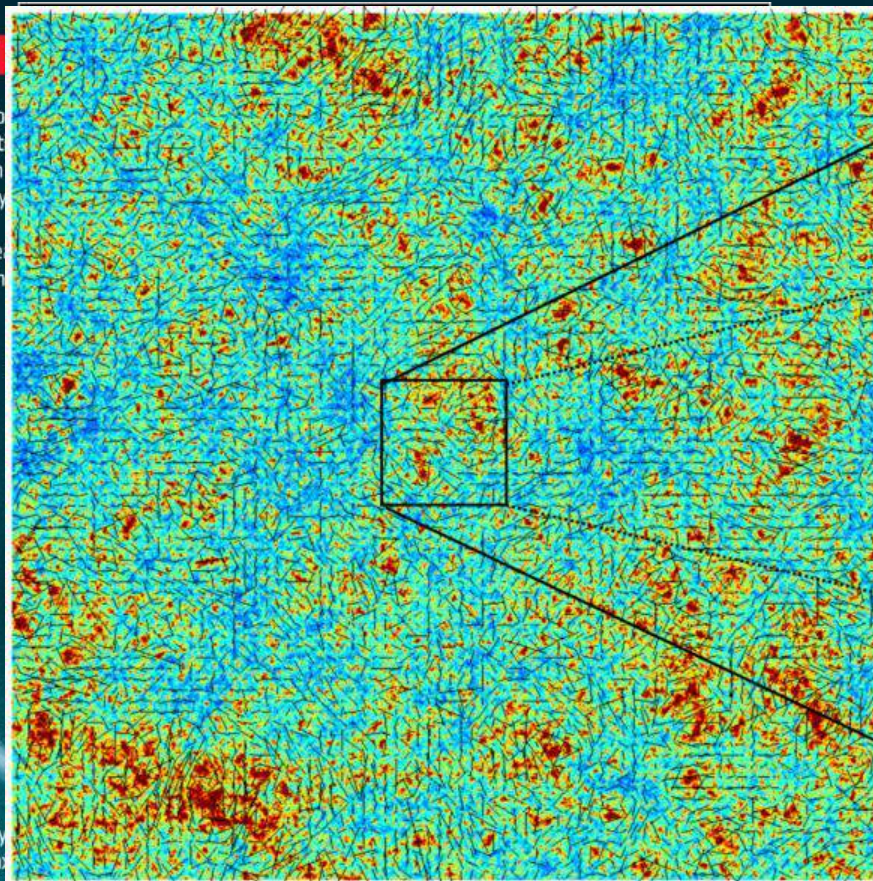


Gravitationally distorted image of galaxy

WHAT

A concentration
Scientists dist
lensing, when
statistical way

Euclid will me
The map of the



between the
and galaxies

and galaxy
(exaggerated)

Gravitational
image of galaxy



VERA RUBIN OBSERVATORY





VERA RUBIN OBSERVATORY



VERA RUBIN OBSERVATORY

- LSST Collaboration (NFS, DOE)
- Cerro Pachón, Chile
- Large Synoptic Survey Telescope
- 2019 -> Vera C. Rubin Observatory
- LSST = Legacy Survey of Space and Time

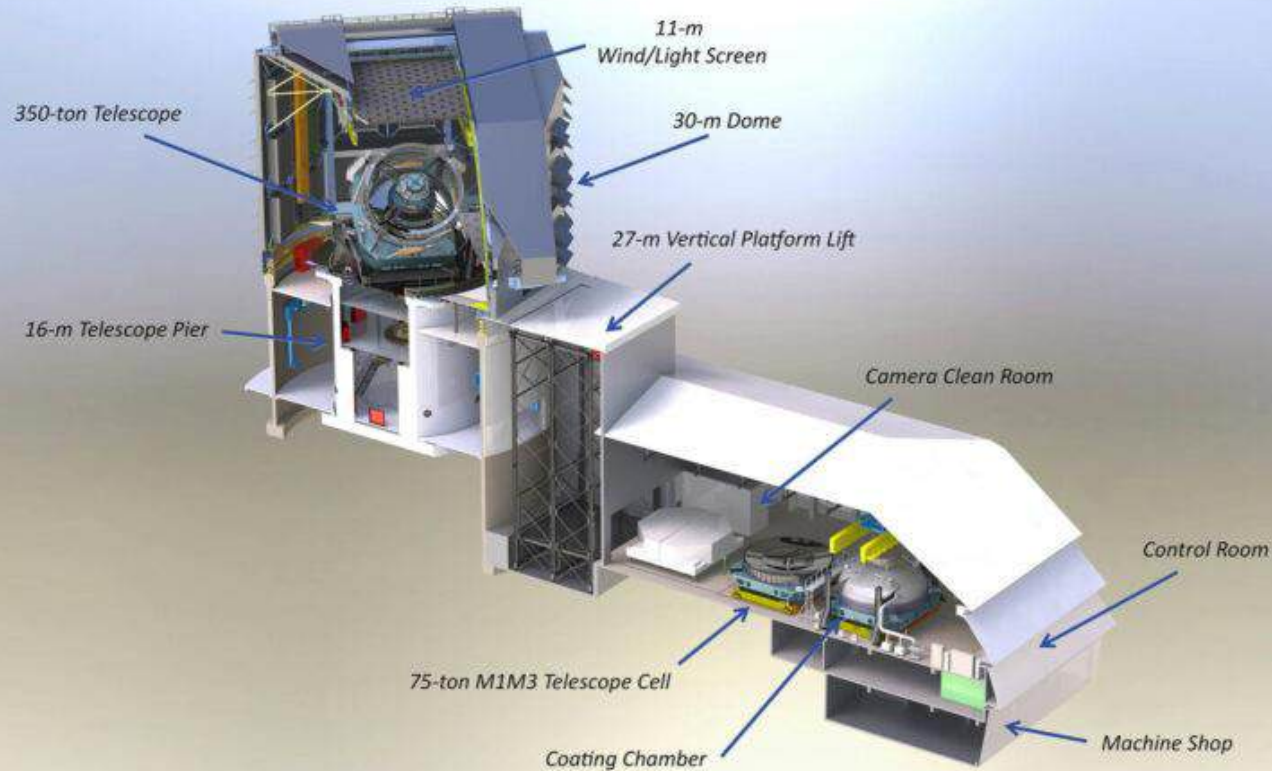


VEDECKÉ CIELE

- Pochopenie podstaty tmavej hmoty a tmavej energie
- Mapovanie Slnecnej sústavy
- Mapovanie Mliečnej cesty
- Skúmanie tranzientných objektov - ktoré sa pohybujú alebo menia jasnosť







350-t

16-m







PARAMETRE

- 8,4m zrkadlo
- 3200 mpx kamera
- 2,8 tony
- 3,5 stupňa veľké pole



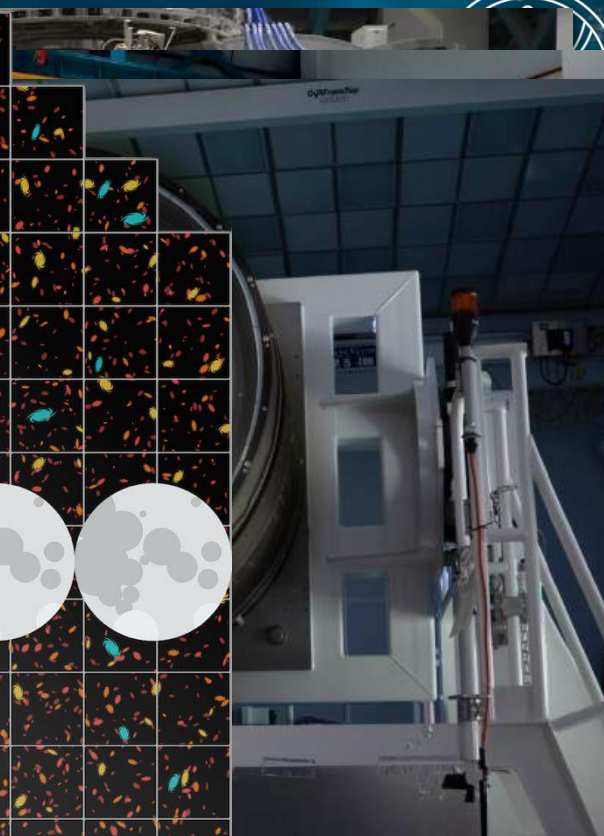
PARAMETRE

- 8,4m z

- 3200 m

- 2,8 ton

- 3,5 stu

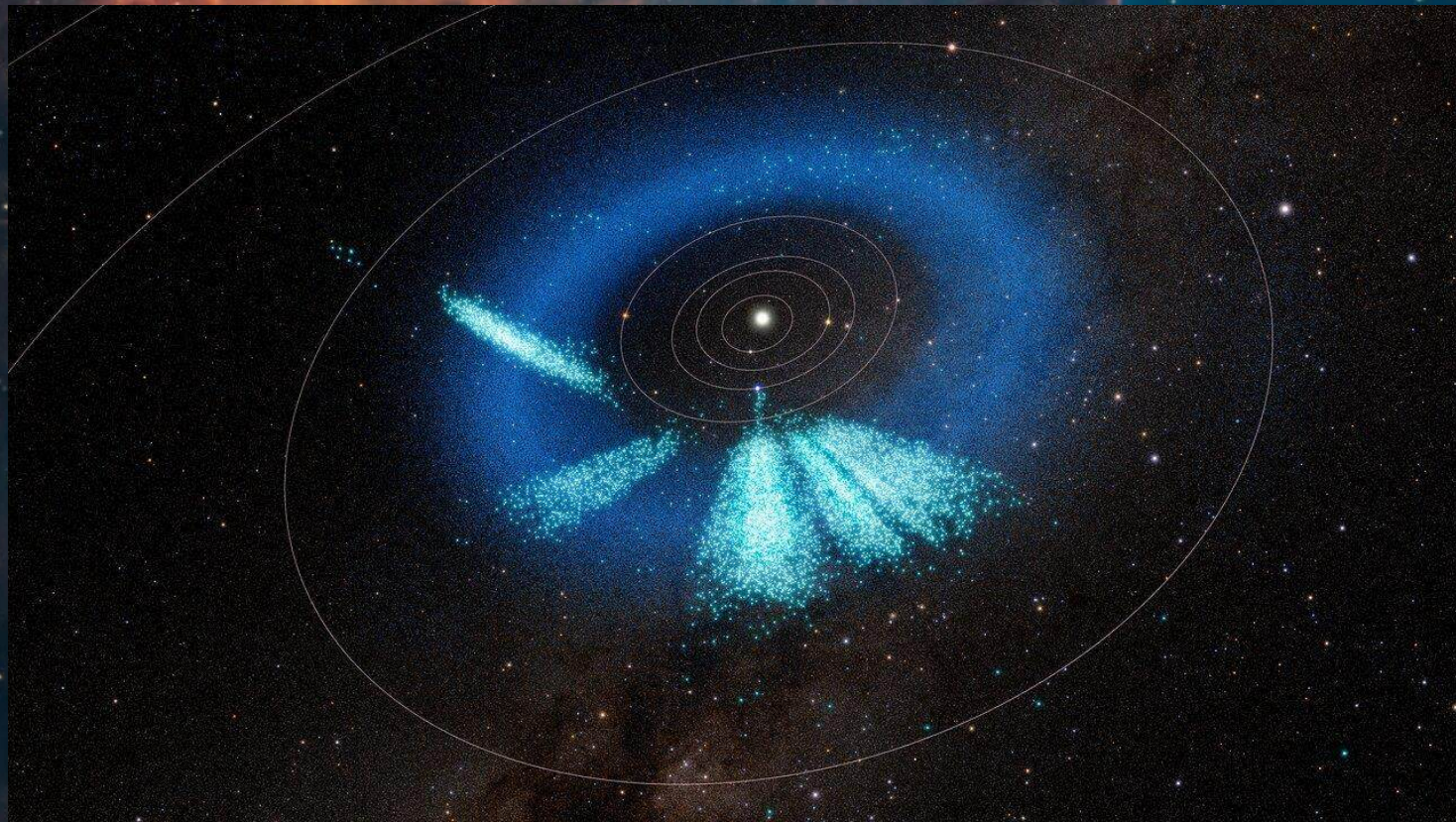


PARAMETRE



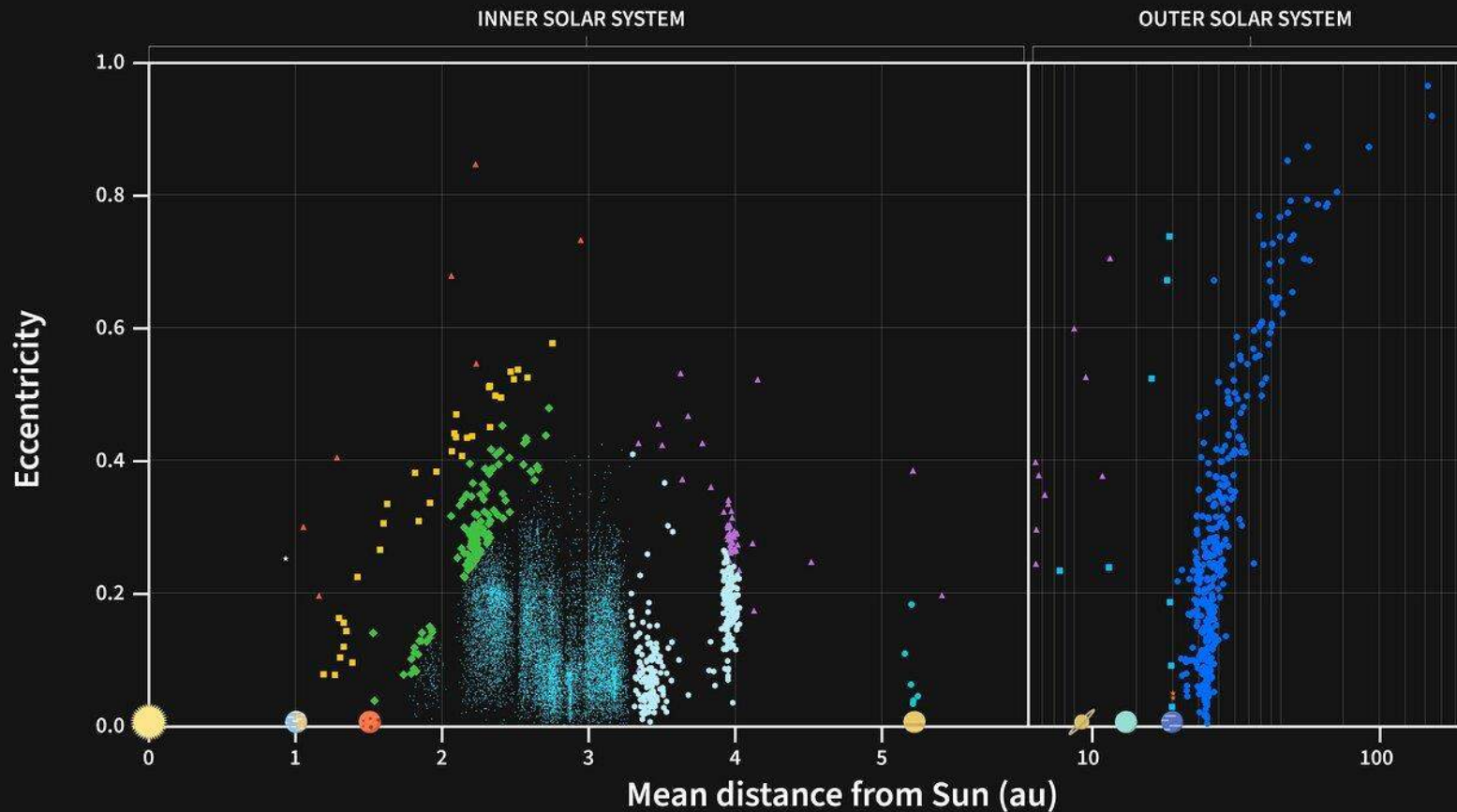
- 20 TB dát za noc (Gaia, Euclid, JWST rádovo 100 GB za deň)
- 37 mld pozorovaných objektov (odhad)
- 23. júna 2025 (verejnosť)
- mozaika 678 pozorovaní za 7h pozorovaní
- Vesmírna truhlica pokladov
- skyviewer.app







NSF-DOE Rubin Observatory's Growing Solar System Inventory



- Amor asteroids
 - ▲ Apollo asteroids
 - Aten asteroids
- Near-Earth Objects

- ◆ Mars-crossing asteroids
- Main belt asteroids
- Outer main belt asteroids
- ◆ Jupiter trojans
- ▲ Jupiter-family comets
- Centaurs

- Neptune trojans
- ◆ Trans-Neptunian objects





NASA NANCY GRACE ROMAN SPACE TELESCOPE



NANCY GRACE ROMAN SPACE TELESCOPE



- Nancy Grace Roman Space Telescope — pomenovaný po Nance Grace Romanovej, NASA prvá žena v riadiacej pozícii v NASA
- NASA infrared wide-field survey observatórium
- Štart 30. augusta 2026 (SpaceX Falcon Heavy z Kennedy Space Center)
- Umiestnenie v L2 (~1.5 mil. km od Zeme)
- Primárne zrkadlo o veľkosti 2,4 m priemer
- Plánovaná misia 5 + 5 rokov



NANCY GRACE ROMAN SPACE TELESCOPE



- Nancy Grace Roman Space Telescope — pomenovaná po
- Romanovej, NASA prvá žena v riadiacej pozícii
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NANCY GRACE ROMAN SPACE TELESCOPE



- Nancy Grace Roman Space Telescope — pomenovaná po Romanovej, NASA prvá žena v riadiacej pozícii



VEDECKÉ CIELE



- Mapovanie galaktických klastrov, meranie tisícov supernov typu Ia a sledovanie vývoja expanzie vesmíru v čase.
- Presné meranie slabého gravitačného šošovkovania na miliónoch galaxií → detailné mapy distribúcie normálnej aj tmavej hmoty naprieč kozmickou históriou.
- Monitorovanie miliónov hviezd v galaktickom strede → očakávaný objav >1200 exoplanét mikročočkami a >100 000 tranzitmi.
- Pokročilý koronograf schopný potlačiť svetlo hviezdy o faktor 1 miliarda → priame snímky Jupiter-like planét a testovanie technológií pre budúce misie zamerané na kamenné planéty v obývateľnej zóne.
- Vďaka veľkému zornému poľu a citlivosti v NIR Roman umožní identifikovať a študovať veľmi vzdialené, slabé galaxie z prvých epoch po Veľkom tresku.





Tracking how stars change in brightness over time to discover distant worlds

Galactic Bulge Time Domain Survey

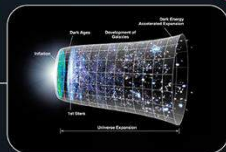
This survey will provide one of the deepest views ever into the heart of our galaxy, making observations every 15 minutes for over a year. Tracking how hundreds of millions of stars change in brightness will unveil 100,000 more exoplanets including rogue planets, find isolated black holes, and help astronomers probe stellar interiors.



Making a 3D map of the universe

High Latitude Wide Area Survey

This survey will combine the powers of imaging and spectroscopy to uncover hundreds of millions of galaxies. Roman's wide and deep 3D images will allow astronomers to explore the structure and expansion of the universe as it evolved through cosmic time.



Revealing the dynamic universe beyond our galaxy

High Latitude Time Domain Survey

Astronomers will use this survey to probe our dynamic universe by observing the same region of the cosmos every five days for two years. Stitching these observations together will create "movies" that will uncover a wealth of transient objects, like exploding stars, stars being swept into black holes, and neutron star mergers.



NANCY GRACE ROMAN SPACE TELESCOPE COMMUNITY SURVEYS

The Roman Space Telescope's exquisite sensitivity and fast survey speed will unveil the evolving universe in ways that have never been possible before.

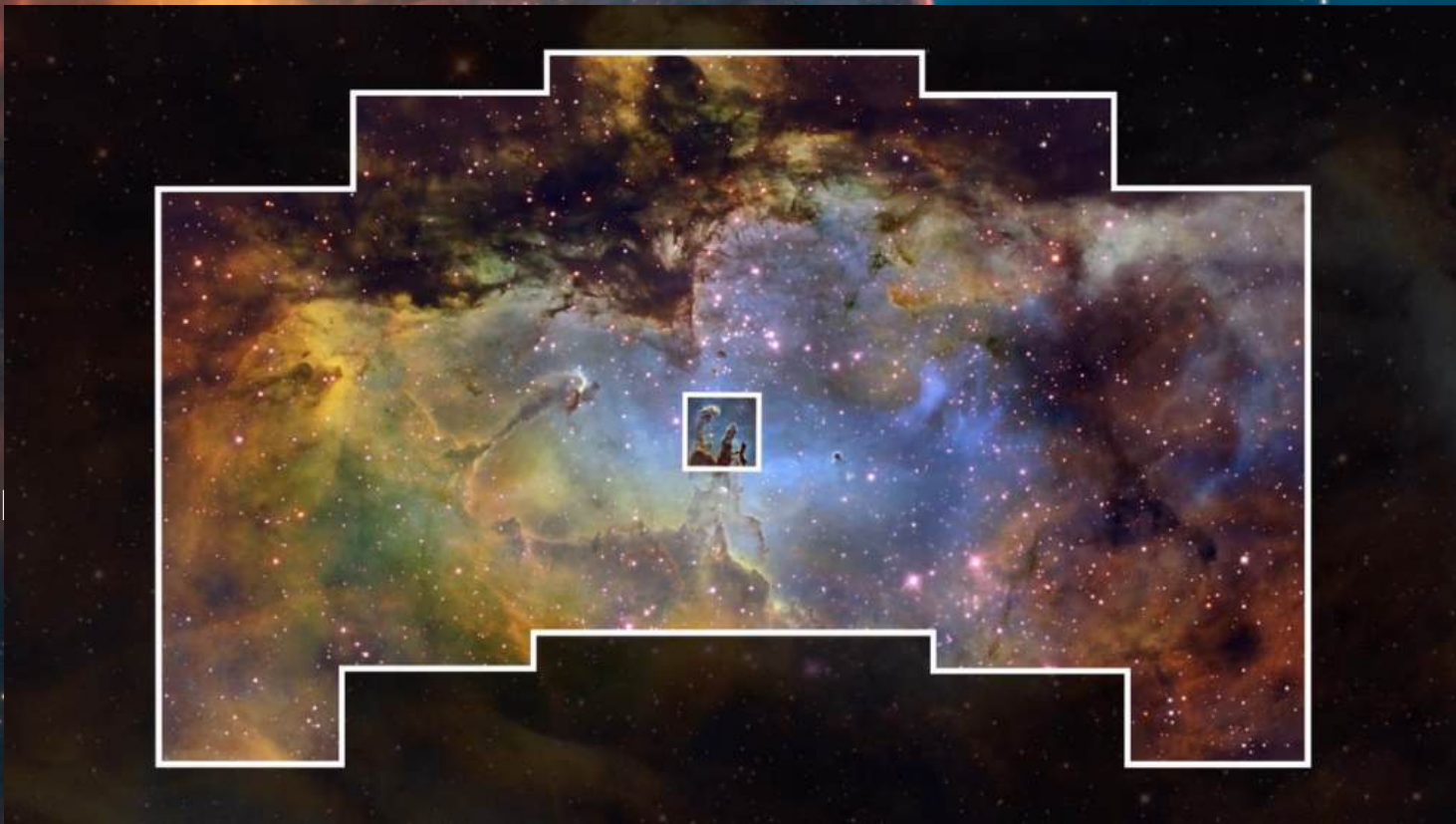
TECHNICKÉ ÚDAJE



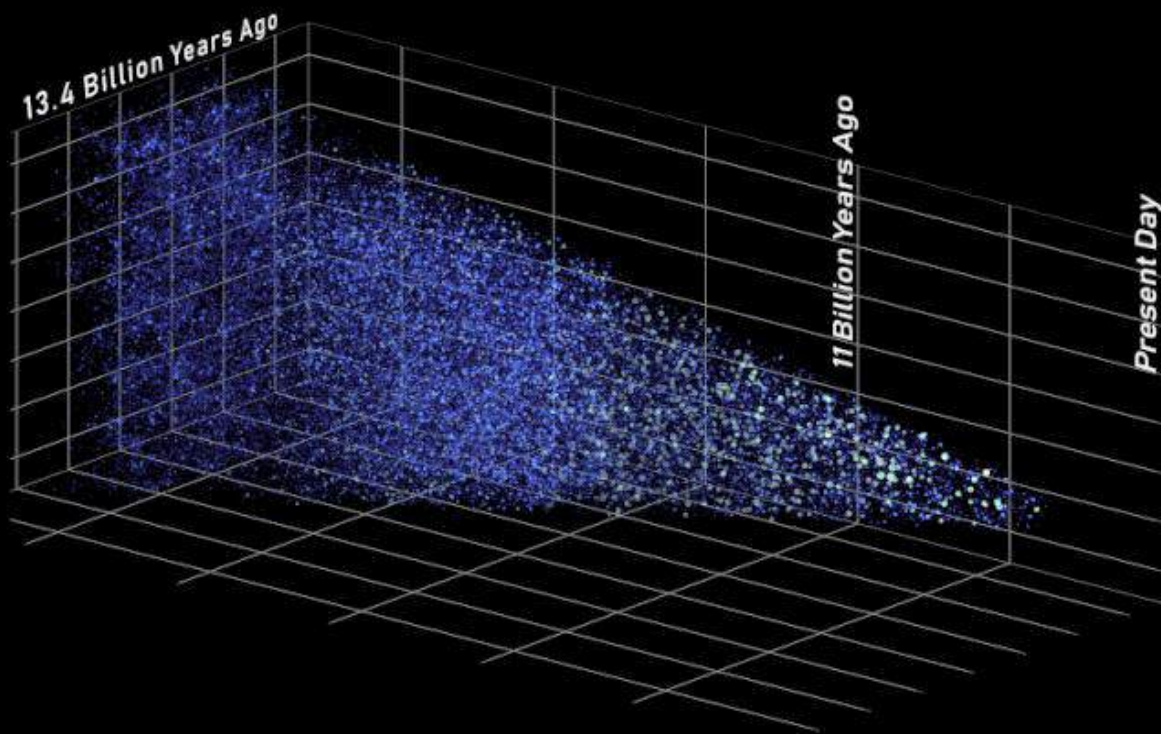
- 2,4 m primárne zrkadlo (rovnaké ako Hubble)
- Wide Field Instrument (WFI) – 300 Mpx VIS+NIR kamera, 0,28 deg² FOV
- Koronograf – potlačenie svetla hviezdy, priame snímky Jupiter-like exoplanét
- Európske príspevky – EMCCD detektory, star trackery, batérie, 35 m anténa New Norcia
- Dátový objem: ~20 PB počas primárnej misie
- Rýchle širokoplošné prieskumy – 200x väčšie IR zorné pole než Hubble



TECHNICKÉ ÚDAJE



TECHNICKÉ ÚDAJE





ČASŤ 3

DÁTA a čo s nimi?

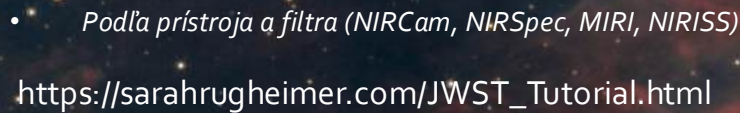


JWST DÁTA



- **MAST** (Mikulski Archive for Space Telescopes) — archive.stsci.edu / mast.stsci.edu
- Webové rozhranie MAST Portal — vyhľadávanie podľa objektu, RA/Dec, Proposal ID, prístroja, ...
- Programový prístup cez Python balík **astroquery.mast** (funguje priamo v Jupyter/Colab)
- Všetky verejné dáta sú voľne dostupné (po uplynutí exkluzívneho obdobia, typicky 12 mesiacov)
- **Ako vyhľadávať?:**
 - *Podľa mena cieľa alebo súradníc (RA/Dec)*
 - *Podľa čísla vedeckého programu (Proposal ID)*
 - *Podľa prístroja a filtra (NIRCam, NIRSpec, MIRI, NIRISS)*
- https://sarahrugheimer.com/JWST_Tutorial.html





FORMÁT SÚBOROV



- Primárny formát je **FITS** (Flexible Image Transport System) [jwst-docs.stsci](#)
- Doplnkové formáty: **ASDF**, **JSON**, **ECSV** (kalibračné a metadáta) [jwst-docs.stsci](#)
- **Štyri stupne spracovania:**
 - **Uncalibrated** (*_uncal.fits*) — surové dáta priamo z detektora
 - **CountRate** (*_rate.fits*) — korekcia detektora, jednotky DN/s (Data Number za sekundu)
 - **Calibrated** (*_cal.fits*) — fotometrická a astrometrická kalibrácia (WCS)
 - **Combined** (*_i2d.fits* obrázky, *_s3d.fits* spektrálne kocky, *_x1d.fits* extrahované spektrá) — finálny kombinovaný produkt
- Väčšina vedeckej analýzy pracuje s už kalibrovanými dátami
- Spektroskopické dátové kocky (IFU) → *_s3d.fits* (3D: dve priestorové + jedna spektrálna os)
- Spracovanie robí automaticky **JWST Science Calibration Pipeline** (open-source, Python)
- Dátový objem je rádovo desiatky GB na jeden vedecký program



FITS



- **FITS** = Flexible Image Transport System — štandardný dátový formát celej astronómie (obrázky, spektrá, tabuľky, ...), zavedený už v 80. rokoch a dodnes univerzálny
- Používajú ho všetky veľké observatóriá a misie — Hubble, JWST, Gaia, Euclid, Vera Rubin, Roman
- **Štruktúra súboru:**
 - **Header** (hlavička) — textové metadáta: súradnice cieľa (WCS), expozičný čas, filter, prístroj, dátum pozorovania
 - **Data** (dátová časť) — samotné číselné hodnoty: pixely obrázka, hodnoty spektra, alebo binárna tabuľka
- Jeden FITS súbor môže obsahovať **viacero HDU** (Header/Data Units) — napr. obrázok + chybová mapa + maska kvality dát v jednom súbore
- **Typy obsahu:** 2D obrázok, 1D spektrum, 3D dátová kocka (IFU spektroskopia), binárna/ASCII tabuľka (katalógy)
- **Otvorenie v Pythone:** knižnica `astropy.io.fits` (`fits.open("file.fits")`) — štandardný nástroj v celej komunite
- Prípona súboru: `.fits` alebo `.fit`; dá sa čítať aj voľne dostupnými nástrojmi ako **DS9** (vizualizácia) bez programovania



GAIA DÁTA



- **Gaia Archive** (ESA) — archives.esac.esa.int/gaia — jediný centrálny bod prístupu k dátam Gaia
- Dáta zrkadlené aj u partnerských centier: CDS (Strasbourg), ARI, AIP, ASDC, Flatiron
- Aktuálna verzia: **Gaia DR3** (2022) — DR4 plánovaný december 2026
- **Tri spôsoby dotazovania**
 - **ADQL Search** — webový formulár priamo v archíve (grafické rozhranie, žiadny kód)
 - **ADQL** = *Astronomical Data Query Language* — rozšírenie SQL o astronomické funkcie (CIRCLE, DISTANCE, CONTAINS) na definovanie oblasti na nebi
 - **TAP+ protokol** — programový prístup cez astroquery.gaia v Pythone (funguje priamo v Google Colab) <https://tapvizier.cds.unistra.fr/adql/>



GAIA DÁTA



- **Gaia Archive (ESA)** — archives.esac.esa.int/gaia — jediný centrálny bod prístupu k dátam

-- output format : csv

- Dáta zriadené aj u partnerských centier: CDS (Strasbourg), ARI, AIP, ASDC, Flatiron

- ```
SELECT "I/354/starhorse2021".Source, "I/354/starhorse2021".dist50,
"I/354/starhorse2021".XGal, "I/354/starhorse2021".YGal,
"I/354/starhorse2021".ZGal
```

*ADQL Search — webový formulár priamo v archive (grafické rozhranie, žiadny kód)*

```
FROM "I/354/starhorse2021", "I/355/gaiadr3"
```

*ADQL — Astronomical Data Query Language — rozšírenie SQL o astronomické funkcie (CIRCLE, DISTANCE, CONTAINS) na definovanie oblastí na nebi*

```
WHERE "I/354/starhorse2021".Source="I/355/gaiadr3".Source AND
"I/355/gaiadr3".RUWE<1.4 AND "I/355/gaiadr3".Solved>30
```



# GAIA DÁTA



- **Formáty výstupu**
  - *VOTable* (XML, štandard Virtual Observatory)
  - *FITS* (binárne, kompatibilné s astropy)
  - *CSV / ECSV* (Enhanced CSV — CSV s YAML metadátami v hlavičke)
- Voľba formátu priamo pri stiahnutí výsledku dopytu
- **Hlavné tabuľky v archíve**
  - *gaia\_source* — hlavný katalóg: pozícia, paralaxa, vlastný pohyb, fotometria (~1,8 miliardy zdrojov, 757 GB)
  - *astrophysical\_parameters* — odvodené parametre: teplota, polomer, extinkcia (~1,6 miliardy zdrojov)
  - Špecializované tabuľky: radiálne rýchlosti (RVS spektrá), premenné hviezdy, kvazary, galaxie





# GAIA DÁTA



- Kompletný DR3 dostupný ako súbor komprimovaných **ECSV** súborov organizovaných podľa **HEALPix** indexov (Hierarchical Equal Area isoLatitude Pixelisation)
- Pre väčšinu vedeckých úloh postačuje cielený ADQL dotaz — nie je potrebné sťahovať celý katalóg
- `pip install astroquery` → `from astroquery.gaia import Gaia` → `Gaia.launch_job()`
- Dátový model (popis všetkých stĺpcov) je voľne dostupný v online dokumentácii archívu



## GAIA DR4



- **Dátum vydania: 2. december 2026**
- Prvý **hlavný** (principal) release Gaia — na rozdiel od DR3 založený na **66 mesiacoch** pozorovaní (júl 2014 – január 2020), teda celej nominálnej misii
- Referenčná epocha: **J2017.5**
- Celkový objem dát: **~400 TB**





## GAIA DR4



- **Epoch data (Level 2)** — po prvýkrát publikované úplné časové rady jednotlivých meraní (nie len priemerné hodnoty): astrometria, fotometria, radiálne rýchlosti, BP/RP a RVS spektrá (650 miliónov)
- Samotné epoch\_astrometry: **~130 miliárd** meraní, **62 TB** dát
- Epoch\_image (surové CCD snímky): **113 TB** — najväčšia jednotlivá zložka DR4
- **Žiadne kvalitatívne filtrovanie** — DR4 publikuje výsledky pre všetkých ~2,5 miliardy zdrojov z interného zoznamu (DR3 filtroval slabšie/menej isté zdroje)







## EUCLID DÁTA

- **Euclid Science Archive (ESDC)** — [eas.esac.esa.int/sas](https://eas.esac.esa.int/sas) — hlavný bod prístupu k dátam (test napr objekt HOPS 221)
- Aktuálna verzia: **Q1 (Quick Release 1)** — publikovaná 19. marca 2025, ~63,1 deg<sup>2</sup> pokrytia, ~26 miliónov detekcií
- Q1 pokrýva 4 hlboké polia: EDF-North (22,9 deg<sup>2</sup>), EDF-Fornax (12,1 deg<sup>2</sup>), EDF-South (28,1 deg<sup>2</sup>), LDN1641 — celkovo ~**35 TB** dát
- Časť dát dostupná aj cez **ESA Sky** ([sky.esa.int](https://sky.esa.int)) — jednoduchý vizuálny prehliadač bez potreby registrácie





# EUCLID DÁTA



- **Webové rozhranie (SAS)** — vyhľadávanie, cutout služba, priame stiahnutie
- **ADQL dotazy** — IVOA Table Access Protocol (TAP), rovnaký princíp ako Gaia
- **Python: astroquery.esa.euclid** — dedikovaný balík na dotazovanie katalógov aj stiahnutie FITS súborov
- **ESA Datalabs** (datalabs.esa.int) — cloudová e-science platforma s predinštalovanými nástrojmi a tutoriálmi — odporúčaná pre veľké FITS súbory





# EUCLID DÁTA



- Všetky dáta (obrázky aj katalógy) sú vo formáte **FITS**
- Katalógy dodatočne uložené v archívnej databáze — dotazovateľné cez ADQL bez nutnosti stiahnuť FITS [pubs.euclid-ec](https://pubs.euclid-ec.europa.eu/)
- **Hlavné typy produktov**
  - **VIS/NIR kalibrované snímky** — *multi-extension FITS (science image + flag map + noise map v jednom súbore)*
  - **MER mozaiky** — *zlúčené veľkoplošné obrázky (background-subtracted mosaic, RMS, flag, PSF model)*
  - **Katalógy:** **MER** (fotometria, morfológia), **PHZ** (fotometrické red-shifty), **SPE** (spektroskopické red-shifty, spektrálne čiary)
  - **1D spektrá** — *kombinované spektrá zo SIR (Spectroscopic Instrument)*





# EUCLID DÁTA



- `pip install astroquery` → `from astroquery.esa.euclid import Euclid`
- Výstupné formáty dotazu: `votable`, `fits`, `csv`, `json`
- Kvôli veľkosti FITS súborov (obrázky = niekoľko GB) je pre praktické cvičenia vhodnejšia práca s katalógmi (ADQL) než hromadné stiahnutie obrázkov





# EUCLID DÁTA



- **Aktuálne:**
  - *Early Release Observations* — máj 2024
  - *Q1 (Quick Data Release 1)* — marec 2025,  $63,1 \text{ deg}^2$ , ~26 mil. *Q2 (Quick Data Release 2)* — 24. jún 2026, Euclid Galactic Bulge Survey — 60 miliónov hviezd v galaktickom bugu,  $4,8 \text{ deg}^2$
- **Nadchádzajúce release:**
  - *DR1-Foundation* — 12. november 2026 — surové dáta + kalibrované obrázky, katalógy, spektrá,  $\sim 1900 \text{ deg}^2$
  - *DR1 (kompletný)* — polovica 2027 — doplnenie vyššej úrovne produktov pre kozmologické analýzy (galaxy clustering, weak lensing)
  - *Q3 (Quick Release 3)* — 2028
  - *DR2* — jún 2029
  - *Q4 (Quick Release 4)* — 2030
  - *DR3 (finálny)* — október 2031 — rok po ukončení hlavného prieskumu
- Celkový plánovaný objem dát za misiu: **>170 petabajtov** surových snímok spracovávaných deviatimi Science Data Centrami





# VERA RUBIN OBSEKRVATORY - DÁTA



- **data.lsst.cloud** — cloudová platforma (Google Cloud) integrujúca dáta, výpočtový výkon a nástroje priamo "pri dátach" [cloud.google.com](https://cloud.google.com/)
- Tri rozhrania v jednom prostredí: [rtn-o11.lsst](#)
  - **Portal** — grafické webové rozhranie, dotazovanie bez programovania
  - **Notebook** — hostované JupyterLab prostredie pre skriptovanú analýzu
  - **API** — vzdialený prístup cez IVOA štandardy (TAP protokol), rovnaký princíp ako Gaia/Euclid





# VERA RUBIN OBESERVATORY - DÁTA



- Rubin/LSST **nie je voľne verejne prístupné** ako Gaia či Euclid — vyžaduje sa štatút *"data rights holder"*
- Prístup majú: vedci/študenti na akreditovaných US inštitúciách, vedci/študenti v Chile, členovia medzinárodných partnerských tímov (in-kind)
- **Výnimka — verejne dostupné bez obmedzenia:**
  - **Alert stream** — okamžite verejný, do ~60 sekúnd od záberu (prenášaný cez "alert brokers")
  - **Prompt Products Database** — verejná bez nutnosti data rights
  - Po **2-ročnom embargu** sa aj Prompt-processed obrázky a ročné Data Releases stávajú plne verejnými







# VERA RUBIN OBESERVATORY - DÁTA

- **Data Preview 1 (DP1)** — dáta z komisioning kamery (LSSTComCam), koniec 2024, **3,5 TB**
- **Early Data Preview 2 (EDP2)** — dáta z LSST Camera, apríl 2025 – január 2026, obmedzené obrázkové produkty (len deep coadd), ale kompletne katalógy
- Plný start hlavného 10-ročného prieskumu: **30. jún 2026**
- **Formát dát:**
  - **Obrázkové produkty:** *FITS — raw, calibrated single-epoch (visit\_image), coadded (deep\_coadd), difference images*
  - **Katalógové produkty:** *uložené v databáze **Oserv**, dotazovateľné cez TAP (ADQL) alebo **Data Butler** — nie sú to samostatné súbory na stiahnutie*
  - **Data Butler** — *Python knižnica špecifická pre Rubin, spája obrazové dáta (pixels, masky, PSF model) s metadátami do jedného objektu*







# ROMAN SPACE TELESCOPE - DÁTA

- Roman dáta **nemajú žiadne proprietary (exkluzívne) obdobie** — na rozdiel od Hubble/JWST (typicky 12 mesiacov) sú dostupné verejne **okamžite po spracovaní**
- Dôvod: obrovský objem dát (20 PB za 5 rokov) robí exkluzívny prístup nepraktickým — cieľom je maximalizovať vedeckú návratnosť
- Archív: **MAST** (rovnaký ako Hubble a JWST), hostovaný na **Amazon Web Services (AWS)**







# ROMAN SPACE TELESCOPE - DÁTA

## ○ Systém úrovni spracovania dát (L1–L4)

- **L1** — surové, nekalibrované obrázky (FITS) s telemetrickými metadátami — do 72 hodín od pozorovania
- **L2 (Prompt)** — kalibrované jednotlivé 2D snímky WFI (Wide Field Instrument) — do ~48 hodín od stiahnutia dát
- **L3 (Prompt)** — mozaikové/kombinované obrázkové produkty — do 5 dní od posledného potrebného L2 vstupu
- **L4 (Prompt)** — katalógy zdrojov — do 7 dní







# ROMAN SPACE TELESCOPE - DÁTA

- **Prompt products** — rýchly prístup ku kalibrovaným dátám čoskoro po pozorovaní, na podporu skorého skúmania a tranzientnej vedy
- **Data Release (DR) produkty** — periodické, jednotne prepracované, plne validované produkty pre dlhodobé vedecké využitie — dostupné približne **6 mesiacov** po zbere všetkých potrebných pozorovaní
- L2 DR produkty **nahrádzajú** staršie prompt verzie (rovnaká prax ako Hubble/JWST); vyššie úrovne (L3, L4) sa zachovávajú ako stabilné referencie







NIE JE TOHO UŽ  
PRIVEĽA?



## NIE JE TOHO UŽ PRIVEĽA?

- Vera Rubin generuje 20 TB dát za noc. Budúce misie (napr. plánované malé satelitné kontštalacie) nemajú kapacitu všetko odoslať na Zem — downlink je limitovaným zdrojom.
- AI modely priamo na satelite filtrujú, kompresujú alebo klasifikujú dáta pred odoslaním







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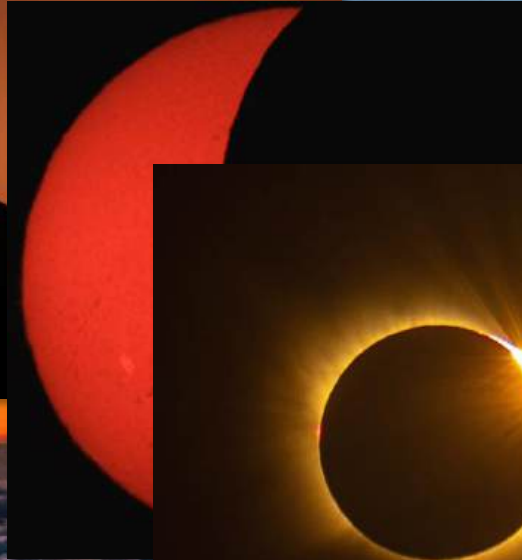


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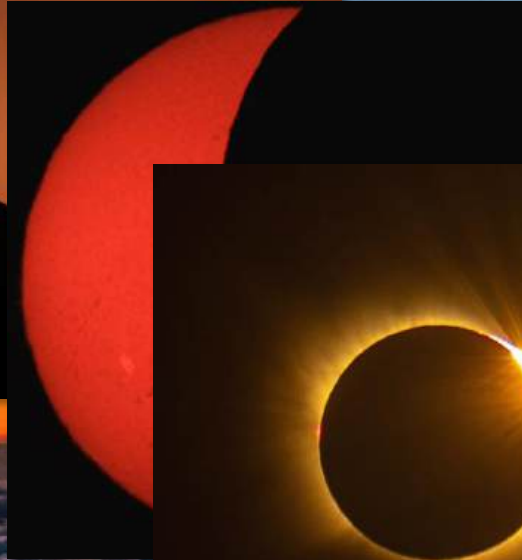
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